

How to build a Passivhaus

GOOD PRACTICE GUIDE



Second edition

How to build a Passivhaus

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Passivhaus Trust
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All the case studies featured in this guide are covered in greater detail on the Passivhaus Trust website, including information about project teams and performance - find out more at pht.guide/howto

Look out for QR codes throughout to take you to more in depth resources.

Find Passivhaus expertise for further support via the links at the back of this guide.

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Introduction

Sarah Lewis

This guide highlights the key aspects of designing to the Passivhaus standard and the dos and don'ts for how to successfully build a Passivhaus.

CLIMATE AND COST-OF-LIVING CRISIS

If we are to prevent climate breakdown, we need to radically reduce energy demand and drive building energy efficiency. We cannot simply generate more zero-carbon energy supply, because of the limitations of grid capacity, peak load, and demands for renewable energy from other sectors.

EFFICIENCY FIRST

When determining the most impactful way to cut emissions in the built environment, we must consider embodied and operational carbon holistically. Passivhaus prioritises efficiency, of both energy and material resources. Passivhaus buildings are optimised for net zero, providing the best route to minimise whole life carbon. Outstanding levels of building performance minimise operational carbon, while the Passivhaus design methodology can also be used to encourage optimisation of embodied carbon. This can be through the optimisation of the building's form factor leading to a more efficient use of materials and through reducing the heating and cooling systems.

"I was working as a physicist. I read that the construction industry had experimented with adding insulation to new buildings and that energy consumption had failed to reduce. This offended me – it was counter to the basic laws of physics. I knew that they must be doing something wrong. So I made it my mission to find out what, and to establish what was needed to do it right."

Prof. Dr. Wolfgang Feist

Passivhaus delivers outstanding levels of energy efficiency, typically using half the energy of a new build home. It enables the levels of demand reduction that we will need to achieve net zero nationally and globally. Passivhaus also provides exceptional levels of interior comfort, health, wellbeing, affordability, resilience, and durability.

PASSIVHAUS BUILDINGS

- Are optimised for net zero
- Consistently perform to design targets
- Are the best fit for a decarbonised grid
- Deliver health and wellbeing benefits
- Tackle fuel poverty

PASSIVHAUS SOLUTIONS

Passivhaus is an international tool, backed by 30 years of evidence, that provides us with a range of proven approaches to deliver new and existing buildings optimised for net zero. The Passivhaus standard provides solutions to the current hurdles to achieving net zero in the UK. This is no time to reinvent the wheel – the tools we need to deliver on our climate pledges exist today, so let's get to work!

What makes Passivhaus different from the plethora of other standards out there is its design and construction quality. This effectively eliminates the 'performance gap', delivering excellent performance in-use. The 'performance gap' is the difference between the assumed energy performance of a building based on its design and the energy performance a building actually achieves. From the monitoring of thousands of built Passivhaus buildings, energy use on average is extremely close to the amount that the modelling predicts. An average non-Passivhaus new home, by contrast, can have a heating demand of 60% more than that forecast using SAP modelling.

Certified Passivhaus projects demonstrate that performance is delivered through high quality design and construction, rather than just through adding more insulation or a high-performance component or two.

Passivhaus is not a trademark. However, if someone claims that a building is a Passivhaus, they are claiming that it meets the standard's strict energy and comfort requirements. Unlike 'eco-home' or 'sustainable building', to describe a building as a Passivhaus is a strong and verifiable claim under Trading Standards. The Passivhaus Trust document *Claiming the Passivhaus Standard* (PHT, 2022) clarifies this.



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claimingPassivhaus

Passivhaus buildings provide a high level of occupant comfort while using very little energy for heating and cooling. They are built with meticulous attention to detail and rigorous design and construction quality according to principles developed by the Passive House Institute in Germany and are certified through an exacting quality assurance process.

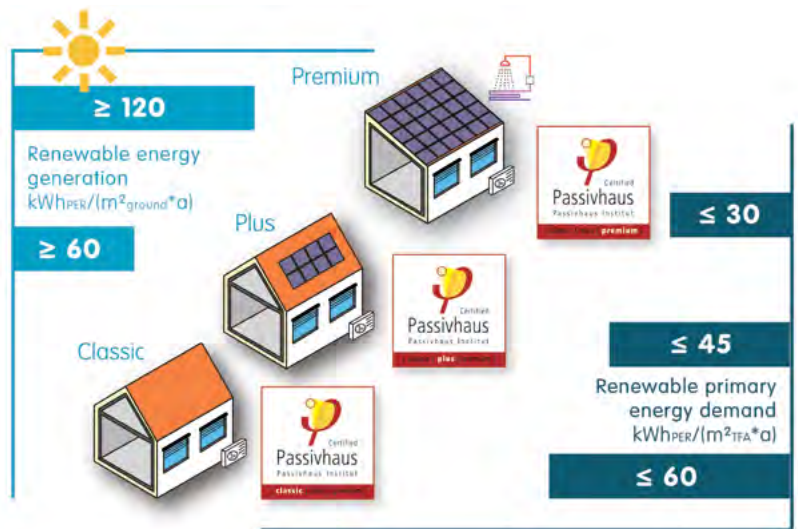


PASSIVHAUS STANDARDS: CLASSIC, PLUS & PREMIUM

As Passivhaus continues to grow in the UK and the rest of Europe, many projects are using the Passivhaus Standard to deliver against net zero targets. A robust way to achieve this is to reduce energy demand to the Passivhaus Classic standard level and source renewable energy to meet the remaining demand. The renewable energy might be generated on-site or at a more economic scale; large scale renewables have the potential for a much lower life cycle cost and lower embodied carbon than domestic building-mounted PV.

For those who wish to demonstrate compliance with emerging definitions of net zero at the building level, **Passivhaus Plus** and **Passivhaus Premium** offer a rigorous methodology. These standards include the use of renewable energy generation in the certification assessment. They also require a further reduction in demand, making them suitable for more ambitious projects.

The **Passivhaus Plus** standard not only drastically reduces energy use, but for a typical two storey family home it will also produce, from renewable sources, as much energy as occupants consume, turning them into Passivhaus powerhouses. By most UK definitions, these would be considered net zero buildings, although this is not the most efficient way to meet national or global net zero targets.



The Passivhaus classes

For a typical two storey family home certified to the **Passivhaus Premium** standard, far more energy is produced than needed, even once storage losses are accounted for. It is therefore a goal for the particularly ambitious: building owners and designers who want to go beyond what economic and ecological considerations already propose. The Passivhaus Classic standard along with these additional classes make the Passivhaus standard the most appropriate target for our net zero future.

Top left: Chiswick Eco Lodge, Passivhaus Classic
Bottom left: Bristol Passivhaus Plus
Below: Erne Campus, Passivhaus Premium



Passivhaus criteria	Classic (new build)		EnerPHit (retrofit)	Plus (new build + retrofit)	Premium (new build + retrofit)
Airtightness n_{50}	≤ 0.6 ACH @ 50 Pa		≤ 1 ACH @ 50 Pa		
Space Heating Demand (SHD)	≤ 15 kWh/m ² .a	–	≤ 25 kWh/m ² .a (variable)		
Heating load	–	≤ 10 W/m ²	–		
Primary Energy Renewable (PER)	≤ 60 kWh/m ² .a		≤ 60 kWh/m ² .a (variable)	≤ 45 kWh/m ² .a	≤ 30 kWh/m ² .a
Renewable energy generation	–			≥ 60 kWh/m ² .a	≥ 120 kWh/m ² .a
Summer overheating	Max 10% at > 25°C			PER varies by building use, and, in retrofit, by allowance for larger heating and cooling demand compared to a new build. In addition, all classes allow for ± 15 kWh/m².a deviation from the PER criteria, with compensation through additional generation.	
Surface temperature	> 17°C				
Ventilation	> 20 m ³ /hr.person				

HEADLINE CRITERIA

The table above shows headline criteria for all Passivhaus classes. Readers must refer to the full [criteria](#) to ensure they have the correct values for their specific project – variations may apply.

As a measure of energy efficiency, Passivhaus offers a choice from two alternative criteria for heating – peak heating load in W/m² or space heating demand in kWh/m².a. By reducing heating and cooling load to a very low level we achieve excellent comfort, simplified building services and very low energy use.

For EnerPHit, the heating demand target varies according to climate zone, of which the UK spans three, most falling into 25 kWh/m².a.

As well as heat demand, Passivhaus also has strict limits for total energy demand. This is evaluated using the Primary Energy Renewable (PER) metric, which takes into account the characteristics of a future fully renewable grid, including distribution and storage losses and seasonality. It thereby incentivises optimum building fabric performance and energy efficiency. Read the in-depth explanation on [Passipedia, the Passive House Institute's online encyclopedia for all things Passivhaus](#).

On-site renewable energy generation is assessed in terms of m² of building footprint, so multi-storey buildings which have less roof space in relation to internal floor area are not excluded from the Plus and Premium standards.

Other criteria address comfort, including ventilation. Here, we recommend supply air is set at 30 m³/h.person. The 20 m³/h.person basic criterion is a minimum, but it is not expected to be sufficient for UK homes because of our mild and damp climate.

Overheating is limited to no more than 10% of the year at over 25°C for certification. However, best practice is <2% of the year, and additional risk assessments and stress testing to take into account future climate and occupant behaviour are recommended, following the guidance in *Avoiding summer overheating* (PHT, 2021).

The surface temperature criterion not only addresses indoor comfort – eliminating the sensation of cold from chilly spots – but also protects the building fabric from the dangers of condensation and mould, with their detrimental impact on occupants' health.



“As contractors, the Passivhaus approach is a quality standard helping us to deliver projects profitably by avoiding costly defects and callbacks. At Makar, we prioritise getting things right first time, and to a high standard of quality. This is why we follow Passivhaus principles.”

Neil Sutherland, Architect & Director, MAKAR

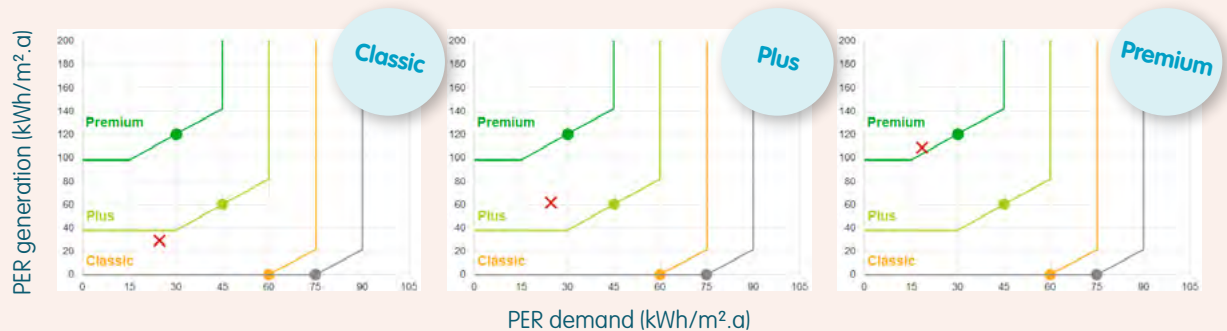


Certified Passivhaus plaque, Lucy Cavendish student accommodation. Image: Tom Foggins

CLASS AMBITIONS

Get the basics right, and upgrading your Passivhaus Classic to Plus or Premium could be as easy as adding solar power and improving your appliances. The graphs below show the energy demand of a 2-storey detached dwelling (the red x) against the Classic, Plus and Premium criteria. To meet Passivhaus Classic, the house is already optimised

for building performance and is heated with an efficient heat pump. Simply filling the south roof with PV upgrades the house to Passivhaus Plus, and a hypothetical step up to Passivhaus Premium would need only the addition of more PV on the north roof, best efficiency appliances and shower water heat recovery. This is one example - there are many ways to meet the higher classes.



CASE STUDY: Agar Grove



Above: Agar Grove Phases 1a and 1b aerial photo

Below: Phase 1b with Phase 1c under construction in the background.

Images: Architype



Agar Grove is set to be the largest Passivhaus development in the UK so far. Replacing existing 1960s H-shaped housing blocks, 345 affordable Passivhaus homes are being built for new and existing tenants in the London Borough of Camden.

Delivered over a total of 6 phases, each iteration is offering scope for learning and improvement for the delivery teams. A different approach to services and heating systems has been developed to respond to both post occupancy evaluation (POE) from earlier phases and to the wider decarbonisation of the UK's electricity supply. Phase 1A saw a communal MVHR system, while Phase 1B provided individual MVHR units for each apartment. Phase 1C is currently under construction (projected for completion in 2024) and will be all-electric, using an innovative ambient loop heat pump system.

Monitoring data on the first phase has shown that internal temperatures rarely dropped below 21°C in winter, and only rose above 26°C in peak summer months, with temperature peaks being rare. All respondents felt the air quality was good. The feedback on summer temperatures was more mixed and has prompted Camden to help residents understand how to best keep their flats cool.

Agar Grove exemplifies the collaborative approach and knowledge sharing that is typical of Passivhaus projects. Its success proves the relevance of Passivhaus in the UK at scale, and as an approach that aligns with local authority requirements on creating sustainable neighbourhoods, not only one-off sustainable developments.

RETROFIT

We need to massively reduce operational energy in our buildings, but it is essential that we think retrofit first, new build second. Retrofitting will play a vital role in mitigating the climate crisis. There are substantial embodied carbon savings made in repurposing existing buildings, compared with the ultra-high embodied energy costs of demolition and rebuilding.

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RETROFIT PAPER AND PRIMER

EnerPHit has a well defined process, very similar to the Passivhaus new build process, which helps to avoid the energy performance gap. The process is helpful, even when a lower standard is being targeted. It is a whole building approach which sets an exemplar standard for the energy and comfort performance of retrofit.

How far should our retrofit strategy go? Achieving net zero, improving health, eliminating fuel poverty and realising broader societal benefits are most likely to be achieved by a deep level of retrofit resulting in significant energy demand reduction.

“Despite having emissions reduction targets, government action is still lagging behind the policy ambition, and progress on retrofitting of existing buildings is flagged as ‘significantly off track’”

CCC

There are two ways to meet the EnerPHit standard, the **heating demand method** or the **component method**.

The heating demand method uses the same criteria as Passivhaus Classic new build, but with a small relaxation for certain metrics. Achieving EnerPHit using the heating demand method will result in a retrofitted building with a space heat demand 80% lower than the national UK building stock average. The component method is an alternative approach that sets limits for the thermal performance of the building elements alongside the same targets for airtightness, ventilation, and surface temperatures. The component method option was developed to cater for buildings where the orientation, form factor, and glazing preclude achieving the required space heating demand, even when the fabric has been upgraded to levels commensurate with Passivhaus performance.

ENERPHIT OPTIONS

- **Heating demand method**
This approach typically delivers buildings with a space heating demand of ≤ 25 kWh/m².a in the UK
- **Component method**
This approach typically delivers buildings with a space heating demand of 30-40 kWh/m².a in the UK
- **Step-by-step approach**
Work carefully planned & spread over multiple stages. Suitable for both heating demand and component methods

Shallow retrofit

30% space heating demand reduction leads to:

Reduced carbon emissions
Large renewable energy demand
Large peak demand
More grid storage required
Little change in energy bills
Limited health benefits

Deep retrofit

75% space heating demand reduction leads to:

Reduced carbon emissions
Reduced renewable energy demand
Reduced peak load
Less grid storage required
Significantly lower energy bills
Improved health & comfort

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NON-DOMESTIC AND LARGE & COMPLEX BUILDINGS

As the adoption of Passivhaus continues to grow in the UK, we are seeing more sizeable projects emerge with increasing complexities and challenges. This guide is aimed at those new to the delivery of Passivhaus projects and is not expected to cover the complexities of large-scale Passivhaus projects. Experienced Passivhaus Designers and Consultants are on hand around the UK to assist with these more challenging projects.



St Sidwell's Point, Exeter – Gale and Snowden © Tom Hargreaves

WHOLE LIFE VALUE

Lower heating running costs mean that Passivhaus homes offer significant whole life cost savings over a current building regulations home. A Passivhaus building is already optimised for net zero and will not need future retrofitting to meet zero carbon targets. By contrast, a building constructed to current building regulations compliance may need substantial investment in often disruptive fabric upgrades over time. Additional financial benefits such as better mortgage rates and lower maintenance costs result in even quicker payback periods and larger whole life benefits for Passivhaus buildings.

PASSIVHAUS BENEFITS

With outstanding levels of building performance, Passivhaus delivers exceptional levels of comfort, health and wellbeing, affordability, resilience and durability, as explored in detail in *Passivhaus benefits* (PHT, 2021).

Individual homeowners will be interested in comfort and the excellent indoor air quality, as well as the lower energy bills.

Local authorities may focus on tackling fuel poverty, improving health outcomes, and meeting zero carbon pledges.

Government and energy suppliers may be attracted to the sound economics of lowering peak demand, minimising budgets for large scale infrastructure costs, and amplifying the resilience of a national grid that is fit for the 21st century.

SUMMARY

Passivhaus is a tried and tested 'fabric first' building standard and methodology, backed by over 30 years of international evidence. It produces buildings with a high level of occupant comfort that use very little energy for heating and cooling.

Analysis shows that reducing energy demands to Passivhaus levels is the only realistic way to achieve zero carbon without massive renewable energy expansion coupled with a significant, and expensive, investment in grid capacity and storage.

"The City of Edinburgh Council has adopted certified Passivhaus as a proven approach. It provides clarity around design and construction expectations, thereby ensuring building performance and user comfort expectations are delivered."

**Patrick Brown, Head of Capital Programme Team,
City of Edinburgh Council**



pht.guide/benefits

2

Design approach and system choices

Jonathan Hines

There are many factors influencing your approach to designing a building, including: location and site; your client's brief, budget, and programme; funding requirements and regulatory standards; and your own design philosophy.

Passivhaus must be seen as one of the fundamental factors influencing design, as it cannot be achieved simply by adding together separate technical standards such as airtightness or window U-values. Passivhaus should become an integrated part of your design approach, underpinning, and even inspiring, all other factors influencing your design.

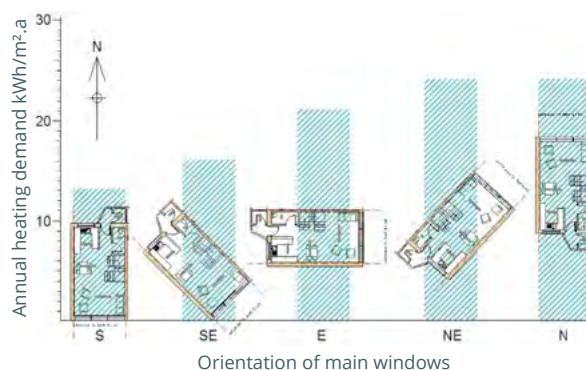
CHOOSING THE DESIGN APPROACH

To ensure that Passivhaus is integrated into the design of the very earliest ideas, its principles and standards must be understood by all members of the design team.

Key early decisions that may be difficult or impossible to change later – including siting, orientation, building form, and fenestration – have a major impact on the viability and economy of achieving Passivhaus.

It is recommended that the design is modelled using the PHPP at the earliest opportunity, in order to test and shape these key early design decisions. PHPP should be seen and used as a powerful design tool, helping you to develop your design approach, rather than as a procedure for proving compliance later, when it might be too late.

Test the influence of altering the building's siting and orientation, or the total and relative quantity of glazing on each elevation, and you will be able to understand the way in which simple environmental principles influence the energy efficiency of a building's performance.



Achieving good design requires not only the solving of a multitude of design challenges, but also the development of creative synergy between them. Good Passivhaus design requires a fully integrated design approach, out of which an even greater creative synergy can be developed. This will produce buildings that not only meet the client's brief, and look and feel good, but do actually work with proven energy efficiency and excellent comfort.



PHPP 10 - the latest version of the Passivhaus Planning Package

Explore the impact of the building's heat loss form factor (the area of external envelope through which heat will escape compared to the area of usable internal floor area), and you will quickly discover that while almost anything can be made to work, the better this ratio is then the more economic the solution. Aim for a ratio of 3 or less. This is a simple calculation and can be done quickly and easily before modelling your design in the Passivhaus Planning Package (PHPP).

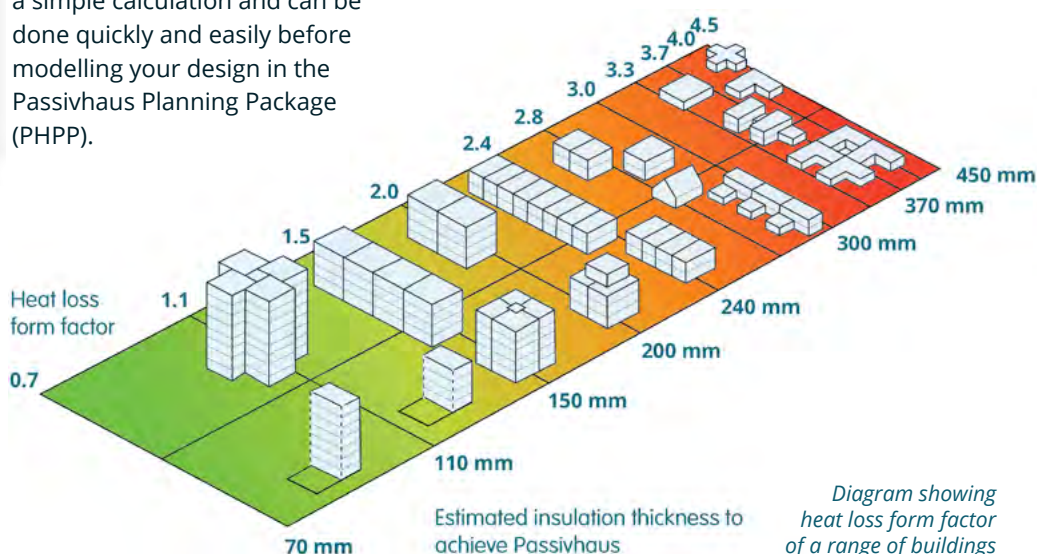


Diagram showing heat loss form factor of a range of buildings

Passivhaus should inform and influence every aspect of your design approach and choice of construction system. This need not be seen as limiting creativity of design nor as restricting what construction system to adopt. To achieve an economic solution, Passivhaus must be understood and integrated into the design approach and system choice from the outset.



SYSTEM CHOICES

Passivhaus does not dictate any particular construction system. Indeed, practically any construction system can be used and adapted to achieve Passivhaus, though each will have its own advantages and challenges for your particular project.

The important issue to consider is how the chosen construction system can achieve the stringent requirements of Passivhaus – including airtightness, elimination of thermal bridging and appropriate U-values – given the particular building type you are designing, its function and the form you are developing.

As a general rule avoid mixing different systems within one building, as this introduces interfaces that will make achieving Passivhaus challenging and more expensive than it needs to be.

SYSTEM CHOICES FOR PASSIVHAUS DEVELOPMENTS

Construction systems can generally be classified as lightweight with less thermal mass (such as timber or steel frame) or as heavyweight with more thermal mass (such as masonry or concrete frame), although elements of thermal mass can be introduced to timber or steel.

The high levels of fabric performance and carefully controlled solar gain in Passivhaus lead to internal environmental conditions remaining very stable. Passivhaus does not generally require either light or heavy thermal mass. Other considerations such as function, occupancy and climate may determine which approach might work better.



An important factor in choice of system is the likely procurement route, and the availability of an appropriate contractor. Thus if you are working on a domestic project and local builders are experienced in traditional masonry, choose masonry, as you are more likely to achieve Passivhaus by working with the available skill base.

*On site for Harris Academy in Sutton.
Image: Architype*

On larger projects there may be pressure to use steel frame due to its familiarity to large contractors and consequent cost advantages, so you may need to choose steel frame and adapt it to suit Passivhaus.

Details and junctions need to be designed with construction and assembly on site under realistic conditions (eg weather!) in mind. This is to avoid site compromises or later changes under pressures such as time, cost and materials availability during construction that can easily compromise design and cost.

Key Passivhaus issues with each of the main construction systems are summarised overleaf.

do

- integrate Passivhaus into your design approach from the earliest possible point
- use PHPP as a design tool to understand, influence and improve design
- make the building itself do all the hard work in achieving Passivhaus economically
- ensure that orientation, form and fenestration are optimised
- select a construction system that is appropriate for your building type, form and function, procurement and contractor

don't

- leave Passivhaus until late in the design process or treat PHPP as a compliance check late in the design
- assume that Passivhaus dictates any particular construction system
- combine unnecessarily different construction systems within one building
- create design problems such as thermal bridges or challenging airtightness details by using unnecessary cantilevered structures or structural penetrations

TIMBER FRAME

- Generally easy to build on a concrete raft 'floating' on EPS insulation, to eliminate thermal bridging.
- Consider the wall build up carefully, to avoid thermal bridging caused by solid studs, and use I-beams or a two layer wall with an inner structure and outer insulation wrap.
- Airtightness can generally be achieved with a membrane or racking board with tape over joints; racking boards are more robust. If a membrane is used on the internal face of the wall it is beneficial to have a service void to protect the membrane.
- Avoid elements of structure crossing the thermal envelope, for example, avoid roof overhangs with projecting joists that act as thermal bridges and make airtightness tricky.
- Avoid heavy cladding that might require a separate foundation.

CONCRETE FRAME

- Address potential thermal bridges at the connection of columns to foundations.
- Ensure concrete structure does not pass through the thermal envelope at floors, roofs or for ancillary elements of the building.
- Airtightness can be achieved with parge or plaster where external walls are infilled with masonry, or with membranes or board and tape, where infilled with lightweight construction such as timber or steel stud.
- Avoid heavy external cladding hung off the structure, or cantilevered structures such as balconies, as this will create unnecessary thermal bridging.

TRADITIONAL MASONRY

- Achieving Passivhaus with traditional masonry requires very wide insulated cavities between inner and outer masonry and thermal bridge free ties.
- Structural openings require a double or thermally broken lintel – consider large openings carefully.
- Voids under cavity barriers can be difficult to insulate and need extra attention on site.
- Airtightness can generally be achieved with parge or plaster.
- Works well with strip foundations.
- When using a concrete raft 'floating' on EPS insulation, a thermal break will generally be required to eliminate thermal bridging, as the external skin requires support.



Passivhaus projects under construction using different methods of construction – clockwise from top:

*Denby Dale Passivhaus (cavity wall) Image: Green Building Store
Oakmeadow Primary School (timber frame) Image: Architype
Two Rivers Primary School (steel frame) Image: Willmott Dixon
Agar Grove (concrete frame) Image: Hill Group*



STEEL FRAME

- Address potential thermal bridging with connection of columns to foundations.
- Ensure steel structure does not pass through the thermal envelope at floors, roofs or for ancillary elements of the building.
- Achieving airtightness on the inner side of the wall will be challenging, if not impossible, due to the number of structural penetrations. Ideally locate the airtightness line within the wall and on the outside of the structure where it can be continuous and unbroken by the structure.
- Avoid heavy external cladding hung off the structure, or excessive cantilevered structures, as this will create unnecessary thermal bridging.

WHOLE BUILDING SYSTEMS INCLUDING MODERN METHODS OF CONSTRUCTION (MMC)

Passivhaus encourages designers to address all aspects of the design at an early design stage, including the types of materials/systems that will be used. A certified whole-building system approach i.e. one where the materials have been pre-selected and developed into a holistic system, can help designers make their design decisions with confidence that the final outcome will meet the Passivhaus requirements. Many whole-building systems use MMC, with the factory controlled environments assisting in delivering to the level of quality required for Passivhaus certification.

- A certified whole-building system can remove some of the complexities of ensuring that different materials and components can be combined to achieve Passivhaus requirements. However, effective design decisions are still required and the system(s) may not work for all projects and design choices.
- The incorporation of MMC and prefabricated components can speed up construction time, resulting in lower labour costs. The construction programme is less affected by bad weather and materials are protected from exposure to the elements during construction. Space is required to transport manufactured modules or components to their site.
- The mechanisation used in MMC construction increases conformity to the Passivhaus standard, with quality control and factory airtight sealing ensuring better quality control.
- Affordability may be an issue, particularly for smaller projects; increased production volumes on larger projects should offer greater affordability.
- Certified whole-building systems do not remove the need for effective design input. Orientation, services, design and other factors will all have

a significant effect on the cost, efficiency and successful delivery of a project using a whole-building system.

OTHER CONSTRUCTION METHODS

The list given here is not exhaustive. Certified Passivhaus buildings in the UK have been completed, for example, using insulated concrete formwork (ICF), straw bale construction, and solid masonry with insulating clay blocks. Other innovative approaches are also possible – ultimately the Passivhaus standard can be applied to any construction type and materials.

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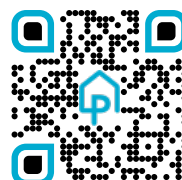
Masonry construction and a range of timber frame options are explored in further detail in chapter 7 – Building fabric.

Steel frame and concrete frame options will usually only be considered for larger projects. It is essential that significant Passivhaus experience is brought to bear on these more challenging projects, and we recommend exploring the Passivhaus Trust guidance on large and complex buildings, steel frame Passivhaus, and the Passivhaus Overlay to the RIBA Plan of Work to gain an idea of the challenges involved.

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LOW EMBODIED CARBON DESIGN

In this section we have mostly been examining different construction typology choices from the point of buildability and suitability for a project. However, a vital part of the decision-making process obviously also relates to the embodied carbon impacts of different construction materials. Embodied carbon looks at the carbon emissions associated with building construction, including those that arise from extracting, transporting, manufacturing, and installing building materials on site, as well as the operational and end-of-life emissions associated with those materials.

The choice of materials, the nature of their manufacture, and the thoughtfulness of the structural and architectural design, will have a greater impact on the whole life carbon footprint than the performance upgrade to meet the Passivhaus standard. Indeed, when the Passivhaus standard was being developed, embodied energy analysis was an integral part of the background that informed it, especially insulation choices.

The additional elements needed for a compliant Passivhaus house design (triple glazing, MVHR) only account for a small percentage of the overall whole life carbon footprint. Other factors, such as reducing steel and concrete and embedding circular design principles, are more important.

Two of the most common materials in construction, steel and concrete, are responsible for 15% of global carbon emissions. Reducing the quantity of these materials will impact on embodied carbon emissions. Carefully designed structurally efficient lightweight steel systems*, along with use of lower emission concrete and steel, can contribute to lower embodied carbon solutions.

**Carbon Assessment by Collida (a Willmott Dixon Company) and Architype, 2021*

Irrespective of the choice of materials, what will reduce embodied carbon is designing and building resource efficient buildings through Passivhaus design principles such as:

- Rationalising the form factor to reduce the amount of materials used overall in the building**
- Designing in structural efficiency, to simplify airtightness and thermal bridges, which also reduces material use.

WHOLE LIFE CARBON

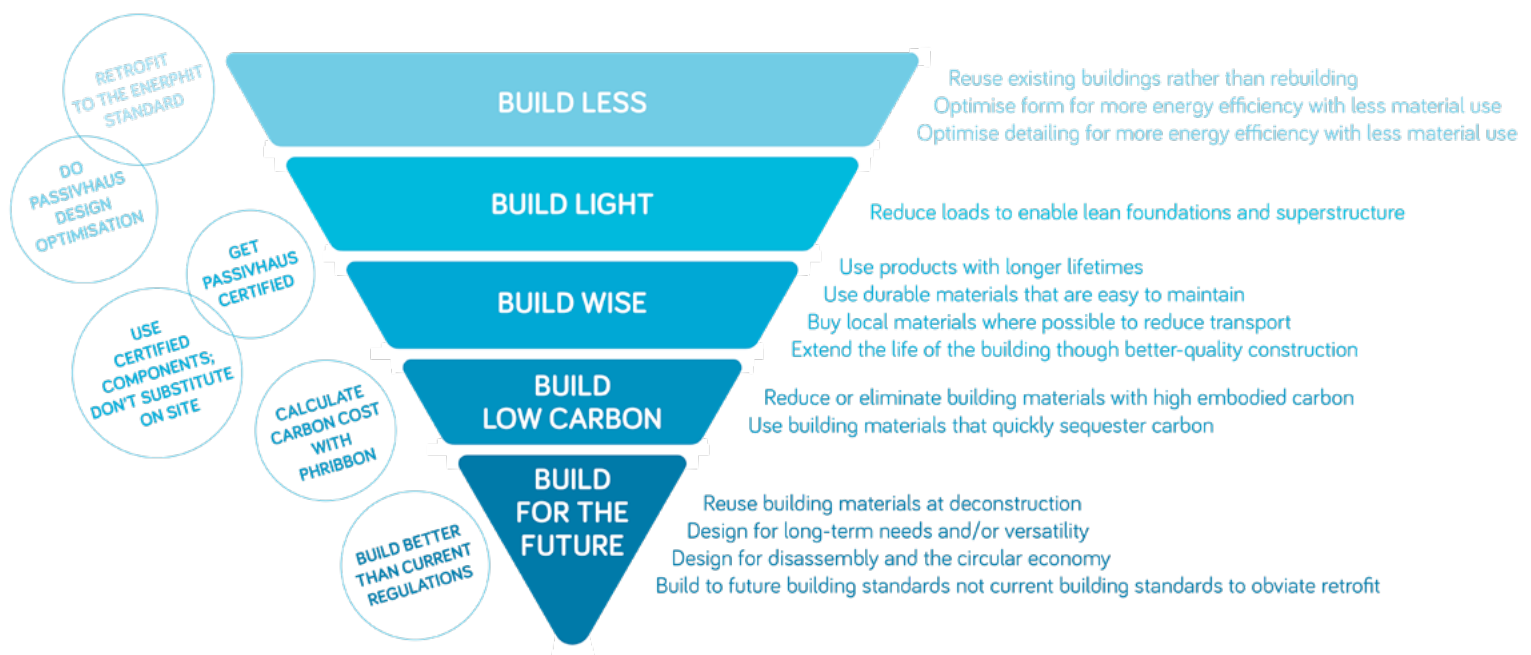
To address the climate emergency, we must tackle the whole life carbon of buildings, addressing both operational and embodied carbon of buildings simultaneously. Whole life carbon calculations are important when considering retrofit as well as new build. Improving the energy performance of a building will add to the embodied carbon (additional insulation and building services), but this can be offset against operational carbon improvements or the embodied carbon cost of demolition and replacement.

CALCULATING CARBON

We would recommend that Passivhaus Designers utilise carbon calculating tools to examine the embodied carbon of construction material choices.

- **PHribbon** is an embodied carbon calculator that works as a plugin for PHPP www.phribbon.co.uk
- **ECCOLAB** is a carbon counting tool that has BIM/ REVIT integration capabilities and can be used in parallel with PHPP eccolab.co

***Form factor study for South Yorkshire Housing Association, Qoda Consulting, 2021*



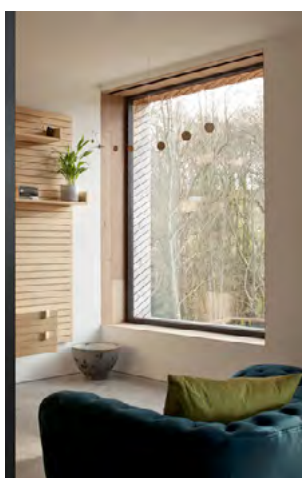
CASE STUDY: Ostro Passivhaus



In March 2014, owner and architect Mhairi & her partner Martin, from PHT member Paper Igloo Ltd, embarked on an ambitious self-build project in the historical conservation village of Kippen, near Stirling, Scotland. Doing most of the building work themselves while also running their architecture practice, they completed the building over the course of six years and documented their progress in a detailed blog.

The 'box in a box' timber frame home received certification in October 2017, becoming the first certified Passivhaus in Stirlingshire. It has also achieved Scottish Technical Standards Section 7: Sustainability Gold Level.

The house takes the form of a Siberian larch-clad box with a flat felt roof system incorporating a sedum green roof and a dramatic south-facing double height space. The orientation and balanced proportion of windows both maximise the views and minimise the risk of overheating in the summer months. The windows are all triple-glazed and the doors aluminium-insulated. The timber frame twin wall features a continuous thermal break and external layer and is totally insulated with wood fibre. The house also features renewables such as solar and PV panels.



Images: Paper Igloo (Mhairi Grant and Martin McCrae)



3

Design approaches to retrofit

Sarah Lewis

The building physics principles behind Passivhaus apply to any building – but you may not be able to get quite the same results from an existing building where you aren't really in control of the building's orientation, structure, shape, heritage significance, moisture strategy or amount of glazing. The Passivhaus standard, therefore, includes a retrofit standard called EnerPHit which takes these limitations into account and relaxes some criteria in recognition of the challenges involved. However, it is still a very demanding standard and will result in a building that outperforms a typical UK new build property both in terms of energy and comfort.

RETROFIT DESIGN APPROACHES

Significant energy savings of between 75% and 90% can be achieved in existing buildings when retrofitted to the EnerPHit standards. The following measures are particularly effective:

- improved **thermal insulation** (based on the principle: if it has to be done, do it right)
- **reduction of thermal bridges**
- considerably improved **airtightness**
- use of **high-quality windows** – in the UK this normally means triple-glazed
- **ventilation** with highly efficient heat recovery (referred to as MVHR)
- efficient **heat generation**
- use of **renewable energy** sources

These improvements can be achieved on a wide range of existing buildings, from older tower blocks and housing schemes seen to no longer be adequate to heritage properties with celebrated architectural features. The former have frequently been earmarked for demolition and replacement. But the squandering of embodied carbon through demolition, added to the emissions costs of replacement construction, invariably results in a much higher total carbon emissions budget for a project than would a refurbishment, adaptation or reuse solution – as well as costing more.



Erneley Close in Manchester, certified EnerPHit - before and after deep retrofit

Woodside multi-storey retrofit (not certified). Image: Collective Architecture



The refurbishment of three high rise towers in the Woodside area of Glasgow, left, targeting the EnerPHit standard, demonstrated how successful deep retrofit at scale can be. Queens Cross Housing Association's Cedar Court and Cedar Street, built in the 1960s, implemented external wall insulation (EWI) paired with high performance triple glazing, MVHR and an airtightness strategy. They now look like new, and perform better than most new builds, providing warmth and comfort to the tenants of 314 homes.

We need to massively reduce operational energy in our buildings, but it is essential that we think retrofit first, new build second. Retrofitting will play a vital role in mitigating the climate crisis. There are substantial embodied carbon savings made in re-purposing existing buildings, compared with the ultra-high embodied energy costs of demolition and rebuild.



Heritage projects, such as the Niddrie Road project in Glasgow pictured right, have also aimed to meet the EnerPHit standard. Heritage considerations meant that this project needed to adopt a hybrid strategy combining internal wall insulation (IWI) to the front and party walls, and external wall insulation (EWI) strategy at the rear and gable walls. Triple-glazed timber windows were chosen that were deemed acceptable for the building's heritage requirements and were carefully integrated into the hybrid wall airtightness and insulation strategies. Careful detailing was undertaken to ensure that the IWI and EWI strategies connected well, maintaining the continuity of insulation and airtightness.

ENERPHIT MANAGEMENT PLAN

Irrespective of the certification method (heat demand or component) or the approach (single stage or step by step) all retrofit projects should start with an EnerPHit Management Plan. This plan includes an assessment of retrofit strategy and certification strategy, surveys to gather data about the existing building, and strategies to address the specific demands of the building and site. This whole building retrofit plan includes modelling the existing building in PHPP to provide detailed calculations of the existing fabric performance of the building. Importantly, all approaches to retrofit, including EnerPHit, require some augmentation to deliver robust solutions addressing all aspects of the retrofit performance gap and unintended consequences.

Even where EnerPHit is too challenging a target for your project, the Passivhaus Trust recommends an EnerPHit-informed Retrofit Plan (EiRP) is created for all retrofit projects. This allows all retrofits to benefit from the distinct advantages of the EnerPHit methodology. Where projects adopt the EnerPHit standard they will also benefit from the performance assurance provided by the EnerPHit certification process.

ONE STEP AT A TIME

A radical whole house retrofit is likely to achieve the best results both in terms of energy reduction and comfort. However, this is a significant undertaking which could take place in stages. Decisions in the early phases, and getting the sequence right, are critical to ensure that the full potential of the retrofit is realised and that the project doesn't inadvertently result in 'carbon lock-in'. In this context the term refers to moderate improvements in energy efficiency which are undertaken on retrofit projects.

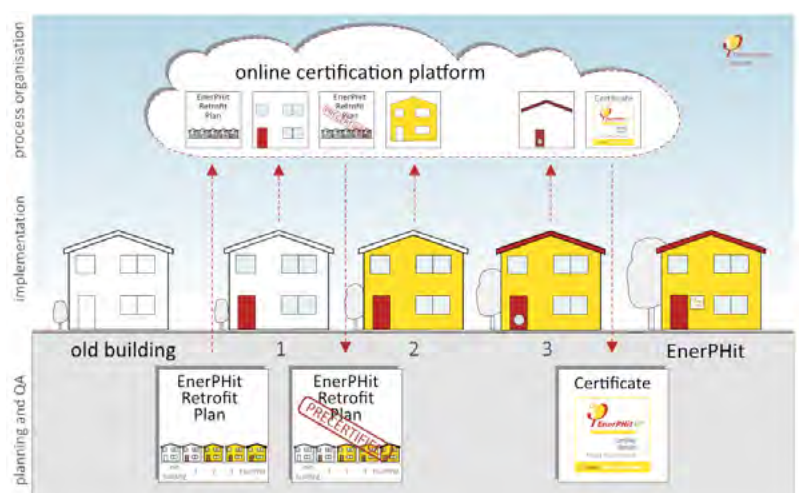


Niddrie Road - an archetypal Glasgow tenement undergoing deep retrofit, targeting the EnerPHit standard.

Once such measures have been undertaken it becomes unsustainable, uneconomical and impractical to change the building, leading to substandard performance for at least 30 or 40 years. To avoid lock-in, the standard of energy efficiency achieved by each measure must be future-proof. The EnerPHit process, therefore, includes tools to structure the retrofit in a step-by-step way, thus ensuring that the implications of each step are understood and properly planned out right at the start of the project.

MANAGING MOISTURE

A critical element in any retrofit project is addressing moisture risks, particularly when considering energy efficiency measures and additional insulation. Detailed guidance on this from the Passivhaus Trust is due out in 2023.



ENERPHIT AND PAS 2035/2030

PAS 2035/2030 is a process for the retrofit of dwellings. EnerPHit is an energy standard. There are more similarities than one might think, and the two complement one another. Both came about after evidence about poor building performance emerged.

PAS 2035 is a framework and guidance document for delivering retrofit projects, published by the British Standards Institute (BSI). It promotes a fabric-first, whole house approach. It focuses on proper retrofit planning and quality assurance and requires the appointment of accredited professionals – including a Retrofit Coordinator – to oversee the retrofit project. PAS 2035 does not set energy efficiency targets or define how ‘deep’ one should go. Its aim is to avoid the unintended consequences, defects and performance gap of poor retrofit. All retrofit projects receiving central government funding are required to be PAS 2035 compliant. PAS 2035 works in tandem with the PAS 2030 standard for the installation of energy efficiency measures in existing buildings. Taken together, PAS 2030 and PAS 2035 lay down the steps domestic retrofit projects can follow to gain compliance and ensure consumer satisfaction in accordance with the recommendations of the Each Home Counts review.

PAS 2035 can be used in tandem with EnerPHit and PHPP is an approved calculation tool for use with PAS 2035. An EnerPHit Management Plan can be used to form the PAS 2035 Improvement Options Evaluation, and also form the basis of a PAS 2035 Retrofit Plan.



Internal to external wall junction receiving airtightness treatment – timber dowels driven into airtight mastic filled holes; mini trowel used to fill gaps between joists and wall (see case study opposite)

Image: Harry Paticas

EFFICIENCY & RETROFIT FIRST

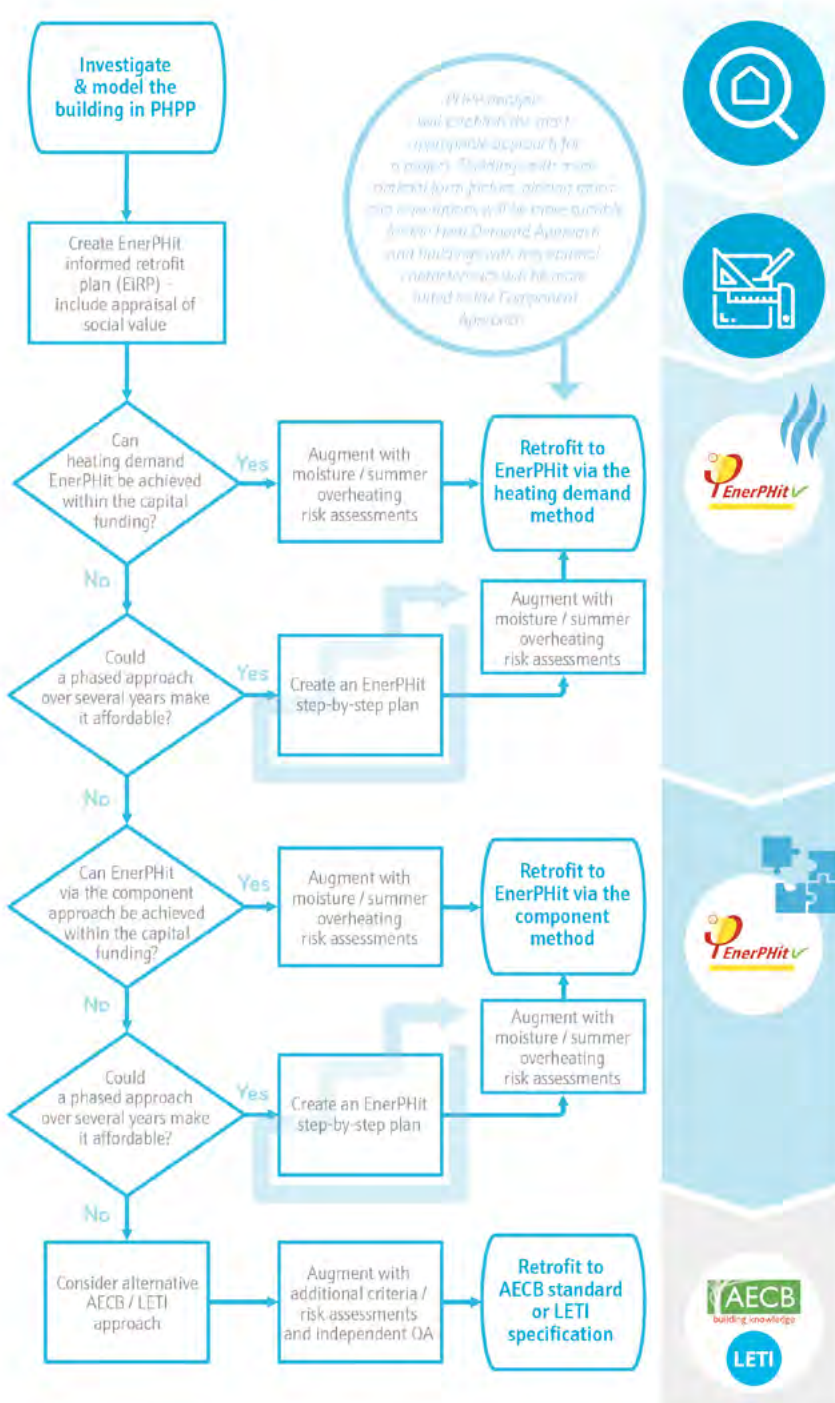
If we are to live within the means of our global system we need to radically reduce energy demand and improve efficiency. This is what the Passivhaus retrofit standard, EnerPHit, delivers. We cannot simply generate more zero carbon energy supply, due to the limitations of grid capacity, peak load & demands for renewable energy from other sectors.

EnerPHit gives us a robust methodology to deliver the changes we need in our existing building stock. This holistic approach helps to eliminate the performance gap and avoid unintended consequences, making the EnerPHit methodology really helpful even when a lower standard is being targeted. The following flowchart sets out our recommended approach to all retrofit projects.

LEARN MORE



pht.guide/retrofit



don't

- undertake shallow retrofit measures, introducing carbon lock-in
- take a piecemeal approach, putting the building at risk of unintended consequences

do

- think retrofit first
- undertake surveys and investigations to understand your building
- create a whole building plan
- model the building in PHPP to understand existing energy balances
- consider all EnerPHit options

CASE STUDY: Bowman's Lea



A 1978 mid-terrace townhouse in London is the first certified step-by-step EnerPHit retrofit in the UK. The 3-storey property was purchased in 2016 though only a very small budget was available to carry out building work. The first step in the phased renovation was completed in May 2018 with a resulting reduction in the space heat demand of 53%. The retrofit has a predicted energy demand, when fully completed, of 22 kWh/m².a.

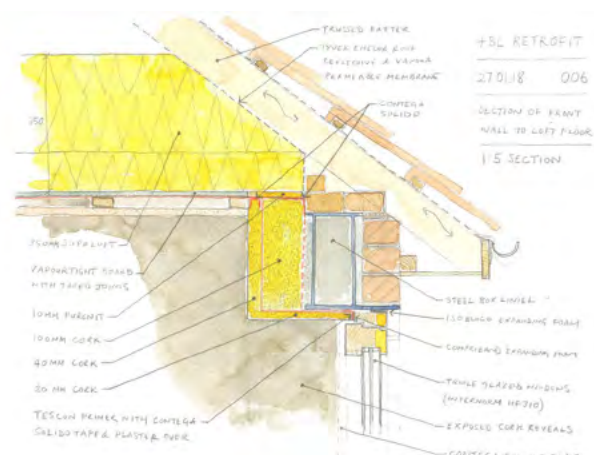
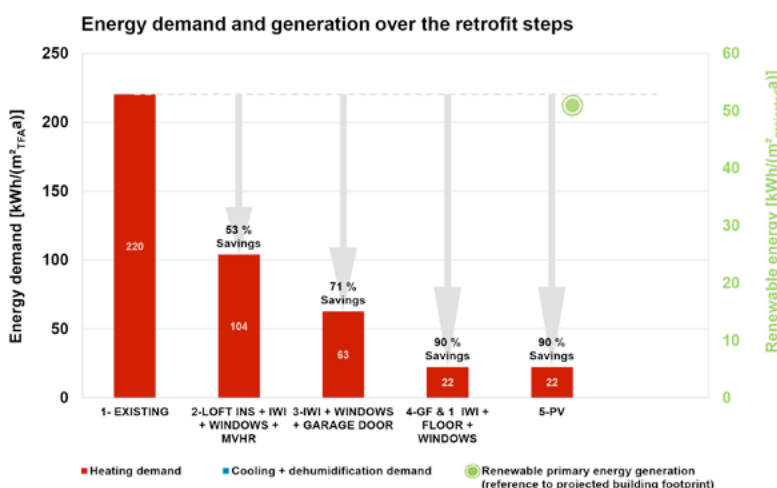
Step one included loft insulation, new high performance windows, internal wall insulation using exposed black cork, extensive airtightness works to the second floor ceiling and front elevation, and the installation and commissioning of a Zehnder Comfoair 350 MVHR system.

The work has been largely carried out by the owner, architect Harry Paticas, and has focused on exploring the complexity of step-by-step retrofit, on learning by doing as a way of informing best building practice, on specifying materials with low embodied carbon, on reducing construction waste, and on improving indoor air quality.

Left: Energy savings over the retrofit steps

Below, left to right: cork IWI internal finish, airtightness detailing, detail drawing showing loft and window head.

All images: Harry Paticas



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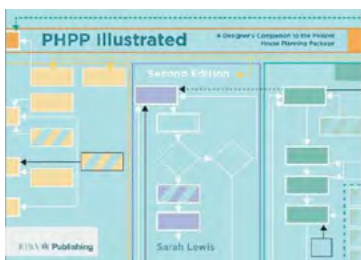
Use of PHPP and quality assurance

Sally Godber

INTRODUCTION

Training in the use of PHPP is strongly recommended, in order to be fully competent with the software. This section is aimed at those who have undertaken training and are embarking on one of their first Passivhaus projects.

The PHPP manual has developed into a great resource to ensure your building is modelled correctly. The comments below are meant to be in addition to the manual, so have it close by!



PHPP Illustrated: A designer's companion to the Passivhaus Planning Package by Sarah Lewis

(RIBA Publishing, 2017)

You would also likely find the book *PHPP Illustrated* a useful companion when building your first PHPP models. The book provides an excellent set of step-by-step instructions, clear examples of the sorts of supporting evidence required for certification, and tips to avoid common pitfalls.

CLIMATE AND ALTITUDE

Check carefully that you have identified the right weather data set. See the maps on page 118-119 in the PHPP 10 manual. The data sets generally follow county boundaries – if you're unsure use additional county maps to be conservative. If you have a Certifier appointed you can ask for further guidance. Entering the altitude of your site is just as important as location. Most of the data sets are

from locations less than 50 m above sea level, and if your site is significantly higher than the weather station this could easily add 1 kWh/m²a or more to the heating demand. Conversely some sets are at quite high altitude – the Severn set, for instance, uses data from RAF Lyneham, which is at 150 m.

HEAT LOSS AREAS

Heat loss areas in Passivhaus are measured to the outside of the thermal envelope. This can be less straightforward than the SAP internal measurement methodology, as what constitutes the outside of the thermal envelope may not be immediately clear. Put simply, the thermal envelope ends at the last element used in your U-value calculation.

VARIANTS

The variants tab is a brilliant design tool, so use it!

A CRITICAL EYE

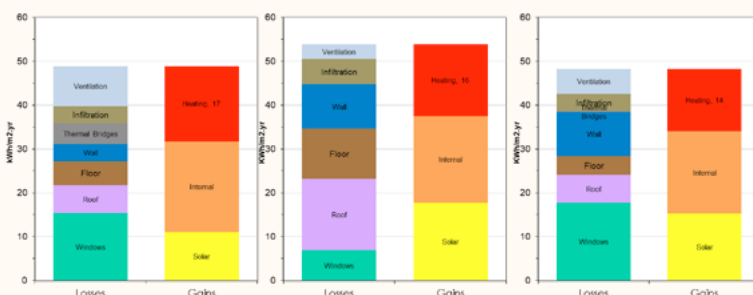
Passivhaus Certifiers will be keeping a watchful eye out for things that will affect the building performance in practice. These might include:

- Manufacturer/supplier claims for material performance – this could be, for example, the amount of timber in a timber frame or the conductivity of an insulation material
- Correct inclusion of repeating thermal bridges – refer to BS 6946 for guidance
Explore further: pht.guide/steel
- Thermal bypass – is the insulation encapsulated for airtightness & windtightness, and can the insulation be easily constructed without air gaps between or behind the material?
Explore further: pht.guide/thermalbypass
- Products that are likely to be damp (e.g. in contact with ground) – conductivity of porous elements such as open-cell insulation or lightweight blocks can vary significantly with moisture content
- Assumptions about occupant behaviour – is it realistic, for example will windows be opened, or equipment turned off?
Explore further: pht.guide/summer

So, be critical about the information you're given.

PHPP: THE PASSIVHAUS PLANNING PACKAGE

PHPP creates a bespoke energy balance for your project, quickly identifying what aspects are the most significant and what is important for you to work on. This means while the Passivhaus standard is the same for most building types, the particular solution for your building will be tailored.



PHPP energy balances for a school, a village hall and a terrace of houses (l to r)

Be conservative with values, as the thermal performance of elements very rarely gets better during design development. Optimistic inputs will be a problem later!

The Passivhaus Planning Package (PHPP) is a very useful and powerful design tool, but it is possible to misinterpret the conventions, giving erroneous results. Avoiding these simple mistakes can reduce the need to find additional energy savings late on in a project. In a worst-case scenario, such mistakes might mean that a project cannot achieve certification.



WINDOW ENTRIES

Unlike the UK's National Calculation Methods (NCMs), such as SAP and SBEM, where each rough opening can be considered as one window, PHPP needs accurate window entries, which include each individual casement and any mullions or transoms. Then PHPP calculates a bespoke U-value and frame factor for each window. Entering two adjacent window casements as one means that the frame losses would be underestimated, and the solar gains overestimated. Opening windows nearly always have thicker frames than fixed, so it is good practice to assume that all windows are opening ones if this is likely to be the case and the designer hasn't specified yet; this should avoid underestimating frame heat losses.

Using an installation psi-value of 0.04 W/m.K is a good conservative estimate for early design, and can easily be improved upon.* This typically represents a window installed into the middle of a standard timber frame.

PRIMARY ENERGY RENEWABLE, PER

For Passivhaus Classic the PER target should be achievable without the need for renewables. There are two major exceptions: when providing heating and/or hot water by direct electric, or for multi-residential projects using a district heating system/central plant. In these cases further efficiencies elsewhere or additional renewable generation may be required.

Meeting the PER limit will nearly always require best practice design of services such as hot water but should not necessarily come at a higher price, as demonstrated by, for example, Wilkinson Primary School – see the case study below. Choice of equipment will also need care – auxiliary systems such as trace heating for sprinkler systems and server equipment can be very energy hungry.

Make sure you've included all the equipment you can think of and ask lots of questions of the design team about their designs. Reducing auxiliary energy consumption through good design has been a major benefit of Passivhaus, particularly in non-domestic projects.

* This is not necessarily the case for steel frames – learn more at pht.guide/steel

VENTILATION RATES

The PHPP starting assumption is a ventilation rate of 0.3 air changes per hour (ACH), which is based on German property sizes and occupancy rates.

For most UK buildings higher ventilation rates are required. Some typical best practice ranges are shown below. It is important to model the actual design and controls as soon as possible to give realistic feedback. For non-domestic projects the ventilation rates can often be double these figures before optimisation of controls etc:

- Dwellings from 0.3 - 0.6 ACH
- Primary schools 0.4 - 0.6 ACH
- Secondary schools 0.6 - 0.8 ACH
- Open plan office building 0.4 - 0.6 ACH

MVHR DUCT DATA

The fresh air and exhaust ducts from the MVHR to the outside environment are essentially outdoor spaces within the thermal envelope and can add significantly to the heat loss of your design. Be realistic about the duct lengths, diameters and the amount of insulation which will be fitted. Where the MVHR is located right against an external wall, duct lengths of >1.5 m are still likely. 160 mm diameter ducts and 25 mm insulation are typical values for a domestic installation and closed cell insulation should be used to avoid condensation problems.

Where there is more than one MVHR unit in a building you must use the Additional Vent Sheet otherwise the duct losses won't be correct. Watch out though – it's easy to make mistakes, so have the manual to hand.

“The Passivhaus requirements to frontload and complete the design in the pre-construction phase results in many added benefits, including early procurement of sub-contractors and full coordination of the digital model. This should, in practice, enable a more efficient build process resulting in less site modification and waste, and ultimately a smoother commissioning and handover process.”

**Allan Smith, Low Carbon Manager,
Morrison Construction**



SHADING

The shading sheet is probably the trickiest within PHPP to get right, and it's worth reading the PHPP manual carefully so you understand the instructions, especially about averaging reveal figures. It's not sensible to be overly conservative and assume more shading than there is, because this can hide overheating problems. Be clear and reasonable in your assumptions and a Certifier is unlikely to disagree with you. If you are at all uncertain, get a Certifier appointed and agree the shading strategy early on.

For complex shading, we would strongly suggest using DesignPH, although it still needs to be inputted with care. Guidance to ensure compliance with certification is available from WARM (find a link at pht.guide/designsupport). If you have simpler shading but it includes trees or semi-opaque elements you can use the simpler 'Additional Shading Sheet'.

While it is possible to get carried away worrying about shading accuracy, a good design should not be too sensitive to exact shading values as this suggests there is probably too much glazing.

TOO MUCH GLASS

While it is possible to build almost anything to the Passivhaus standard, be cautious about projects with large expanses of glass. Such buildings are highly reliant on the weather for both internal comfort and energy consumption – this does not make it easy to achieve the comfort and energy requirements! And moreover, overheating testing will not include for the comfort impact of lots of hot glass during the summer.

QA PROCESS

It is easy to make mistakes with any modelling software, so having a robust QA procedure within your organisation is key. Ideally there will be someone who can look over your PHPP on a regular basis. Clear mark-ups of treated floor area, heat loss areas, windows and shading make for easy checking, as does keeping previous versions of your PHPP.

If your experience is limited, employ a Certifier early on in a project to check what you're doing, agree any points of contention and provide any hand-holding you may need. While certification usually starts pre-construction, having some experienced input (especially before planning) will ensure you're on the right track.

As a simple starting point for reviewing a PHPP file, we suggest the following steps:

- Check, review and resolve any errors in the 'check' tab within the PHPP
- Review the 'plausibility statements' that are given at the bottom of the check tab. These require you to unprotect the sheet to reveal them. It's an incredibly good tool for spotting simple errors.
- Run the file through a PPP transfer to confirm it gives the same answer in a fresh PHPP. The PPP transfer tool is available with the original material provided when you purchased your copy of PHPP.

LEARN MORE



[pht.guide/
summer](https://pht.guide/summer)

SUMMER COMFORT

The overheating check in PHPP is basic, and as such shouldn't be used for non-residential or complex residential buildings (except as a design tool or check). However, the summer worksheet in PHPP 10 has some useful tools to carry out stress testing for different climate conditions and building uses. In addition, overheating risk key indicators and a detailed risk log can be found in the Passivhaus Trust's free PHPP plugin, available at pht.guide/summer with further guidance.

PHPP considers the building as one zone, distributing the gains evenly and as a result won't identify high solar gains concentrated in one room. Be cautious about assuming high levels of window opening to solve an overheating issue, as this may not be realistic due to security or noise issues; results from social housing indicate that night purge ventilation i.e opening windows is seldom used even with clear guidance in the user manual.

It is recommended for the building to exceed 25°C for no more than 5% of the year, and best practice is less than 2% of the year. The upper limit for certification is for the building to exceed 25°C for less than 10% of the year. Note that this is based on 'current' climate data; this broadly represents the last 20 years and so may not represent the current, let alone future scenarios. For example, a project in London had an overheating result of 6% using the standard dataset within PHPP, but this increased to 15% when replaced with the actual data for 2018. Use the stress tests!

ADDITIONAL RESOURCES & PLUGINS FOR PHPP

There are many useful plugins and resources for using PHPP, including:

- **DesignPH** Sketchup based design tool that allows inputs to PHPP. Particularly good for complex shading.
- **PHribbon** allows for embodied energy calculations along with a huge range of other aspects. A fantastic addition to PHPP.
- **WARM results sheet** provides key results for reports or to keep track of a project. Other tools such as additional shading sheet are also available on the WARM website.

A number of other tools have been developed to complement and extend PHPP – find them listed at pht.guide/designsupport.

LEARN MORE



[pht.guide/
designsupport](http://pht.guide/designsupport)

do

- be conservative with values, as the thermal performance of elements very rarely gets better during design development – optimistic inputs will be a problem later!
- consider the impact on both summer and winter cases – the variants tab is helpful for this
- model your design in PHPP at the early design stages, making sensible assumptions
- leave a clear trail of what any inputs are based on
- keep the model up-to-date throughout the process, and give feedback to designers and client regularly
- get someone else to QA your work – even if you're experienced it's easy to make a mistake

don't

- assume that a building specification from a previous development will suit all future Passivhaus buildings
- forget to consider the design. Just because the PHPP is passing it doesn't mean it's a good design solution – particularly important for glazing design

CASE STUDY: Wilkinson Primary School



Wilkinson Primary School, the third Passivhaus school delivered by Architype, firmly established the pattern for Passivhaus school design, since successfully repeated many times. Innovative services design not only met the Passivhaus primary energy target but also kept costs down. According to Architype, the school 'incurred no additional design or construction costs' over a non-Passivhaus equivalent, while running costs plummeted - just as with its Passivhaus predecessor Oak Meadow Primary where, after moving into the new Passivhaus certified building, the school's annual heating bill dropped from £45,000 to only £5,000.

For this project, the internal heat gain (IHG) factors were adjusted in agreement with the Passive House Institute to reflect the fact that UK schools have a higher density of children than German schools. Modelling assumptions for the certification of UK Passivhaus schools have since been further refined to accurately reflect typical usage, as part of the UK Certifiers Circle ongoing work to bring increased consistency and clarity to the modelling and certification processes for all types of buildings.

*Wilkinson Primary School, Wolverhampton.
Images: Architype*

5

Certification and quality assurance

Kym Mead + Will South

INTRODUCTION

Passivhaus Certification is a quality assurance process for the design and construction of low energy buildings. It is independent and impartial, with the Certifier representing the best interests of the building and building owners, now and in the future. The Passivhaus Certifier may also help the designer navigate more technical or unusual aspects of modelling, and share experience of common issues from across a wide range of projects to help the design team prioritise design development.

THE PASSIVHAUS BUILDING STANDARD

The Passivhaus and EnerPHit building standards apply to both domestic and non-domestic buildings. The standards are based on performance, rather than individually specified construction or technical details. Certification requires that overall design and construction is carried out with a calculated energy assessment within PHPP, with robust evidence including drawing information and supporting documentation on components and services.

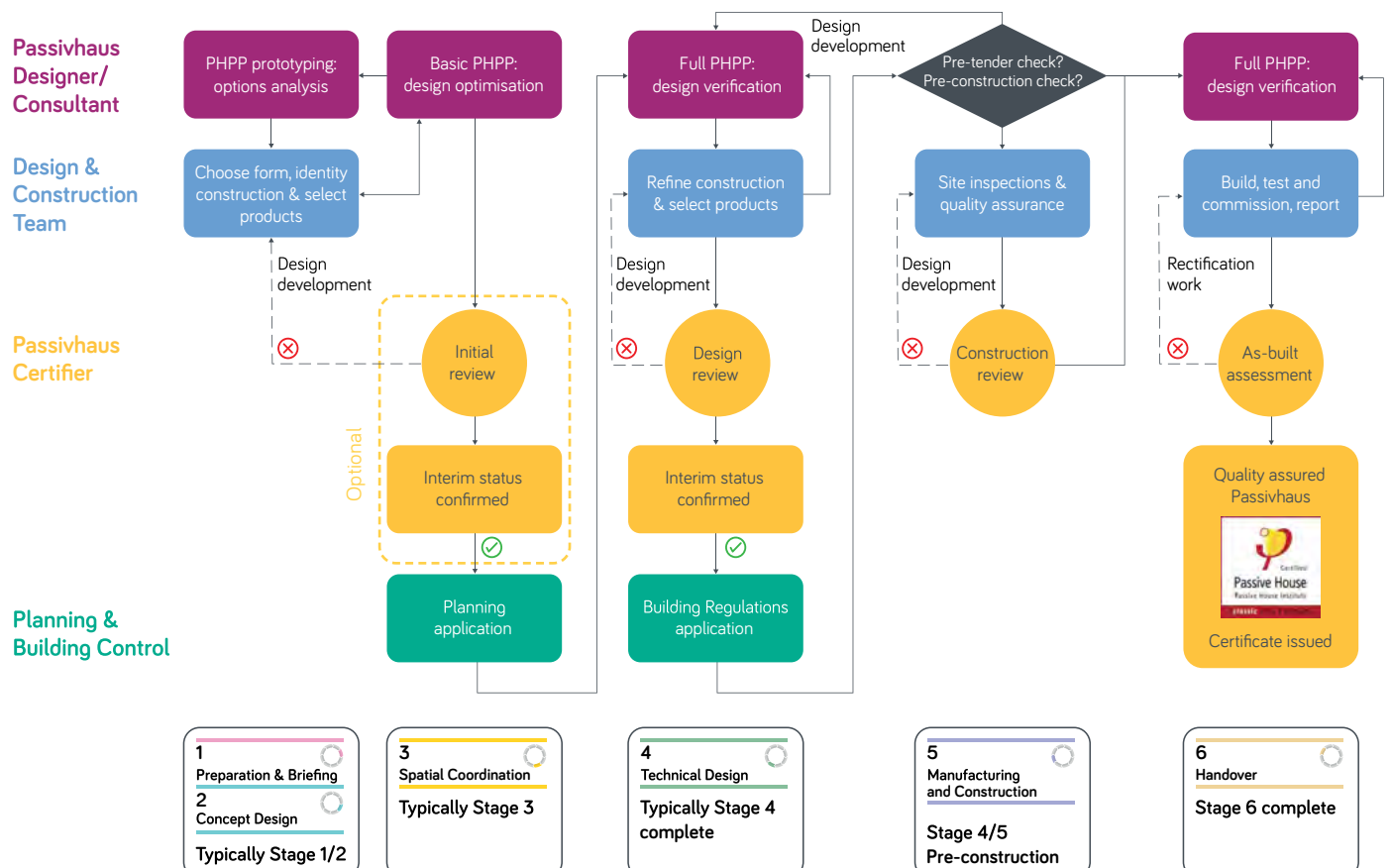
Building types that have straightforward assessment by any Certifier are:

- Any type of residential building, student accommodation, hotel, prison etc.
- Office buildings
- Schools and education buildings

Other types of building require input from the Passive House Institute and may have a slightly more complex assessment, or become a pilot project. Projects including retail, supermarkets, swimming pools, hospitals and healthcare facilities have all been certified to the Passivhaus standard. There are also typologies where certification may just not be practical – for example, aircraft hangers, light industrial garages or sheds.

Passivhaus is a construction standard. Only the completed building can be certified and there are no interim approval points. However, successful projects typically have earlier involvement, at least before tender, and Certifiers will often provide a letter of comfort to demonstrate to the client or contractor that the quality process is being followed, or to record the outcome of checks at key stages.

A typical Passivhaus project workflow for a successful outcome and Passivhaus certification. Find out more in pht.guide/PassivhausOverlayRIBA



Certification to the Passivhaus or EnerPHit building standards is a formal quality assurance process and assessment of the project's year round comfort and energy efficiency. Building certification involves a third party Passivhaus Certifier reviewing the technical design and construction information and PHPP calculation at key stages during the project.



Passivhaus certification is awarded on completion of the building or shortly after, at the “factory gate”. There is no post-occupancy evaluation or measurement required for certification, although it is highly recommended to further enhance building performance. The attention to detail and quality checking principles help ensure that the planned building will actually perform as designed for energy efficiency and comfort.

CAREFUL PLANNING

Certification is intended to ensure a good quality of work. Airtightness, thermal bridges, the quality of the windows and their installation, the ventilation system and other building services will all be checked as part of the project both during design and in construction. The main factor in successfully achieving certification is design, as only careful and considered design will ensure you complete all the required factors to achieve the Passivhaus standard.

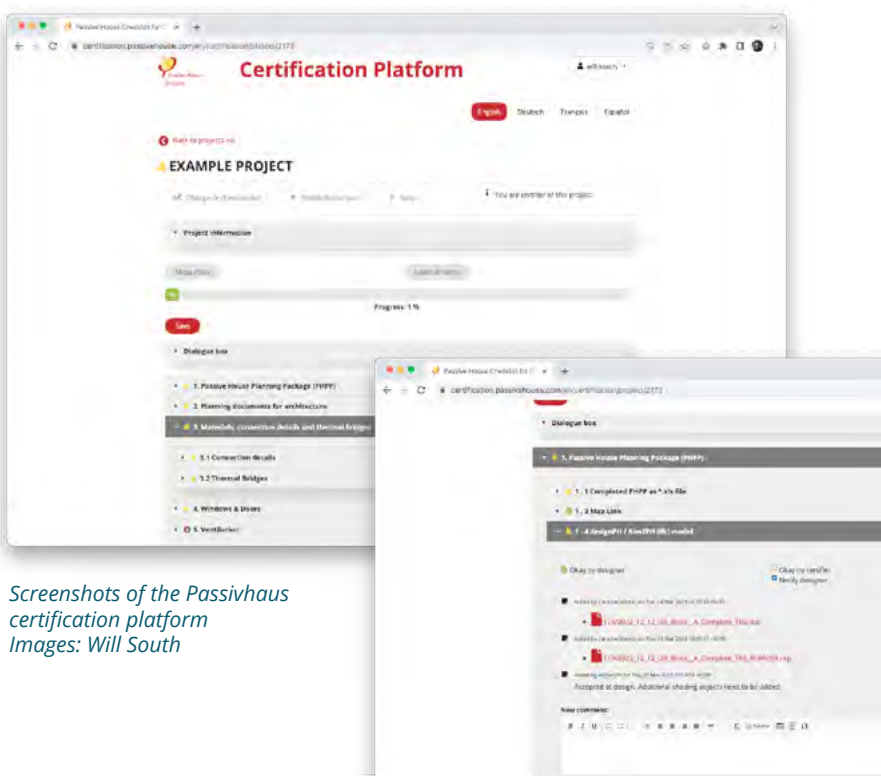
The process for designing a Passivhaus building is detailed in the Passivhaus Overlay for the RIBA Plan of Work (see left). Typically the Passivhaus Certifier will carry out design reviews at two or more points in the design process.

A more experienced design team on a straightforward building may not need as much input from a Passivhaus Certifier in earlier stages. Typically the latest involvement recommended for the Certifier is at technical design stage, tender stage, or pre-contract with the contractor builder.

TRACKING EVIDENCE AND GIVING FEEDBACK

The Passivhaus Certifier will use an evidence checklist for their quality assurance, which is normally provided on their appointment to the project. This checklist(s) should be referenced during the design stages with information collated as the project develops. This will save time when checks need to be made and signed off by the Certifier.

Projects may use the Passivhaus certification platform as an interface for uploading information and keeping track of progress, or a bespoke system may be set up. The Passivhaus Certifier will advise you on the best approach for your project.



*Screenshots of the Passivhaus certification platform
Images: Will South*

LEARN MORE



[pht.guide/
PassivhausOverlayRIBA](https://pht.guide/PassivhausOverlayRIBA)

do

- involve a Passivhaus designer from the start of the project
- model your design in PHPP at the early design stages
- engage with a Passivhaus Certifier once a design is developed to concept stage
- engage a Passivhaus certifier before tender and procurement (when you still have the chance)

don't

- assume that a building specification from a previous development will suit all future Passivhaus buildings
- try to certify a building that is already complete, this loses the majority of the value in the quality assurance process
- leave the collation of information to the last minute as this could cause delays in the building being certified



BENEFITS OF CERTIFICATION

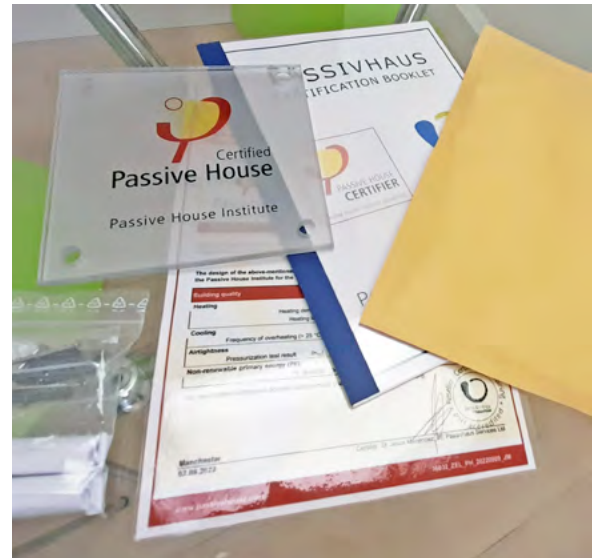
There are many benefits to certifying a Passivhaus for the designer and building owner:

- Passivhaus certification offers quality assurance during design and construction: it acts as a consistent input or golden thread focused on thermal comfort and energy efficiency.
- Certification checks catch mistakes in energy calculations and design before they have serious cost implications, and in construction while the impact can still be mediated.
- Quality checks during design and construction help the client know that they are getting what they pay for, and help ensure the design ambition for energy efficiency is achieved.
- The Certifier can share valuable experience and common issues from a significant number of projects from a wide range of designers and contractors, many more than individual designers or architects will see. This can help improve quality, and how solutions can be simplified and implemented in a better and more cost-effective way.
- The support provided by an experienced Passivhaus Certifier can result in a more economical building.

PERFORMANCE OVER TIME

Passivhaus certification is an assessment of the building at the point of completion and as such does not have a defined validity period. Research has shown that, properly maintained, Passivhaus buildings should continue to perform over time as intended. But bear in mind that Passivhaus is a whole building standard. To continue to enjoy the benefits of living in a Passivhaus, any modifications or extensions should be approached with just the same attention to quality as the original build, and you may want to consider recertification.

Certified Passivhaus Tradesperson training course. Image: WARM



*Passivhaus certification documentation and plaque
Image: Zero Energy*

PEOPLE AND PRODUCTS

As well as buildings, people and products can also be independently Passivhaus certified. Passivhaus projects don't have to use a Certified Designer or certified products, but it can make it easier to find the right team and equipment.

There are a number of accredited Passivhaus qualifications available for industry professionals. Architects, engineers, consultants, and others can attain the Certified Passivhaus Designer/Consultant qualification by one of two routes:

- pass a half day exam
- submit details of a certified Passivhaus building on which they were the person responsible for the Passivhaus design

A Certified Passivhaus Designer/Consultant training course is the best preparation for either route.

The qualification must be renewed after five years, either by collecting continuing education credit points or again, by submitting details of a certified Passivhaus building.

The Certified Passivhaus Tradesperson qualification is available for the site team, and there are also a number of further expert qualifications.

Passivhaus accreditation is also available for components, from windows and doors, through airtightness products, building services systems, connectors and anchors, to whole building systems. Components that achieve Passivhaus certification have been tested according to uniform criteria for each component type, which means they are comparable in terms of their specific values, and are of excellent quality regarding energy efficiency. Although it's not required for a Passivhaus project, using certified components can make the task of achieving certification for the building easier. The Passivhaus Trust recommends using only Passivhaus certified MVHR units.

CASE STUDY: Primrose Park



Certified in 2018, the Primrose Park development of 72 homes for affordable rent and shared ownership in Whitleigh, Plymouth, was at the time the largest residential Passivhaus project in the UK.

The developer Plymouth Community Homes worked in partnership with Plymouth City Council to deliver the fully Passivhaus certified scheme. As both client and contractor were new to Passivhaus, the role of the Passivhaus Consultants (WARM: Low Energy Building Practice) and Passivhaus Certifiers (Etude) working alongside architects with Passivhaus experience, was crucial to help guide the project to certification.



The success of the scheme depended on all members of the team working together and committing to the extra level of care and attention to detail required. The design on paper needs to be realised on site, and there will always be challenges that haven't been anticipated by the architect or consultant so it's essential that communication between designer and contractor goes both ways, and that time is made for feedback from site. At Primrose Park, the contractors on the project, Mi-Space, built mock-ups and sample panels to get details and buildability right before they were used as a model across the scheme.

Chris Herron, site manager on the build, told Inside Housing that, with Passivhaus, "You have greater control over a build. It means every aspect of construction is tied down before you start: the materials, supply chain and price. And the close monitoring means that quality is maintained. It's guaranteed all the way through. I've loved it – everyone got on board."

Left: Site visit to the development under construction

Below: Passivhaus Certifier Will South hands certification plaque to John Clark, CEO of Plymouth Community Homes. Images: Etude



6

Airtightness

Paul Jennings

Airtightness often appears to be the most difficult element to achieve in Passivhaus projects, yet once a robust airtight design is in place, delivery is primarily about rigour and attention to detail.

WHY IS AIRTIGHTNESS IMPORTANT?

Airtightness reduces heat loss, improves comfort and protects the building fabric.

Once the insulation levels in a building have been increased and the thermal bridges dealt with, heat losses from ventilation become significant and can account for a large proportion of all heat losses in a low energy building if airtightness is not considered.

Improving the airtightness of a building is likely to improve comfort. Air leakage through the building fabric, or uncontrolled ventilation, can lead to drafts. When the air outside is colder than inside, this leakage can be very uncomfortable. Air velocity is one of the basic indicators of thermal comfort. Movement of air at just 0.1 m/s can be felt as a draft in cold climates in the winter.

Airtightness also protects all building fabrics from the moisture in the air (note that being airtight is not the same as being moisture closed, 'non-breathable' or vapour impermeable). Movement of moisture by bulk air movement can carry far more moisture than vapour diffusion and if this air enters the building fabric, then interstitial condensation may occur. To minimise this risk, a rule of thumb is for the airtightness layer to be within the first third of the insulation U-value build-up from the warm side. Beyond that third, or for complex builds,

it is advisable for the detail to have hygrothermal modelling (eg. in WUFI) to ascertain the risk.

PASSIVHAUS AIRTIGHTNESS TARGETS

The Passivhaus airtightness targets are:

- **< 0.6 ACH @ 50 Pa**
for new build projects and
- **< 1.0 ACH @ 50 Pa**
for EnerPHit refurbishment projects

This compares with English and Welsh Building Regulations 2021, where a maximum air permeability of 8 m³/hr.m² @ 50 Pa is commonly permitted, and Scottish Building Regulations 2022, which allow a maximum air permeability of 7 m³/hr.m² @ 50 Pa. The Passivhaus airtightness requirements for new build are some 10 times more onerous than UK Building Regulations, while insulation requirements will typically only be 3 to 4 times more demanding than Building Regulations. So it is not surprising that UK contractors often find the stringent Passivhaus airtightness requirements particularly challenging to meet.

The quoted targets are the final acceptance value; it is sensible to leave a margin to allow for the leakage often found around services that are installed after the first airtightness test. This margin should be at least 25%, which means aiming for preliminary test results of:

- **< 0.45 ACH @ 50 Pa**
for new build projects and
- **< 0.75 ACH @ 50 Pa**
for refurbishment projects.

AIRTIGHTNESS OR AIR PERMEABILITY?

Building Regulations across the UK set targets for **air permeability** (q_{E50}), which is determined by dividing the airflow in m³/hr as measured during a pressure test by the **envelope area** (the total surface area of floor, roof and walls, including party walls, in m²) – given in units of m³/hr.m².

For Passivhaus, by contrast, the relevant metric is the air change rate, which is determined by dividing the measured airflow by the **internal volume** of the building (m³). The unit of measurement is ACH (air changes per hour), alternatively expressed as m³/hr.m³. This is the building's **airtightness** (n₅₀).

Passivhaus uses V_{n50} for the airtest volume, which is typically 20% less than used for UK building regulations compliance, and so more demanding*.

It's not possible to directly convert between ACH and m³/hr.m², because the relationship varies depending on the surface-to-volume ratio of the building. This can range from 1.2 or more for a complex detached house, to under 0.3 for a large tower block. For large buildings such as these (≥1500 m³), an air permeability measurement must also be taken to achieve Passivhaus certification. In addition, the airtightness and air permeability tests for the various standards follow different protocols.

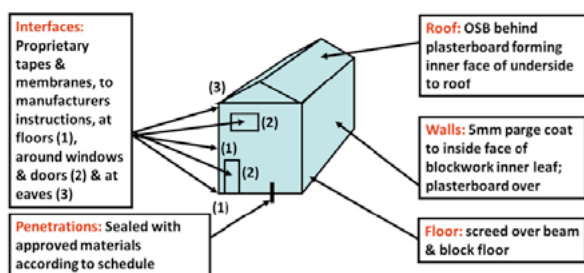
*https://passivpedia.org/planning/building_services/ventilation/differentiation_between_vv_and_vn50_values

In this chapter we discuss the Passivhaus requirement for an excellent level of airtightness and explain the principles of airtightness testing, highlighting where dwellings leak and how to go about meeting the airtightness target for a project.



AIR BARRIER STRATEGY

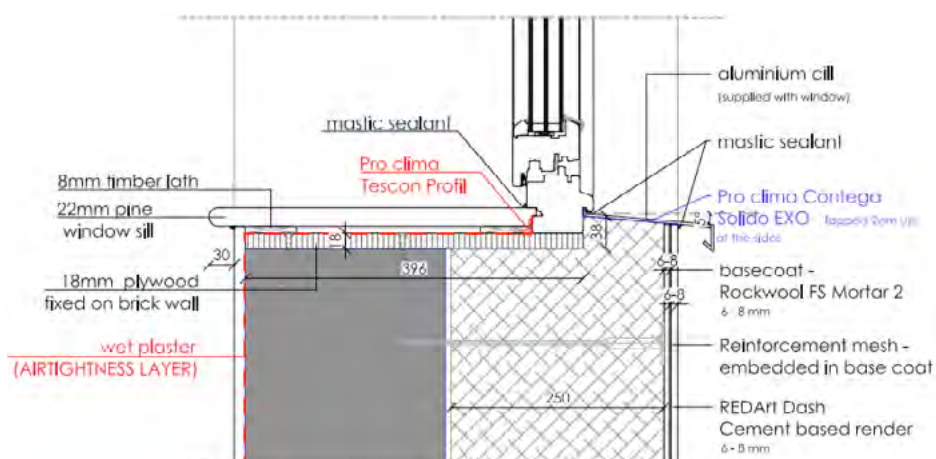
Deciding what sealing products and processes are required to deliver the target airtightness leads to the development of a formal air barrier strategy. For clarity and ease of communication, we suggest that this is summarised in the manner shown below, tailored for an individual project:



The figure above is an example of a summary of the air barrier strategy to deliver Passivhaus airtightness in a new build warm-roof dwelling. For a specific project the manufacturer of sealing tapes and airtightness membranes should be detailed, and a list of penetrations and how they are to be sealed would also be needed. Your project would be individual and almost certainly different, but summarising it in this way is an essential step in communicating how the Passivhaus target is to be achieved. The next step should be to prepare a set of airtightness drawings, detailing the line of the airtightness barrier on plans and sections - see example right.

SPECIFICATIONS FOR AIRTIGHTNESS

When specifying airtightness for a building where the target is to achieve Passivhaus or EnerPHit certification, it is essential to specify the airtightness process as well as the airtightness target and the sealing materials and process to be used.



QUALITY ASSURANCE AND AIRTIGHTNESS CHAMPION

This is the key role in ensuring an airtight design and quality assurance is satisfactorily delivered on site. Sealing failures can lead to costly additional works and delays.

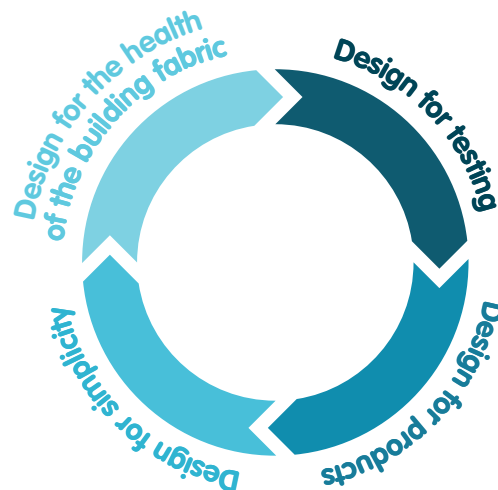
SIX KEY THINGS TO GET RIGHT

- 1 Keep designs simple
- 2 Choose robust materials and don't substitute on site
- 3 Think about junctions in 3D, visualise how they will be built and produce clear drawings that illustrate this
- 4 Communicate verbally, ensure all parties have understood the design, use site meetings, workshops, phonecalls
- 5 Put sensible site management processes in place, employ an airtightness coordinator and educate subcontractors
- 6 Undertake leakage tests whilst the air barrier is still accessible

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Above: Concept design principles for airtightness

Left: A detailed drawing showing elements of the airtightness barrier in red

AIRTIGHTNESS TESTING

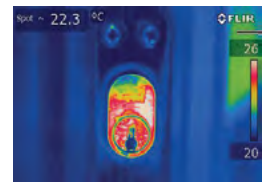
Interim testing is essential for any airtight building. The building should be tested at a stage where most of the air barrier is still accessible, so that remedial works can be undertaken. Smoke pens or thermal cameras can be used to aid leak detection. This can be done by an experienced airtightness tester, or it is possible to rent fans and train up site operatives in leak checking, which may allow for more flexibility and save money on larger sites.

The final test is undertaken upon completion of the building or retrofit and in compliance with Passivhaus procedure (ATTMA TSL4) for Passivhaus certification evidence.

LOCATING LEAKS

During airtightness testing, the pressure differential imposed on a building, usually during depressurisation testing, accelerates the airflow through leakage points, making them easier to identify. We use a variety of tools – thermal imaging equipment, vane anemometers, noise detectors and non-toxic smoke – to find these leaks, and can then of course use the door fan equipment to check the effectiveness of remedial sealing works.

Another common tactic is to fix a square metre section of polythene or similar membrane over the surface of a wall that we suspect is leaking. Under depressurisation, if the membrane bulges into the property, there is clearly a leak through that wall construction. A common gross leak that we routinely find is through door locks, even in very expensive Passivhaus-certified doors.

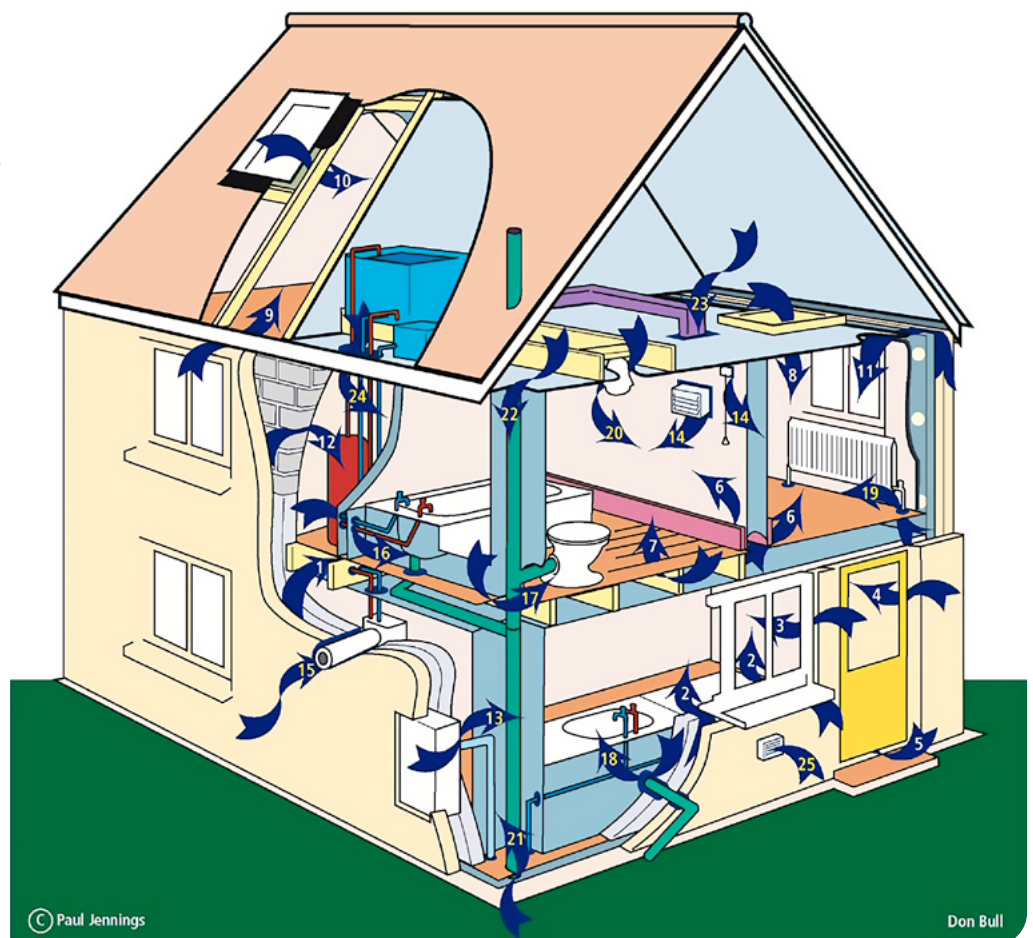


Thermographic image of a leaky door lock, where the blue key hole is much cooler than the adjacent metal lock.

WHERE DWELLINGS LEAK

This diagram illustrates many of the leakage sites that have been found in new and existing housing. One key point to remember is that wherever there are voids, there is the potential for complex connecting air paths to form, allowing leakage to occur in some very surprising locations.

- 7 Around floor joists
- 8 Beneath windows sills
- 9 Around windows
- 10 Around doors
- 11 Beneath door threshold
- 12 At edges of suspended floors
- 13 Between floorboards
- 14 Loft hatches
- 15 At eaves
- 16 Around rooflights
- 17 Behind drylining
- 18 Through poor masonry
- 19 Via meter boxes
- 20 Around and through sockets
- 21 Around boiler flue
- 22 Plumbing to bath
- 23 Around waste from WC
- 24 Kitchen waste
- 25 Around radiator pipes
- 26 Around spotlights
- 27 At base of soil vent pipe
- 28 At top of soil vent pipe
- 29 Around duct connections through ceiling
- 30 Around pipe connections to tanks in unheated loft
- 31 Around tumble drier vent





Airtightness testing underway at Fishleys Passivhaus.
Image: Liz Taylor & Mike Hill

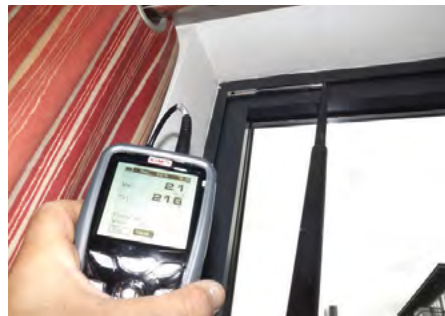
TESTING PROCEDURES

When carrying out an airtightness test, a steady-state condition is established, when the air blown out of the building by the fan is balanced by the air re-entering the building through various cracks, gaps and openings, and vice versa - respectively positive (pressurisation) and negative (depressurisation) testing. The reference pressure is 50 Pascals (Pa), about the same as a 20 mph wind acting upon the volume under test. The current standard referred to in the Building Regulations is CIBSE TM23 (2022) which recommends both positive (pressurisation) and negative (depressurisation) testing, where this is practicable. Passivhaus testing, in conformance with ATTMA (Air Tightness Testing & Measurement Association) TSL4 (2021), requires both pressurisation and depressurisation testing.

The result calculated from the series of readings, usually generated by proprietary software, is an airflow at a pressure differential of 50 Pa, with readings often being spread between 20 Pa and 80 Pa. For comparison, a 20 mph wind generates a pressure of 50 Pa on a building. In high-rise buildings in particular, the stack effect caused by warm air rising can make testing problematic, often requiring test equipment to be installed on multiple levels.

REQUIREMENTS FOR A MULTI-POINT TEST

- a minimum of 7 readings, typically 11 or 12 readings, at ≈ 5 Pa intervals, 15 or more readings recommended in windy conditions
- at least one reading at a pressure differential of magnitude > 50 Pa
- no readings at pressure differentials of magnitude > 100 Pa
- the zero-flow pressure differential (e.g. wind-induced) shall have a magnitude of no more than 5 Pa
- the correlation coefficient, r^2 , must be > 0.980
- the airflow exponent, n , (a measure of the turbulence of the airflow through the various leaks) must be: $0.5 < n < 1.0$.



Below and left: using chemical smoke to check for leakage

Above left: Extending vane anemometer showing a high air velocity of 2.1 m/s through a gap at the head of a balcony window



do

- develop a formal air barrier strategy
- appoint or nominate a quality assurance and airtightness champion
- the airtightness testing organisation must be acceptable to the certifying body
- plan staged testing – early testing when the air barrier is visible makes rectification easier, quicker and cheaper to complete, helping to de-risk achieving the Passivhaus targets for contractors
- ensure testing follows the ATTMA TSL4 protocol, including both pressurisation and depressurisation testing
- perform the final acceptance test after the building is completed. No temporary sealing, other than to exclude the ventilation system, is permitted

don't

- assume that something is sealed – always check
- forget to calculate air volume to the Passivhaus methodology (V_{n50}) – this has much more detailed calculation requirements and the resulting volume can be 20% smaller, making it harder to pass the airtightness test
- allow the construction program to override quality requirements – remedying airtightness defects at the end of a build can be disruptive, costly, and significantly delay completion

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SOME AIR BARRIER APPLICATIONS



All images: Mhairi Grant and Martin McCrae, Paper Igloo



CASE STUDY: Larch Corner



Larch Corner, an all-timber Passivhaus in a sleepy village in the Cotswolds, is believed to be the most airtight house in the UK and the third most airtight house in the world.

Although this was a first time Passivhaus project for both contractor and client, the team's attention to detail and commitment to quality delivered a record-breaking airtightness result of 0.048 ACH @ 50 Pa with relatively little effort. To accurately measure such a low level of air leakage, airtightness expert Paul Jennings of Aldas was compelled to recalibrate his fan, and ATTMA attended the test as an independent witness. This detached home is twelve times more airtight than required by Passivhaus and 244 times more airtight than Building Regulations – if all the points of leakage were combined into a single area, they would fit onto a one penny coin.

The building's poor form factor (the result of planning requirements, and already fixed before Passivhaus was considered for the project) didn't make it easy to hit the Passivhaus targets – a wall U-value of 0.092 W/m².K and roof 0.122 W/m².K was needed. The project is exemplary in tackling upfront carbon – the adoption of an all-timber construction, using sustainably sourced CLT, has resulted in a reduction in embodied carbon of 40% over more traditional masonry.

All images: Mark Siddall



Testing carried out by:

Paul Jennings, Aldas

Target air changes, ACH @ 50 Pa:

0.6

Achieved air changes, ACH @ 50 Pa:

0.048

Achieved air permeability, m³/hr.m²:

0.041

Data consistency, r² (requirement: r² > 0.98):

0.995

Slope, n (requirement: 0.5 < n < 1.0):

0.89

All construction types can achieve the Passivhaus standard. The decision over construction type often comes down to:

- building form and planning restrictions
- ease of achieving a good thermal envelope
- cost restraints, buildability, speed, material availability, risk, client/ contractor preference and local skills.

BUILDING FORM

Compactness of building form makes it easier to achieve Passivhaus standards, because the heat loss area (envelope) is minimised in relation to the building's volume. As a rule, terraced housing will need less insulation than detached houses, and a block of flats less still. Integral garages, ornate dormer windows, bay windows, or an intricate building footprint, all worsen the ratio between a building's envelope and its volume, and make it more difficult to achieve Passivhaus.

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ESSENTIAL ELEMENTS OF THE THERMAL ENVELOPE

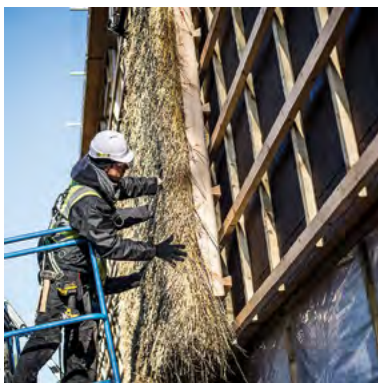
- 1 Rainscreen
- 2 Windtight vapour open layer
- 3 Super insulation
- 4 Airtight layer
- 5 Structural zone
- 6 Service zone

Examples of these essential elements of the thermal envelope are illustrated below. Note that this list does not imply that the elements will be arranged in this order, or that they will necessarily be separate elements.

CONSTRUCTION COSTS

In the context of other factors which result in higher build costs (design form, ground conditions, choice of construction type, finishes etc), the extra cost of building to the Passivhaus standard can be only a minor uplift, and results in a far superior product in terms of running costs, carbon emissions, comfort levels and health benefits. A detailed analysis of factors particular to Passivhaus that have an impact on construction cost, and tips on how to keep costs down, are available in *Passivhaus construction costs* (PHT, 2019). Passivhaus may not deliver the lowest cost, but it can deliver the best value.

- 1 thatch being installed as rainscreen at the Enterprise Centre (Image: Architype)
- 2 windtightness membrane at Larch Corner Passivhaus (Image: Mark Siddall)
- 3 full fill masonry cavity wall at Steel Farm Passivhaus (Image: Mark Siddall)
- 4 airtightness membrane (Image: ProClima)
- 5 I-beam portal frame structure (Image: Conker Conservation)
- 6 service cavity at Silverdale Road (Image: Etude)



The fabric, or envelope, of a Passivhaus needs to achieve high thermal and airtight performance while providing structural stability and weather protection. For building durability, the design also needs to eliminate the danger of interstitial condensation forming within the structure. This section will look at general principles to consider in design and construction of the building fabric.



PRINCIPLES OF FABRIC

The essential principles of fabric, both in the design and the build processes, to be considered by the whole team throughout the whole process in delivering a Passivhaus are:

Airtightness

Passivhaus buildings need to achieve an airtightness of 0.6 ACH @ 50 Pa. The airtightness strategy must include a continuous layer joining each of the fabric components, using an appropriate material robust enough to last the lifetime of a building. Not only does an effective airtightness layer stop draughts and reduce energy consumption (by limiting heat loss), it also protects the fabric of a building by preventing internal warm humid air from passing into the fabric, where it could form interstitial condensation. See chapter 6 above.

Insulation

A Passivhaus building has a very high quality insulated envelope to achieve the strict space heating demand target, but the exact U-values required for each element of the construction will be determined by the whole building modelling in PHPP. Quality of design and installation is key – there has to be a consistency of ‘wrapping’ the whole building, with particular attention to the junctions of the different fabric elements, in order to prevent thermal bridging and thermal bypass.

Minimising thermal bridging and bypass

The need for structural integrity, and the need to allow light and access into a building leads inevitably to the use of different materials with different thermal properties. Good design of each of these details to minimise the thermal loss is crucial both to keep heating demand down and to avoid cold spots where condensation might form.

Thermal bypass is the wicking of heat from a building by air movement around and through its fabric. A breather membrane layer on the cold side of the building's fabric will help avoid some of the more common causes.

Windows

In the UK, Passivhaus buildings almost always require triple glazing. The approach to glazing should start by getting the design right, considering both heat loss in winter and overheating risk in summer, before moving on to detailed specification of glazing products. The details and execution of installation can have a big impact on performance.

Ventilation

Mechanical Ventilation with Heat Recovery (MVHR) is essential for any airtight building. These units extract warm but potentially stale and humid air from wet rooms and pass this over a heat exchanger which transfers the heat to air from outside, providing 100% fresh and filtered air to the building. MVHR will be explored in chapter 8 – "Building services".

LEARN MORE



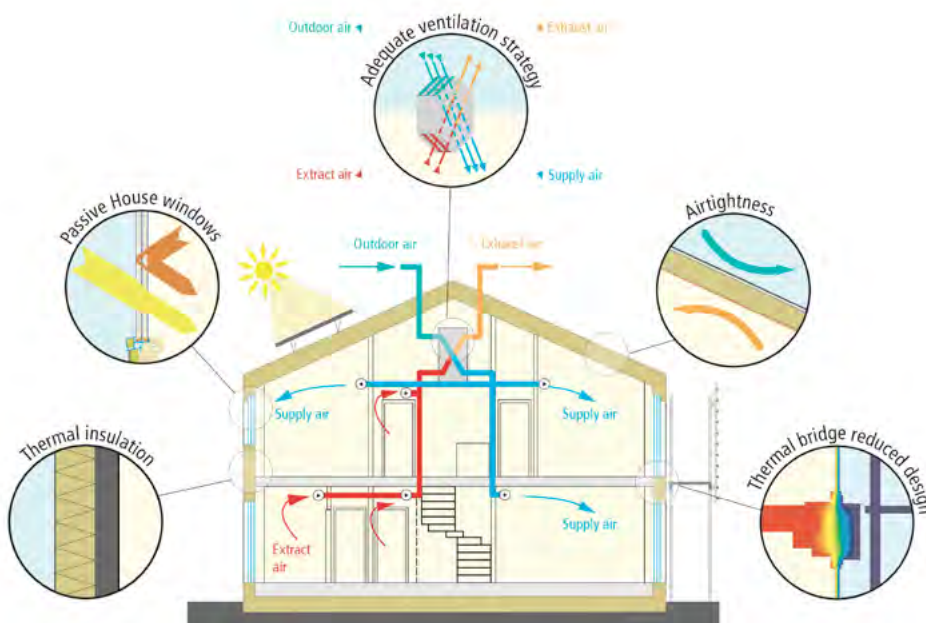
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airtightness](https://pht.guide/airtightness)



[pht.guide/
insulation](https://pht.guide/insulation)



[pht.guide/
thermalbypass](https://pht.guide/thermalbypass)



“There’s a lot to learn and hurdles to go through but I genuinely don’t think it’s so hard, it’s not that different. Once you’ve got your design, there comes a point in the project where it just becomes a normal build again. Get your substructure right, get your thermal envelope right, and then really after that it’s just a normal building.”

**Euan McDermott,
Pre-Construction Manager,
Fleming Buildings**



Walls

INTRODUCTION

Typically wall, roof and floor elements in Passivhaus buildings should aspire to: a U-value of between 0.10 W/m².K to 0.15 W/m².K, whatever the construction method employed; minimal thermal bridging; and ideally an internal service void so that services do not penetrate the airtight/vapour barrier. The thickness of the wall construction needed to achieve the required thermal performance will vary greatly, possibly from 300 mm to 500 mm, depending on the chosen insulation and construction type used. This section looks at the advantages and disadvantages of typical UK wall construction techniques.

TIMBER CONSTRUCTION

There are a range of timber frame construction options available. Timber frame construction can either be site fabricated (known as 'stick construction') or prefabricated in a factory and delivered to site in panels.



*Bristol Passivhaus Plus under construction.
Image: Greenheart Construction*

Timber stud

The most common timber framing system would typically use 144/44 mm timber studs at 600 mm centres filled with insulation. This would not normally achieve a recommended U-value of 0.10 – 0.15 W/m².K, particularly when extra studs have to be inserted for structural reasons around window openings. Therefore, additional insulation is required, usually applied externally over the whole frame, covering junctions and studs.

Double timber studs

This is a relatively simple way of achieving the depth of insulation needed, while providing the building's structure or skeleton.

Timber I-joist system

Timber I-joists (or TJI joists) have a lot of advantages and offer a relatively economic way to attain low U-values. They are available in many sizes and are structurally strong, with minimal thermal bridging due to the slim compressed timber web.

Cross laminated timber (CLT)

CLT has become very popular with architects due to both its aesthetics and structural integrity. It has high levels of airtightness and is quick and easy to use. The material consists of planks of sawn, glued, and laminated wood.

Larsen trusses

Though not offering structural support to a building, Larsen trusses offer a useful method of containing insulation and giving support for a cladding board and rain screen, such as rendering or timber cladding. Larsen trusses can be used in conjunction with timber framing or masonry walls in new build or retrofit projects. They are very simple to use and can be built on-site or prefabricated.

Closed panel systems

Closed panel systems are fully insulated wall panels that are structurally engineered off-site, and together with the floor joists, roof trusses and roof panels, form a complete structural frame. Typically they will comprise of: an exterior breathable membrane; exterior insulating breathable board, timber I-beam studs which minimise thermal bridging; insulation within the panels (such as recycled cellulose or mineral wool batts); an internal airtight vapour barrier anti-racking board. Once the structure is installed on site, external finishes can be fitted, such as render system, brickwork or wooden cladding.

Eden insulation / Kirsty Maguire



Portal frame systems

Portal frame kits include timber I-joists and plywood gussets that are strong enough in their own right that they don't need cross-members to support them. They arrive on-site in kit form and are assembled in situ by joiners. They are particularly advantageous for sites limited on space as, unlike closed panel systems, they don't require large lorries and cranes to offload the panels.

Structural insulated panels

Structural Insulated Panels (SIPs) are an off-site, easy-to-build wall system commonly made up of rigid insulation, such as PU or XPS, sandwiched between 15 mm OSB board. Some SIP panel systems need additional insulation to achieve the U-values necessary for Passivhaus, particularly the thermal bridging criteria. This additional insulation should preferably be installed externally to minimise the thermal bridging of solid timber sole and wall plates, and to compensate for the additional timber commonly used around window and door openings.



ADVANTAGES AND DISADVANTAGES OF TIMBER FRAME CONSTRUCTION

The key **advantages** of timber framing are the speed of erection and the ability to easily monitor the quality of insulation and membranes. Airtightness testing can be carried out at an early stage of the build process, when any faults can be rectified relatively easily.

Disadvantages of timber frame walls include the need for additional studs to perform a structural role (e.g. around window openings), which create extra thermal bridging. When timber frame is used with a brick or stone rainscreen, there is a danger of moisture build-up in the timber frame, particularly during the summer months when strong sunshine can drive moisture from a rain-soaked masonry façade into the cavity (known as 'solar-driven moisture'). This situation can be prevented and controlled by ensuring that there is an adequately ventilated cavity, but research in this area is still in its infancy.



Above: Beltane Drive Passivhaus Plus under construction.
Left: Timber I-beams. Images: Passivhaus Homes

COMPONENT MATERIALS WITH TIMBER FRAME

This section looks at the types of materials used with timber frame construction to make up the rest of the walls.

The **windtight layer** in a timber frame wall construction (also known as the breather membrane) can be a rigid board or membrane, which will need taping to eliminate thermal bypass. Modern 'vapour open' but windtight and water-resistant materials allow any moisture within the fabric to evaporate harmlessly to the outside.

Insulation in timber frame walls can be of any type. Flexible insulation can be used between studs to cope with the thermal movement of the timber, minimising the possibility of air pockets which can lead to thermal bypass. Alternatively, if rigid board is used between timber studs, careful installation is essential to ensure all gaps are fully-filled, and this should be checked whether onsite or in the factory.

The **airtightness**, or vapour control, layer can be achieved using a variety of means. These include intelligent membranes with variable permeability, plasters/ render, concrete, specialist gap-filling airtightness liquids, and boards/ woods, such as specialist melamine-faced chipboard. OSB is not usually recommended as even at a thickness of 16 mm and above it can vary in its effectiveness. Attention to detail is needed at joints and connections around airtightness materials, for example when using specialist tapes, grommets, and butyl mastics.

MASONRY CAVITY WALL

Cavity wall is still probably the most common form of construction in the UK, so is familiar to most builders, but the approach needs to be adapted to achieve the more rigorous Passivhaus requirements.

Insulation in the cavity

For a masonry cavity wall to meet Passivhaus standards, we need the insulation to be able to cope with the natural rough surfaces of masonry walling without leaving air passageways for thermal bypass. One way to do this is to use soft fibrous insulation, such as mineral wool, where mineral wool cavity batts are woven in vertical layers and are designed to be able to get wet in the first 10-15 mm from any wind-driven rain penetration through the naturally porous brick or stone. The use of well-installed cavity trays over window and door openings, and vigilance in keeping insulation batts clean from mortar droppings ensures thermal performance is as designed.



Airtightness

Airtightness can be adequately dealt with by traditional two-coat plasters. Where the plaster meets another material, such as timber, airtightness membrane, OSB, plywood, etc, cracking will occur through differential movement, seriously affecting the airtightness strategy. Specialist airtightness tapes are needed to address and cope with this issue. Components such as intermediate timber floors, staircases, SVPs, etc that are installed before plastering require a preliminary simple weak sand, cement/lime parge coat applied to the blockwork. The parge coat should give adequate airtightness and blend with the plastering later. Occasionally some building contractors are opposed to parging and two-coat



Kirkburton Passivhaus. Image: Green Building Store

Wall ties

Thermal bridging from wall ties should be minimised by using specialist wall ties made of lower conductivity materials such as basalt and resin or nylon. Where stainless steel wall ties are used, the repeating thermal bridging of the ties diminishes the overall U-value of the wall and needs to be compensated for, e.g. with thicker insulation.

Below left: Cavity wall construction at Kirkburton Passivhaus

Below right: Cross section at the first cavity wall Passivhaus in the UK, Denby Dale. Images: Green Building Store



wet plaster approaches, due to the time they might take to dry and/or plastering skill shortages etc. However, the quicker method of dot and dab plasterboard does not deliver the requisite airtightness levels for a Passivhaus. An alternative option could be use of specialist airtightness paints, which can be spray-applied, have a quicker drying time and can sometimes be used as a suitable substitute for parging and two-coat wet plaster. Such products are especially suitable for hard-to-reach areas such as ensuring airtightness around the back of electrical sockets. However, caution is needed to ensure that the substrate is appropriate to receive these materials and that they are carefully applied to achieve an effective airtightness barrier. A test section should be carried out.

ADVANTAGES AND DISADVANTAGES OF MASONRY CAVITY WALL

Advantages Materials are readily available and the construction method is familiar to the wider construction industry, though not necessarily to the stringent Passivhaus standards. There is no danger of interstitial condensation and holes in the plaster airtightness layer can always be repaired as part of redecoration work. Masonry cavity wall can be the cheapest construction method and is still the most common house building construction method in the UK, so its use will avoid taking builders outside of their experience and 'comfort zone', and hence reduce the risk of over-pricing due to perceptions of risk.

Disadvantages Masonry cavity wall is a slower construction method than timber frame and it is more difficult to check the quality of the continuity of insulation (eg, the cleanliness from mortar dropping, etc). Airtightness testing occurs later in the building programme which can make it more difficult to rectify hidden faults. The absence of a service void means that more care is needed in services design and execution. Although electric cabling can be buried in the plaster effectively, badly installed back boxes can create weak points. A longer structural drying out period will affect the first season's heating demand.

SOLID MASONRY WITH EXTERNAL INSULATION

In theory, this is the easiest method to attain the Passivhaus goals for walling, and can be achieved with a wide choice of rigid and semi-rigid insulation materials. The structure, usually of concrete or blockwork, is inside the thermal envelope providing good thermal mass. The airtightness layer is usually applied to the outside of the structure in the form of a flattening render coat, in preparation for the adhesive layer for the insulation. Thermally broken mechanical fixings are usually required. The rainscreen is commonly a render system, with integral keying membrane. It is crucial to inhibit air movement between the insulation and the structural walling, as any thermal bypass would seriously diminish the thermal performance. It is important to have a flat surface on which to fix the insulation and to apply adhesive evenly with a notched trowel.

ADVANTAGES AND DISADVANTAGES OF SOLID MASONRY WITH EXTERNAL INSULATION

Advantages Insulation and structure are completely separated, meaning that thermal bridging can be minimised. With this system, quality of workmanship is, on the whole, visible, making quality assurance easier. There is also no need for cavity trays with this method of construction. All trades can proceed within the building without the worry of damaging the airtightness layer.

Disadvantages External rendering can be easily damaged because there is no structure behind it. If heavier cladding (such as brick slips) is used there may be significant thermal bridging problems if supported off the internal structure. Contractors in the UK do not appear to be experienced yet in the care needed to minimise thermal bypass, and external insulation systems to reach Passivhaus standards seem expensive in the UK at the present.



Clay block walling at Chester Long Court in Exeter

NEW MASONRY MATERIALS

It is worth noting that new innovative products are coming onto the market, offering reduced embodied carbon compared with the traditional concrete blocks used for masonry construction. For example:

- Cement-free dense concrete block
- Clay block walling systems

If exploring the use of materials that are being promoted as having lower embodied carbon, we recommend the use of a rigorous carbon calculating tool such as PHribbon. It is also obviously important to check that the thermal properties of these new materials can help achieve the U-values required for the Passivhaus standard.

INSULATED CONCRETE FORMWORK (ICF)

ICF is a simple construction system that uses hollow polystyrene block components. These blocks are dry stacked together to provide a formwork system, which is then filled with concrete to provide a solid wall. The concrete bonds to the insulation on both sides and as such the air barrier can be either the concrete or the insulation layer. ICF is useful in creating curved walls and other non traditional shapes, so long as it can be formed out of insulation. The structure is always insulated on both sides, which can create a complicated thermal envelope, especially when ICF is used for internal structural walls. Be careful to avoid structures that penetrate insulation on both sides of the ICF and opposite, as this will create a thermal bridge.



ADVANTAGES AND DISADVANTAGES OF INSULATED CONCRETE FORMWORK

Advantages ICF is a relatively easy-to-use system often favoured by self-builders.

Airtightness is inherent in the system since the bonded insulation-concrete-insulation structure is airtight, and there is flexibility in where the air barrier is chosen to be. Internal or external structures connected to the ICF will usually bridge one side of the insulation, but there will always be the other side of the insulation to protect the internal surface temperature and mitigate the thermal bridge.

Disadvantages Thermal bridging can be complex, for example if the ICF is used for internal structural walls. It can be challenging to ensure continuity of airtightness at window reveals with an ICF system when the concrete is seen as the airtightness layer. And for small domestic projects, ICF might not reach the necessary U-values without additional insulation.



Intermediate floors (masonry & timber)

There are two types of intermediate flooring system construction, with the less common 'Balloon floor' design offering distinct advantages to Passivhaus Designers. 'Platform floor' is a very common form of construction in timber frame construction. Wall panels are built to ceiling height and the floor is placed on top, and then another wall for the second storey. The disadvantage of a platform floor approach is that it breaks the

airtightness and insulation strategy, as well as causing thermal bridging difficulties. These can be dealt with using careful detailing and specialist tapes and membranes, but it is more fiddly and complex than the detailing for balloon floors.

Balloon floors offer improved detailing for Passivhaus buildings by making the walls more continuous between the floors and using a wall plate to attach the joists for the intermediate floor. This offers better detailing in terms of continuity of the airtightness and insulation strategy, so long as the area behind the wall plate is treated carefully for airtightness, e.g. with parging, use of membrane, or use of specialist airtightness paint.

Balloon floor being installed at Bristol Passivhaus Plus. Note joints already taped for airtightness behind the joists. Image: Greenheart Construction



CASE STUDY: Old Holloway



A single storey detached home self-build, Old Holloway utilised an innovative pre-fabricated straw-bale/timber construction on an insulated concrete slab. The prefabricated Ecococon timber/straw prefab panel system is a certified Passivhaus component, and allowed for rapid and accurate construction. The external walls were erected in three days and the house was watertight in four weeks. Each Ecococon panel has holes at the top and bottom, and wooden dowels are used to locate the adjacent panels accurately before they are screwed together. Unlike 'traditional' timber frame, the Ecococon system has the airtightness layer on the outside face of the wall, covered with a further layer of woodfibre insulation and, in this case, a timber rainscreen cladding.

Materials used on the project include self-finished clay plaster walls, charred timber cladding from local cedar, and corrugated metal roofing which was local and very economical. Orientation of the house is perfect due south, maximising the far-reaching views. The large south roof overhang in combination with manual external blinds provides a simple way of minimising the risk of overheating.

*Below, l & r: Ecococon straw panels arriving and being assembled
Bottom: Cathedral roof with timber I-joists hung from glulam ridge beam.
All images: George Mikurcik*





Ground floors

INTRODUCTION

Ground floors can be either solid or suspended. Solid floors tend to be the easier and cheaper option, whereas suspended floors tend to be used where the thermal envelope is above a basement, or where planning authorities require a floor void because of flooding risk. Suspended timber ground floors are not advisable for Passivhaus new build, because of the inherent moisture risk and complicated detailing required.

INSULATED SLAB

An insulated slab ground floor system satisfies all the Passivhaus principles of design, including: continuity of insulation, no thermal bridging, airtightness of the concrete slab, and no issues with thermal bypass. In principle, achieving a 'thermal bridge free' junction with the walls is ideal, although the design depends on the walling system used:

- A lightweight construction, such as timber frame, lends itself to being built directly off the slab
- Heavyweight construction commonly needs direct support from the foundation; cost-effective minimisation of this thermal bridge requires the project's structural engineer to understand the fundamentals of 'thermal bridge free' construction.



Retrofit of an existing suspended timber floor - specialist membranes and tapes provide windtightness, airtightness and vapour control, and a vapour-permeable insulation is installed between the rafters. Image: Ecological Building Systems

BEAM AND BLOCK

On its own, beam and block floor construction will not satisfy the principles of Passivhaus design mentioned previously. By its very nature, ventilation is created under the system allowing air movement in and around the floor structure. The simplest strategy is to treat the beam and block floor as providing support only. To achieve Passivhaus performance levels, a taped windtightness membrane can be laid over the beam and block and carefully joined in with the wall windtightness layer. Insulation should then be added to the required depth on top, with the top screed over. The screed will also become the floor's airtightness layer but will need taping into the wall airtightness layer, as the screed will shrink back from the internal walls during drying out.

INSULATED RAFT FOUNDATIONS

Insulated raft foundations are designed to dispense with the need for complex concrete strip foundations and work especially well with lightweight timber frame systems. The raft is designed to thermally encase and super-insulate the foundation floor slab in expanded polystyrene (EPS). It simplifies the thermal bridging detail at the junctions between foundations, ground floor, and the walls. The polystyrene raft with upstands is laid down onto hardcore blinding, into which the reinforced concrete floor for the building is poured. The concrete floor is therefore completely surrounded by insulation so there is minimal heat loss and thermal bridging from the ground floor.



An insulated raft foundation (Isoquick) being installed, in preparation for rebar and the concrete pour. Images: Paper Igloo

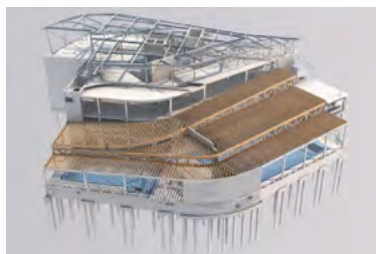
PILED FOUNDATIONS

Pile foundations are usually used in situations where the bearing strata is so deep that it is uneconomic to do standard foundations. There are several ways that pile foundations can be formed including reinforced concrete or steel columns. A foundation is described as 'piled' when its depth is more than three times its breadth. Piled foundations usually have a concrete ringbeam and integral reinforced floor slabs on top of the piles. As thermally broken piles are not an option, for detached dwellings a thermal bridge free ground floor is typically achieved by using an insulated raft over a ring beam or other pile supported structure.

For larger buildings where piling is more likely to be required, the thickness of floor insulation required is much less and is easiest applied to the top of the slab rather than under the slab and around beams and piles. Load bearing masonry can be thermally broken at the wall slab junction with insulating blocks but a concrete frame will introduce thermal bridges. For a large building these can be tolerated as long as surface temperatures are acceptable which would be the same for any Building Regulations compliant building. Vertical perimeter insulation will help raise surface temperatures and reduce heat loss.

INSULATION MATERIALS IN DAMP CONDITIONS

Care is needed in choosing insulation for ground floor and foundations that are outside the damp-proof membrane, as the thermal performance can be substantially diminished depending on which material is used. Extruded polystyrene is probably the most appropriate insulation material in ground floors and foundations, where installed below the damp proof membrane.



Above and left: Piled foundations at St Sidwell's Point Leisure Centre in Exeter. 250 mm of high-density EPS was installed around the pile caps and 1000 mm down the side of the piles to reduce thermal bridging. Images: Exeter City Council, Gale and Snowden

CONCRETE-FREE FOUNDATIONS

A few Passivhaus Designers and structural engineers are also now exploring ways to build Passivhaus buildings using concrete-free foundations, to reduce embodied carbon and for situations where site access makes concrete delivery difficult. It is obviously important to get advice from an experienced structural engineer if exploring this option.



Right: limecrete utilised for shallow foundations. Images: John Butler





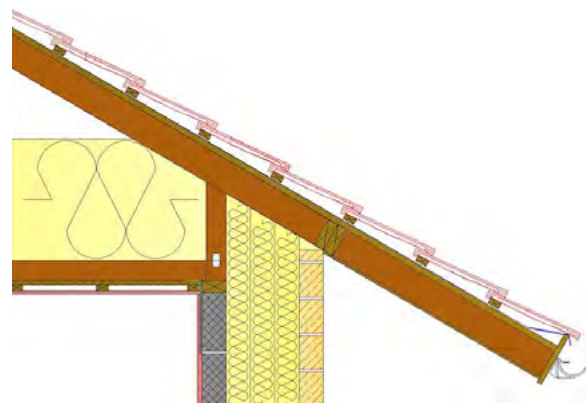
Roofs

INTRODUCTION

Roofs for residential buildings are predominantly pitched and designed as either a 'cold' roof with insulation at horizontal ceiling level, or a cathedral roof where the insulation follows the line of the pitch of the roof.

COLD ROOFS

Timber trussed rafters are very familiar to the construction industry and are inexpensive and fast to erect. To create a 'thermal bridge free' junction, 'dropped ceiling' or 'bob tail trusses' allow for the continuity of insulation between the roof void and wall. Airtightness is dealt with at ceiling level with airtight board or other robust membranes. Thermal bridging is minimised with only the timber bracing members of the truss penetrating the insulation layer. There should not be any air movement in the roof void, to avoid thermal bypass. This is achieved by using modern high quality 'vapour open', underslate membranes taped at laps and carefully detailed at the eaves and verge. A key point of this cold roof detail is to have continuity of windtightness. To achieve this, false eaves detailing will be required (see drawing and photo right).



Eaves detail at Denby Dale Passivhaus, as drawn and during construction. Images: Green Building Store



CATHEDRAL ROOFS

Timber I-joists can help cathedral roofs meet Passivhaus performance requirements. The joists come in various depths for soft or blown insulation, and thermal bridging is minimised thanks to the slender timber hardboard web. Thermal bridging can further be reduced by using either a vapour open insulation sarking board externally (also acting as the windtightness layer), and/or using insulation board internally, immediately followed by the airtightness membrane and before a service void for lighting and cabling. The roof/wall junction needs careful detailing to provide minimal thermal bridging and to satisfy aesthetic demands.



While usually seen as an aesthetic choice, insulation in the roof plane can allow lower eaves or a dry loft for storage. If future conversion of a roof space to accommodation is being considered then the insulation and airtightness strategy needs to anticipate this. Traditional attic trusses are sometimes used to allow this flexibility but are very difficult to properly insulate and make airtight especially if this is to be done as a future retrofit with initial insulation at ceiling level.

External and internal views of a 'cathedral' roof under construction. Images: Green Building Store

FLAT ROOFS

The basic principles for flat roof design apply to a Passivhaus in the same way as any other building, and generally fall into two categories: ventilated roofs or warm roofs. A ventilated flat roof allows any unwanted interstitial moisture to evaporate away. There should, of course, be less danger of this happening with Passivhaus levels of airtightness as long as the vapour control/ airtightness layer is positioned correctly on the warm side of the insulation. Timber I-joists are, again, very useful for giving good structural strength with minimal thermal bridging. Warm roofs tend to cause less potential interstitial moisture problems by separating the roof structure and placing the rigid insulation on top

of the decking. Another version of the warm roof is the inverted roof. This is where the insulation is placed on top of the roofing membrane to give it protection from sunlight and subsequent differential thermal movement. This places the insulation in a wet position which is not ideal for its thermal integrity, as the thermal performance of all insulation materials is affected by moisture, particularly running water. The designed U-value can be seriously compromised in flat roofs if there is any potential for moisture in or around insulation material, whether interstitial or from precipitation. Parapet walls are potentially a very bad thermal bridge if not designed very carefully. Flat roofs and parapets add risk that is not specific to Passivhaus.

CASE STUDY: Carrowbreck Meadow



Carrowbreck Meadow is a development of 14 Passivhaus homes on the outskirts of Norwich. The design response is a contemporary rendition of a Norfolk vernacular – defined by several references to the historic barn seen throughout the county.

Main contractors RG Carter worked closely with Metsä Wood to propose a commercially and structurally viable roofing solution. A Finnjoist I-beam roof, made from the only fully engineered timber I-joist to be manufactured in the UK, designed at a depth of 400 m, achieved a low U-value of 0.101 W/m²K with timber fraction within the roof of only 7.2%. Low carbon materials were incorporated in the project wherever possible – the roof insulation, for example, was recycled newspaper (Warmcel). Warmcel's high density gives superior decrement delay, so reducing solar overheating and making it a very good choice for projects like this one with rooms in the roof.



Above and left: Carrowbreck Meadow, certified Passivhaus. Images: Hamson Barron Smith

Below: Roof under construction. Image: Metsä Wood





Thermal bridges & thermal bypass

Mark Siddall

INTRODUCTION

In a building that meets the Passivhaus standard the junctions and interfaces require specific attention. As with all buildings it is at these points that heat loss can occur through thermal bridges; furthermore, low surface temperatures can result in surface condensation and mould growth. Unaddressed thermal bridges can increase fabric heat loss by over 30% ($>4.5 \text{ kWh/m}^2\cdot\text{a}$).

Attention must also be paid to address any risk of thermal bypass, which occurs when there is air movement across, within and behind insulation. It increases heat loss, causes discomfort and enables mould growth, and is a significant contributing cause to the performance gap in construction.

ZERO THERMAL BRIDGING CONCEPTS

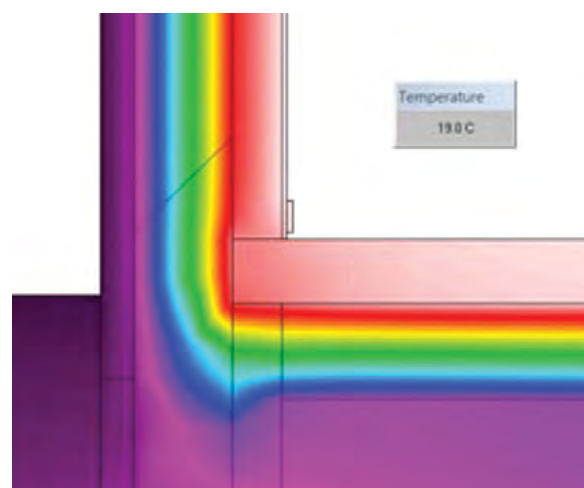
The concept of 'zero thermal bridging' can only be used when using external dimensions. Due to geometry, it is not possible to achieve zero thermal bridging using internal dimensions. The Passivhaus standard considers a detail to be 'thermal bridge free' when the ψ -value is $<0.01 \text{ W/mK}$.

TYPES OF THERMAL BRIDGE

There are essentially three types of thermal bridge:

- Repeating thermal bridges
- Linear thermal bridges: The Ψ -value (ψ -value)
- Point thermal bridges: The χ -value (χ -value)

Linear and point thermal bridges are determined in accordance with BS EN ISO 10211. Linear thermal bridges occur at the junctions between two building elements. Point thermal bridges arise when three building elements intersect (eg, a corner between two walls and a floor) or where a single element (eg, a steel beam) penetrates the insulation.



Heat flux at ground floor/wall junction. Image: Mark Siddall

RULES OF THUMB

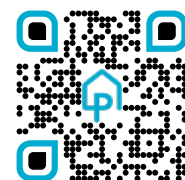
The use of external dimensions allows Passivhaus Designers and Certifiers to conduct a visual assessment of thermal bridging. If they are both satisfied, then thermal bridging calculations may be avoided. When designing 'thermal bridge free' details the following rules should be followed:

- **Prevention** Disruptions to the thermal envelope are avoided
- **Penetration** Where an unavoidable disruption is made to the insulating layer, the thermal resistance of the insulation should be as high as possible
- **Junction** At the junction between building elements, insulating layers should join without interruption or misalignment
- **Complexity** Minimise complexity – this reduces the chance of error and can assist buildability
- **Geometry** Pay specific attention to corners, ground floor/wall interfaces, eaves, verges, gable ends, and structural openings (head, jamb and sill)

- **Buildability** Ensure that simple robust details are developed with adequate construction tolerance
- **Construction tolerance** Think of a reasonable dimension for a construction tolerance, and then double it (or calculate it precisely)
- **Protection** Protect insulation materials from mechanical damage and the weather
- **Workmanship** Avoid gaps and discontinuities and ensure a clean working environment.
- **Improvisation** Do not improvise, read the drawings and specification or ask the design team if unclear
- **Inspection** Inspect details prior to closing any openings, to ensure that there are no gaps and discontinuities

Avoid steel work penetrating the insulation. Where steel is included within or penetrates through the insulation, refer to pht.guide/steel, make suitable allowances at an early stage in the design and undertake thermal bridging calculations.

LEARN MORE



pht.guide/steel

THERMAL BRIDGES AND ENERGY CALCULATIONS

The Passivhaus standard uses external dimensions while UK conventions utilise internal dimensions. As it is possible to calculate thermal bridges using internal and external dimensions, this difference has an impact on the results from thermal bridging calculations. When calculating building heat losses the two conventions for calculating thermal bridges should not be combined. This is because the use of internal dimensions tends to underestimate heat loss, while external dimensions tends toward the safe side by overestimating heat loss. Historically heat loss calculations have used internal dimensions as it was more convenient. As the external heat loss area is greater than the internal area the heat loss is underestimated. As insulation

got thicker this became significant and so SAP and other calculations now must include a correction for all external junctions. By using external dimensions we can avoid having to calculate these so-called thermal bridges due to geometry, as long as there are no physical thermal bridges such as structure through the insulation.

don't

- allow disruptions to the thermal envelope - where an unavoidable disruption is made to an insulating layer, the thermal resistance in the insulation should be as high as possible
- underestimate construction tolerances.

do

- minimise complexity – this reduces the chance of error and can assist buildability
- at junctions between elements, ensure insulating layers meet without any gaps
- ensure insulating layers join without interruption or misalignment
- design edges and corners to have as obtuse angles as possible
- during the design process, ensure that simple robust details are developed with adequate construction tolerance
- avoid steel work penetrating the insulation. Where steel is included within or penetrates through the insulation refer to pht.guide/steel.

THERMAL BYPASS

One of the major contributors to the energy performance gap is faulty, ill-considered design and construction that results in a thermal bypass. This has been defined as heat loss that bypasses the thermal insulation layer between two areas of the construction, and it is caused by a combination of conductive and radiative heat loss mechanisms resulting in uncontrolled air movement. A 2012 study showed that U-values can be reduced by as much as 80% as a result of this phenomenon.

If thermal performance gaps are to remain within an acceptable tolerance, then continuous insulation must be encapsulated on all sides by uninterrupted, unbroken air and wind barriers. By properly addressing thermal bypasses, heat loss can be reduced, thermal comfort can be improved, the building fabric can be protected, performance gaps can be mitigated against or

avoided, both carbon emissions and fuel poverty can be reduced, and health, well-being and energy security can be improved.

Some examples of potential points of vulnerability:

- improperly installed insulation, eg. 'tuck ends'
- wind washing at eaves
- wind washing in vertical cavities
- irregular surfaces
- blown fibre insulation
- joints, interfaces and penetrations

An in-depth examination of the evidence around the occurrence and impact of thermal bypass, together with practical guidance on the implications for projects in design and on site, is to be found in *Thermal bypass risks* (PHT, 2022), available at pht.guide/thermalbypass.

LEARN MORE

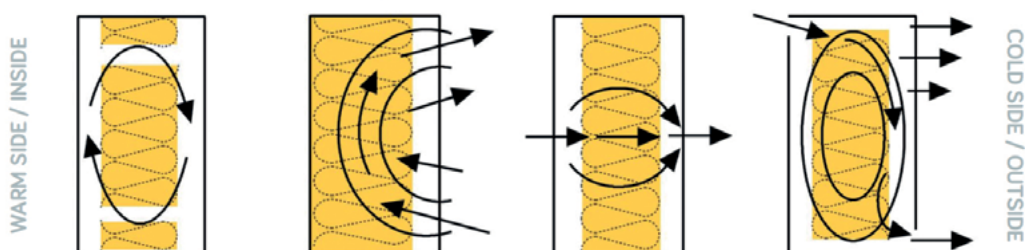


pht.guide/thermalbypass



pht.guide/insulation

Types of thermal bypass - from left: closed loop; open loop (wind washing); open loop (air leakage); reality.
Image: Mark Siddall





Fenestration and shading

Nick Grant

INTRODUCTION

Good fenestration design requires effort, knowledge and experience but following some basic rules of thumb can help avoid the most common pitfalls while challenging some Passivhaus myths.

ORIENTATION

Even Passivhaus windows have about five to eight times the heat loss of a wall. Glazing is also considerably more cost per m^2 . Depending on position, shading and orientation, glazing can lead to net gains or net losses of energy. While solar gain is welcome during the heating season, in hot weather excess net gain is unwelcome and may lead to overheating. An intelligent approach will get the best from Passivhaus glazing in terms of energy, light, outlook and comfort.

Optimised Passivhaus windows facing between due east and due west will generally gain at least as much useful heat as they lose over the heating season. While less beneficial in winter, **east and west** glazing will admit much more solar gain during summer because of the long days and lower morning and evening sun. By contrast, **south** windows can be quite effectively shaded from direct summer sun by simple overhangs from roof and reveals. Theory and practice show that, contrary to popular belief, **north** glazing, and particularly skylights, can lead to overheating if oversized.

PASSIVHAUS IS NOT PASSIVE SOLAR

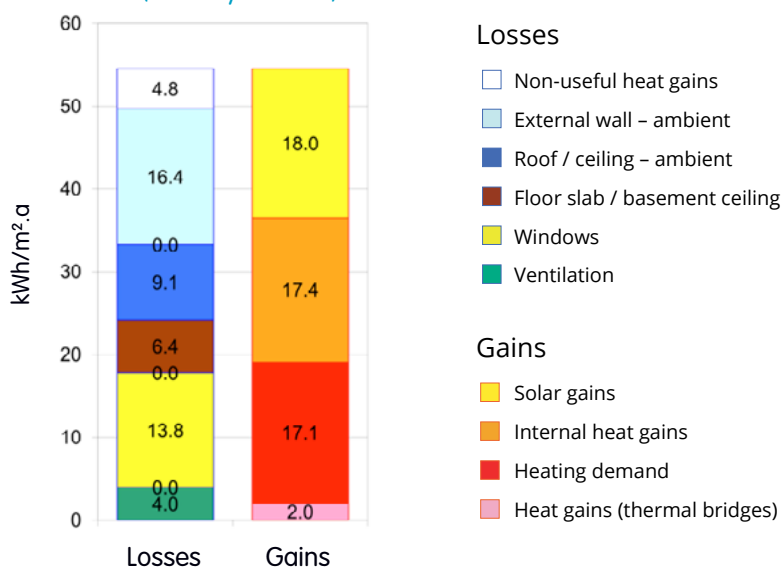
The fantasy of passive solar design requires low density development which faces all habitable rooms towards a beautiful, private, southern view with small, windowed utility rooms to the north. This also requires generous glazing (more than required for daylight), shaded in summer by overhangs. Also, while heat from the sun is free, capturing it with extra window area is far from cost-effective and greatly increases summer overheating risk and upfront carbon emissions. If Passivhaus is to be a widely applicable and scalable solution, this approach is much too limiting.

At the design stage, increasing south facing glazing areas looks like a very easy way to nudge a design under the $15 \text{ kWh/m}^2\text{.a}$ space heating demand target, but it comes at a high price.

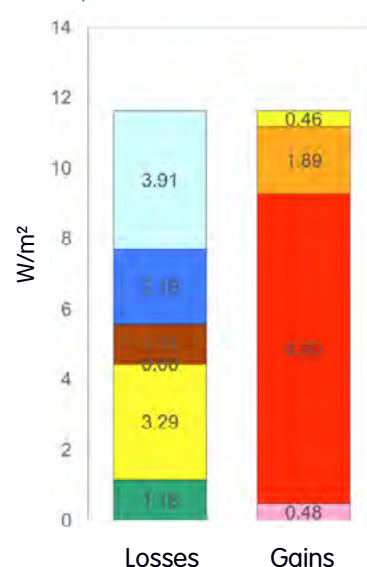
When designing to the 'alternative' 10 W/m^2 peak heat load criteria, solar gains are also considered, but we can see from the bar graphs below that it is the associated losses that are more important – compare the yellow blocks. This means we are penalised rather than rewarded for oversizing the glazing. The result is a more comfortable and cost-effective building. For larger buildings and most urban sites, we need rooms to work regardless of orientation. Optimising the fenestration based on daylight and summer comfort makes designs much less site sensitive.

Additionally, solar shading can be difficult to model accurately and will change over the life of the building as trees grow or new buildings are constructed and demolished. It is best if the energy efficiency and comfort of our building is not overly dependent on such external uncertainties.

Space heating demand energy balance (monthly method)



Peak heating load energy balance (W1 - worst winter condition)



Bar charts showing energy balance according to the space heating demand method and the peak heating load method. Find the PHPP plugin to generate the peak heating load energy balance chart from Max Fordham via pht.guide/designsupport.

GLAZING RATIOS

A better approach is to design for good, but not excessive daylight, with a maximum of about 20% of the internal wall area of any south facing room as glazing. All other orientations should ideally have only as much glazing as is needed for views, daylight and ventilation.

As a rule of thumb total glazing area (excluding frame) of around 15-20% of treated floor area (TFA) is a good starting point for design. The England and Wales Approved Document O, simple method for avoiding overheating, may conflict with good daylight practice but at the time of writing this has yet to be tested. It remains to be seen how Part O influences Passivhaus design in the UK.

SHADING

With modest glazing designed for daylight rather than maximum solar gain, it is usually possible to achieve sufficient summer shading of south glazing via the existing roof overhang or window reveals.

To fully shade south glazing at summer solstice we need a 60° shading angle in the UK. The taller the glazing top to bottom, the deeper the shading has to be to achieve this. As with any building, low sun in autumn and spring may still cause some overheating but we can usually open the window for cool air. On hot days in summer the outside air will be warmer than indoors so throwing open the large glass doors doesn't really help. Full height glazing needs a deeper overhang and that reduces sky view and daylight – a design solution defeating the original purpose.

Another issue with larger areas of glazing is radiant discomfort. Passivhaus quality windows solve this problem in winter by avoiding low surface temperatures. The thermal image below taken at 11.30 am on 21 June 2022 shows east facing bi-fold doors and a south facing window.



The PHPP promises good summer comfort and the air temperature of 23°C confirmed this. Doors and windows had been opened but it was a hot, still day. However, the building occupants were



complaining that the dining area was 'stuffy and airless'. At the other end of the open plan kitchen dining room the air felt pleasant despite having the same temperature, CO₂ and RH. In this and other examples, shading was enough to keep the building cool but diffuse and reflected solar radiation still raised the surface temperature of the glass enough to cause radiant discomfort.

For hotter climates or where the glazing has been overdone, it is necessary to include external moveable shading. However, blinds and shutters are expensive, require maintenance, and obscure the daylight and view that the window was installed to provide.

Internal blinds have much less impact on overheating (including radiant discomfort), as the heat is already in the room, but can control glare and provide privacy.

Either inward or outward opening windows can be used in a Passivhaus building, but inward opening windows are far more common in Passivhaus buildings as they allow external shutters or blinds and fly screens.

Note that it is very difficult to effectively shade rooflights (without using external blinds), so it is important to size skylights very carefully.

SOLAR CONTROL GLASS

Solar control glass has a coating that reduces solar gain but also visible light transmission. Smart design will tune the glazing area rather than how opaque it is. Conversely using high g glazing (close to 0.6 g) will tend to reduce the heating demand of a south facing building designed for solar gain and this seems like an easy win. However, the higher g-value makes summer comfort worse and typically means accepting a slightly worse glazing U-value. Designing for heat load rather than specific heating demand we find the sweet spot is for glazing products with a g-value around 50% and a Ug of around 0.5 W/m².K. Extra layers of glass will further reduce the heat loss but also daylight transmission, and so only make sense in very cold climates. While in some situations double glazing can reduce heating demand because of the extra solar gain, it will increase peak heat loss and will not meet the Passivhaus comfort criteria in the UK climate.

Above: Shading at River Studio EnerPHit. Image: Sjolander da Cruz

RULES OF THUMB

- **South glass area:** maximum ~20% of internal wall area
- **Total glazing:** maximum ~15-20% of TFA
- Limit areas of **skylights** to that required for daylight
- Glazing less than ~850 mm **distance from floor** does little to improve daylight, so should be avoided
- **Window heads close to ceiling** maximise daylight penetration – but remember to leave room for curtain rails, blinds, and ventilation opening for tilting windows
- **Daylight factor:** 2% is a good target for kitchens, 1.5% for living rooms and home office, 1% for bedrooms. Higher than 4% might indicate overheating risk for that room
- Keep fenestration **simple!** Minimise transoms and mullions to reduce the proportion of frame, which will be more than you think



Right, top: carefully positioned glazing allows views out when enjoying a lie in, without passers by being able to look in

Right, bottom: thoughtfully placed windows for views and daylight, designed from the inside not for external symmetry or decoration

Images: Charles Grylls

don't

- rely on trees to provide summer shading
- try and heat the building with the sun – passive solar is hot in summer and cold in winter
- over-glaze
- use solar control glass ($g < 0.5$) to solve excess solar gain – reduce the glazing area instead
- use fenestration to purely decorate the façade
- make opening window lights too large to open easily and safely
- imagine that overheating can be solved by opening the windows
- Ignore radiant discomfort due to large areas of glass (even if shaded) – PHPP and other models do not consider this, but it is still important

do

- consider designing for 10 W/m² heating load in preference to the 15 kWh/m².a specific heating demand target
- choose lower Ug-value over high g-value
- have at least one opening window in every habitable room
- design for daylight and views, design from the inside out
- design for daylight quality and usefulness over brute quantity
- for the UK climate, do try and design without the need for external blinds but ensure they can be retrofitted (i.e. inward opening windows)
- consider glazed doors rather than solid, for extra light and solar gains
- consider fly screens and security for summer ventilation
- use light surface colours to improve daylight without increasing overheating
- consider splayed reveals to reduce contrast and maximise daylight for a given window size

CASE STUDY: The Old Woodyard



The Old Woodyard self-build in Herefordshire, aiming for certification, avoided concrete in its entire construction – including the foundations!

Architects Polly Upton and Kirk Rusby, who both work at PHT Patron Architype, designed and project-managed their own self-build home in rural Herefordshire. Clad in a striking red corrugated metal, the 3-bed house is designed 'upside-down' so that the bedrooms are on the ground floor and the living space above, to take advantage of the views.

Timber frame construction was chosen as a more familiar construction methodology and because it was more practical for delivering to site. Timber I-joists and I-beams are used for the walls, roof and internal walls, insulated with 300 mm recycled cellulose insulation. The timber frame is sheathed externally with a wood fibreboard, and finished with an intelligent breather membrane and metal cladding. Internally, a vapour block board is used as the airtightness barrier.

The foundations comprise a footing of compacted recycled aggregate, with a bed of natural hydraulic lime mortar on top. A foam glass perimeter strip



is used, which has the timber frame soleplate attached to it, on top of the damp proof membrane (DPM) layer.

The use of concrete-free foundations avoided the stress of a concrete delivery to the hard-to-access site, but also reduced embodied carbon for the project. Modelling by Architype's Sonia Brookes has calculated the upfront carbon of the project to have a 40% reduction on LETI targets, and the whole life carbon a 60% reduction.

The strategy for shading and fenestration balances daylighting, views, solar gains and cross-ventilation while avoiding complexity. Two large windows on the south façade are non-opening, reducing cost. Cross-ventilation flows east to west, where the view is captured by glazed lift-and-slide doors at first floor level. Roller shutters here, together with a roof overhang and external shading on other windows, mean that overheating risk is designed out – in the first summer, the house didn't exceed 25°C even when the outside temperature soared to 31°C.



Images: Polly Upton and Kirk Rusby



Windows and doors

Sally Godber

INTRODUCTION

Manufacturers are not used to providing thermal performance data in a format suitable for inclusion in PHPP, so spend time reviewing the exact frame, glazing and spacer specification they provide for each window. This is usually easier for certified windows but still take care. Once you have interpreted manufacturer information and put it into the PHPP, we recommend sending a copy back to the manufacturer for them to confirm the values you have taken. As with most Passivhaus topics, there's great background reading on Passipedia.

PERFORMANCE

It is not necessary to use certified windows as long as manufacturers supply data to fill in the 'Components' worksheet in PHPP and it meets the criteria set out in the Certification guidance document. Take particular care over the following:

- Separate U-values for each frame configuration (head, sill and jamb) to EN 10077-2
- Glazing U-values need to be given to two decimal places to EN 673
- fRsi requirements must be met (internal surface temperature). Note that the required value has changed in PHPP 10. It is now based on a bespoke calculation for each project, and in most cases is more relaxed than PHPP 9. Take particular care where manufacturers are claiming still air spaces for elements of the frame that are ventilated – the area where an opening casement meets the fixed part of the frame tends to be the weakest component.
- Compliance with the comfort criteria for each window is shown in the PHPP windows sheet – see the PHPP manual for further information.
- Spacer psi-value will depend on the frame type it is installed in, so a single number to cover multiple frame types is not suitable. Many spacers are now Passivhaus certified products so the values for these can be downloaded from the certificates on the PHI database. Beware that these may be different from the manufacturer's claimed values, and note that if the spacer isn't certified or haven't got a bespoke performance calculation the Certifier may require you to add a 10% margin.

Whatever door and window options you choose, it must fit the overall energy balance in PHPP. PHPP is a fantastic tool for assessing different frame and glazing options and comparing costs, but be careful to ensure other factors are taken into account.

And if you are picking a certified component, remember to check the details are exactly the same as the certified versions. Manufacturers have been known to specify a different packer at the bottom of the sill, or change to a non-certified version at the last moment without identifying it, grrr.

SUPPORT

Some manufacturers are willing to provide support, for example, help with airtightness detailing, installation psi-value calculations and site visits. Another reason to get manufacturers involved early is to ensure that proposed window sizes are achievable; typically opening windows that are wider than 1.2 m can cause problems.

AIRTIGHTNESS

Passivhaus certification of windows requires no airtightness testing to be undertaken, so tread with care. While most certified windows have a precedent of meeting the airtightness standard, it is recommended to ensure the manufacturer confirms their windows are suitable. It is recommended to use windows and doors that meet EN 12207 class 4, the highest performance. In practice solid frames (wood) tend to have a more consistent performance than profiled (PVC or aluminium).

*Certification document for
Munster Joinery's Passiv AluP+ window.*

CERTIFICATE
Certified Passive House Component
Component-ID 1576ix03 valid until 31st December 2023

Passive House Institute
Dr. Wolfgang Feist
64283 Darmstadt
Germany

Category: **Fixed window**
Manufacturer: **Munster Joinery,
Mallow,
Ireland**
Product name: **Passiv AluP+ Fixed**

This certificate was awarded based on the following
criteria for the cool, temperate climate zone

Comfort $U_{Wf} = 0.80 \leq 0.80 \text{ W/(m}^2 \cdot \text{K)}$
 $U_{Wf, installed} \leq 0.85 \text{ W/(m}^2 \cdot \text{K)}$
with $U_g = 0.70 \text{ W/(m}^2 \cdot \text{K)}$

Hygiene $f_{Rsi+0.25} \geq 0.70$

Passive House
efficiency class

phE

phD

phC

phB

phA

cool, temperate climate

**CERTIFIED
COMPONENT**

Passive House Institute

SPACER REBATE

A good frame should have a rebate >48 mm to allow sufficient glazing options. For triple glazing, 18-20 mm cavities are optimum for Argon – 16 mm or less gives inferior performance. Krypton is generally unsuitable except for special purposes where narrow cavities are required.

FRAMES AND ENERGY BALANCE

Generally frames perform worse than glazing, so small windows, or those with lots of mullions and transoms should be minimised. When picking a window frame, the dimensions should always be considered as well as frame U-value. While it can be tempting to choose the frame with the best U-value, there is often a trade-off between performance and thickness. A slimmer yet worse performing frame can sometimes prove the better option, as the amount of glazing (which has a better U-value) is optimised and solar gains are increased. The new Passivhaus window certificates from the Passive House Institute include an A to C rating which gives an indication of the balance between frames and glazing, because gains are considered as well as losses. However, the best approach is to model the various options in PHPP as the 'best' option can vary between projects.



Window frame at Bowman's Lea. Image: Harry Paticas

Inward and outward opening windows can perform equally well. Inward opening windows allow external shutters, blinds or insect mesh to be fitted, and are usually easier to wrap in insulation, thus helping achieve better thermal performance. Designers also need to consider how the choice of window frame will effect what natural ventilation rates are possible. Opening windows and the free area can have a big impact on summer comfort.



Harris Academy, Image: Architype

GLAZING

Triple glazing gives the best energy balance, even in the UK climate, and is needed to meet comfort requirements and to avoid condensation and mould at the edges. The glazing g-value (Solar Factor) refers to the fraction of solar heat that transmits through the glass and is just as important as the glazing U-value in this optimisation process, so ask for both to two decimal places. Note that Building Regulations only considers losses, so a manufacturer's 'high performance' glazing will often have a low U-value but also very low g-value which is likely to give a poor energy balance in PHPP.

Manufacturers will have a range of glass they can supply with varying costs, so ask for alternatives. Take care to look at the light transmittance of the glass as well – usually the light transmittance percentage is roughly in line with the g-value. Other standards such as Secured by Design or Building Regulations may dictate that security glass is used. Generally a choice of toughened glass does not impact on performance but laminated glass almost certainly will.



Installation of Smartwin windows at Old Holloway. Image: George Mikurcik



ROOFLIGHTS

When installing horizontal glass (rooflights) the glazing performance changes; this must be taken into account in the U-value. For example, glazing with a stated U-value of $0.60 \text{ W/m}^2\text{K}$ should be entered as $0.90 \text{ W/m}^2\text{K}$ in PHPP when it is installed horizontally to take account of this. EN 673 has a calculator to determine this accurately. Note that rooflights are very difficult to shade from overheating in summer. It is very hard to achieve good installation details with a rooflight, although manufacturers have developed installation kits that improve performance.

CURTAIN WALLING

There is a particular method to input curtain walling in PHPP, follow it! A common mistake is not including the glass carrier as thermal bridges in the Areas sheet. Entering a chi-value for these in the Components sheet is not sufficient.



Curtain walling at St Sidwell's Point in Exeter. Image: Exeter City Council

DOORS – SLIDING, FOLDING, GLAZED

While a number of sites have met the airtightness criteria while using sliding doors, their use needs careful consideration as the airtightness often degrades faster than a standard door due to wear and tear. This is particularly important if the door is directly next to a living space where a draft could be felt. Lift-and-slide doors are inherently more airtight, and may be cheaper than parallel slide versions. Bi-fold or folding-sliding doors are generally expensive at the level of airtightness needed for a Passivhaus. In addition, it is very difficult to get as good installation psi-values as a normal window/door because of the frame configuration. Consider using glazed doors, as these have a better energy balance than solid doors.



Lucy Cavendish College. Image: Nick Guttridge

OPAQUE GLAZED-IN PANELS

Opaque glazed-in panels often don't perform much better than glazing of equivalent thickness (i.e. U-value = $0.6 \text{ W/m}^2\text{K}$) unless it is expensive. Since such panels are an absolute vapor barrier, adding significant insulation behind to bolster performance is likely to increase the interstitial condensation risk. In this scenario, bringing the airtightness line inside of the insulation is attractive, but usually incredibly difficult to do, and you can never be certain if it is the glazing or the designated airtightness line that is performing. Much better to replace the section with the standard wall construction.

“The Passivhaus requirements to frontload and complete the design in the pre-construction phase results in many added benefits, including early procurement of sub-contractors and full co-ordination of the digital model. This should, in practice, enable a more efficient build process resulting in less site modification and waste, and ultimately a smoother commissioning and handover process.”

**Allan Smith, Low Carbon Manager,
Morrison Construction**

INSTALLATION & DETAILING

For best thermal performance, position frames near the centre of the insulation line. Optimisation of such details in thermal bridging software can be worthwhile – you can evaluate how much difference it makes in PHPP.

Other considerations are the skill level of the team doing the installation when detailing – a thermally beautiful detail might end up a headache on site. Additionally, deep external reveals increase the solar shading which may or may not be advantageous. The frame dimensions and the way it opens (inward or outward) will determine how much insulation can be wrapped around the frame. For inward opening windows generally insulate externally; for outward opening windows

insulate internally. Thresholds also need careful consideration as they are a difficult area for achieving good thermal performance – note there is a different fRsi requirement here. In addition, airtightness needs to be considered when detailing window installations. Note that there is little or no benefit to wrapping aluminium-clad frames with insulation externally.

Providing secure natural ventilation is usually a key component of summer comfort design and the detailing can have a big impact. Make sure that the assumed opening widths can be achieved – a recent case assumed opening widths of 100 mm, but on completion it was measured at just 18 mm because of a close reveal.

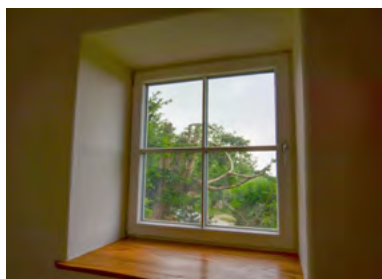
CASE STUDY: Reform Cottage



Reform Cottage is a deep retrofit of an existing building, originally a 1838 stone cottage, which was subsequently extended and modified with a mix of construction methods. The project was EnerPHit certified in 2018, having achieved an airtightness of 0.31 ACH @ 50 Pa.

A number of IWI (internal wall insulation) and EWI (external wall insulation) strategies were explored for the retrofit. In the end, a full 200 mm EPS EWI solution was possible, combined with a top-up of polystyrene beads for the later cavity wall sections, which were already partially filled with mineral wool. This enabled continuity of insulation between wall and roof – designed as a ‘cold roof’ and insulated with blown Warmcel cellulose insulation.

The project used Green Building Store inward opening tilt-and-turn triple-glazed timber windows, achieving a whole window U-value of 0.75 W/m².K. The windows were installed using airtightness tapes and membranes and were located in the ‘insulation zone’ within the external wall insulation to reduce thermal bridging. The installed U-values for the windows varied between 0.94 – 0.83 W/m².K. Glazed doors help with daylighting as well as thermal efficiency.



*Left: deep reveals internally
Bottom left: windows installed on brackets projecting from the wall, inline with EWI to follow*

Below: airtightness detailing around a new door

*Below right: before the retrofit
Images: Gervase Mangwana*



INTRODUCTION

Building services comprises heating, ventilation, plumbing and electrical systems. Energy use by all of these is included in the overall energy use metric in PHPP, Primary Energy Renewable (PER).

Ventilation is critical for Passivhaus to deliver good internal air quality. The importance of heat recovery efficiency means that high quality Passivhaus certified components are required along with a well designed, installed and commissioned system. In-use monitoring has shown the attention to detail paid to Passivhaus ventilation systems pays off – they consistently deliver the best indoor air quality.

Heating is required in a Passivhaus building, just not much, and in practice a modestly sized conventional radiator system can often be the best approach.

VENTILATION

The performance of the ventilation system is an integral element of the primary Passivhaus heating demand calculation, though the rest of the services are not. Heat recovery ventilation is effectively seen as insulation for the fresh air supply. The main intake and exhaust ducts carry air at outside temperature, so form part of the building's external insulated envelope. Thus, in keeping with the emphasis on compact form, the designer needs to be aware of these ducts and how to make them as short as possible at an early stage in design.

The impetus for using MVHR may have come from the opportunity for heat recovery, but now we recognise that the main benefit of these systems is effective ventilation – essential for occupant health. The Passivhaus standard sets detailed criteria for airflow rates, noise levels and commissioning which, if followed, will lead to ventilation that works well. The ventilation section of this guide provides specific details about designing ventilation to the Passivhaus standard.

HEATING AND HOT WATER

A Passivhaus does need some heating – this is implicit in the standard, which is in terms of annual heating demand or peak heat load, after taking account of solar and internal heat gains. However, the amount of heating is very small – much less than a standard new building, and far less than the twentieth century buildings our heating systems have developed in.

It should be possible to heat a Passivhaus by heating the ventilation supply air, but this doesn't mean you should do it this way. There is often a mismatch between where you want air and where you need to supply heat, so it is usually easier to separate them.

The heating system in a Passivhaus is one area where you can reduce costs compared with a conventional building, and still use conventional suppliers and technologies – the system will simply be smaller.

Heat pumps are well suited to Passivhaus buildings – radiators 'oversized' to work at heat pump temperatures (ideally below 40°C) are not large or expensive in a Passivhaus, thanks to the low heat load. Also, the quality assurance which minimises the performance gap of Passivhaus means the system can be sized accurately without risk of under-heating in cold weather. The minimal heating requirements of Passivhaus mean air source heat pumps with radiators are usually the rational choice rather than ground source with underfloor heating combinations, which are a little more efficient but have a high capital cost. Some Passivhaus buildings are suited to air-to-air split systems, which are cheaper still. Larger buildings such as Passivhaus schools are also ideal for air source heat pumps, and here, the much reduced heat load offers large reductions in plant size.

COOLING

With increasing concern about overheating in the UK climate we expect to see more use of active cooling in dwellings. Being a global standard, Passivhaus already has this covered, with an annual cooling demand limit of 15 kWh/m².a (the same as for heating) to ensure a fabric-first approach is used to minimise cooling energy use.



Plant room with ASHP system. Image: Alan Clarke

LEARN MORE



[pht.guide/
caseforMVHR](http://pht.guide/caseforMVHR)

While building fabric is the key to Passivhaus performance, it is important to pay attention to building services. Mechanical ventilation with heat recovery (MVHR) is the least familiar but it is also important to design heating systems appropriate to the level of heat demand, so they run smoothly and economically. This section looks at the principles of building services design for Passivhaus.



RENEWABLES

Traditionally, solar thermal has been a popular choice for Passivhaus Designers to reduce the total energy demand of a building (i.e. the PER). However, big drops in the cost of solar photovoltaic (PV) generation have changed the landscape, and PV has largely replaced solar thermal as a means of harvesting the sun's energy.

To recognise the use of renewable generation in buildings the Passivhaus standard has three classes. "Classic" is the original Passivhaus standard and has no requirement for renewable generation, then "Plus" and "Premium" have increasing requirements for renewable generation along with reduced PER (Primary Energy Renewable) demand.

To get round the fact that compact multi-storey buildings have less space for PV per m² of floor area, yet provide an efficient built form in most other ways, the on-site generation is assessed in terms of m² of building footprint, so that these build types are not excluded from the Plus and Premium standards.

Note that renewable generation is not simply subtracted from energy consumption in the Passivhaus model: all classes allow some generation to be allocated to on-site consumption, subject to an upper limit of 15 kWh/m² PER. Similarly, the Plus and Premium options allow for this range of reduced PER demand, in lieu of generation.



RETROFIT

When retrofitting a building to the EnerPHit standard the same quality building services are required as for a new build Passivhaus. MVHR can be hard to retrofit given existing floor to ceiling heights, but is still achievable with careful design. In larger buildings this tends to lead to decentralised ventilation being a more common choice for retrofit. Heating systems benefit from the very low peak heat load in an EnerPHit building, not much higher than a Passivhaus. Often existing heating systems can be retained while transitioning from a high temperature fossil fuel system to a low temperature heat pump system, taking advantage of the existing emitter sizes.

do

- ensure any subcontracted building services designers (eg suppliers providing designs) are aware of and understand Passivhaus requirements
- think about MVHR design and location at an early stage
- make sure the MVHR is easy to access for filter changes
- get a full ductwork design done in time to integrate with the architectural design
- allow for a heating system, probably heat pump, but keep it simple

don't

- think you have to heat via the air supply
- get carried away with expensive heating systems – they won't be used enough to be worth it
- think that PV can power a heat pump directly – peak heating demand is in winter when PV generation is lowest
- expect that the MVHR will distribute heat from a heated room to an unheated room – it won't

Left: Photovoltaics on the roof of Hackbridge Primary School, certified Passivhaus Plus. Images: DF Roofing



MVHR

INTRODUCTION

Mechanical ventilation with heat recovery (MVHR) provides whole building ventilation with minimal heat loss, and is an essential component of a Passivhaus in our climate.

LEARN MORE



pht.guide/MVHR

MVHR

A domestic MVHR system has a ventilation unit with two fans: one draws outside air in and supplies it to bedrooms and living rooms; the other extracts air from kitchens and bathrooms to exhaust to outside. Inside the unit there is a heat exchanger which transfers the heat from the outgoing air to the incoming air without any recirculation of the air itself.

The fans run all the time, but can be switched to different speeds – usually low, normal and boost – and these rates are adjusted at commissioning to suit the individual house. The unit includes filters which keep dirt out of the heat exchanger. In order to ensure good indoor air quality the intake filter also removes small particulates (PM1, PM2.5 and PM10, which are harmful to health) with high efficiency.

Ducts distribute the air from the ventilation unit to and from the different rooms in the dwelling, and there are also ducts between the unit and outside to bring in fresh air and exhaust stale air.

Large buildings such as schools basically use the same system, though may use a number of ventilation units in one building and may control airflow on a different basis, such as measured CO₂ levels as an indicator of the number of occupants in a space and thus air flow requirements.

Plant room at Old Forge Cottage Passivhaus. Image: Green Building Store



VENTILATION UNIT AND DUCTWORK

Ventilation unit

Ventilation unit location is very important to both the efficiency and also the long term usability of the system. Ducts connecting to outside contain air at the outside temperature, so are effectively outside walls – we use insulated ducts but can't possibly insulate them to the standard of the wall. The ducts therefore need to be as short as possible; basically the unit needs to be located adjoining an external wall, or failing that, within two metres of one. It is also possible to install the unit in a frost free location outside the thermal envelope, such as an enclosed porch, but the connecting ducts still need to be as short as possible and of course the external terminals have to be outside the building altogether. All distribution ductwork on the warm side of the heat exchanger – supply to rooms and extract from rooms – has to be within the airtight thermal envelope.



*Old and new MVHR filters – left to right: intake; exhaust; new.
Image: George Mikurcik*

Access is needed to change filters (every 3–6 months, depending on how dirty the outside air is), so you should be able to walk up to the unit and reach it without needing a ladder. Some units are too noisy to have in a living room or bedroom, so the ideal is a utility room or WC/cloakroom. Allow space for ducts and ancillary equipment when allocating space – look at previous installations for guidance.

Condensate drainage is needed from the ventilation unit – this should be to the internal soil pipe system via a trap and not just straight to outside, otherwise the drain could freeze solid and the unit would flood.

External terminals

External terminals can be close together: it is better to have the exhaust on the same wall as the intake, to minimise impact of wind pressure difference.



Placing the terminals vertically one above the other with exhaust 600 mm below intake is a good solution to minimise wall use on a narrow frontage house – there are also “combi” terminals which combine intake and exhaust in a single unit, designed to avoid recirculation from exhaust to intake. Although people can be concerned about such recirculation, for a single house there is little reason to worry – the transfer from adjacent terminals is minimal, and far less than transfer between rooms inside the house.

The position of the intake terminal is however important. Avoid locating the intake near smells or pollution, eg, bins, flues or cars, or the path of exhaust from a neighbouring dwelling. Put

the intake two metres or more above ground level to minimise intake of particulates. Intake through the roof risks bringing in hot air in summer – even north roofs can get hot, so walls are a better location. Roof exhaust also causes problems – in this case, with condensate from the damp exhaust air collecting in the duct and needing special drainage.

Ductwork sizing

The basic principles of ductwork systems revolve around airflow rate, air velocity, and pressure. For a particular airflow rate (ie volume per second) the air will move faster in a duct with small cross sectional area than a large one. Higher pressure is needed to push the air at higher velocity, and this is accentuated with small ducts where more of the air is close to the duct walls.

High air velocity has a direct down-side, so called “regenerated” noise, which is the whistling you hear when air is forced through small gaps. However the main impact is due to the higher pressure needed. This increases the energy needed by the fans to move the air, which we don’t want, but also some of that energy is converted to noise, which we also don’t want. Hence Passivhaus ductwork systems are designed around principles of low velocity and low pressure.

For practical purposes ductwork size can be determined initially from the desired air flow rates, calculating the size that will not exceed the limits for air velocity at that volumetric flow rate. In domestic design the usual requirement is velocities <2 m/s, and up to 3 m/s for the main ducts at the MVHR and connections to outside.

With these velocities the ductwork system will have low pressure loss, low energy use and low fan noise. For non-domestic projects the situation is a bit different due to larger and longer ducts, and full design is needed to determine the system pressure drop.

Ductwork type

Ductwork can either be rigid – normally steel – in a branched configuration with larger ducts at the MVHR and smaller branches to each room, or ‘semi-rigid’ systems which use plastic ducts that can be bent round corners, but still have a smooth inner bore. For the latter, the ducts are arranged radially running one or two to each room from supply and extract distribution manifolds. ‘Flexible’ ductwork (like old tumble drier hoses) is not acceptable – it gets squashed and has a high resistance to airflow. Flat channel ductwork is also best avoided since the pressure loss at bends is very high. Semi-rigid systems can be easier for shallow ceiling distribution depths.

Since semi-rigid systems use small ducts right from the MVHR they can run the risk of high pressure drops on the longer ducts to more distant rooms. Pressure drop calculations for these systems – which are straightforward – should be done to ensure pressure drops are low and similar for different terminals so that the system is easy to balance.

For the MVHR units used in dwellings, the test figures for PHI-certified MVHR units only apply up to a maximum system pressure of 100 Pa, though this is usually easy to achieve with good design. Larger ventilation units (for flow rates higher than 600 m³/h) are certified at a specified maximum external pressure, which is individual to each unit, though there is some scope to trade pressure against flow rate, so long as the specific fan power criterion of 0.45 Wh/m³ is not exceeded.

Rigid steel (left) and semi-rigid (right) ductwork.
Images: Walled Garden, Nigel Dutt (l); New Forest EnerPHit, Ruth Butler (r)



Noise attenuation

Noise control is needed to make the system unobtrusive – noisy systems get turned down or off. The primary source of noise in MVHR systems is the fans, and the noise level increases with increasing pressure, so well designed low pressure systems are quieter as well as cheaper to run. All systems, whether metal duct or semi-rigid, should include proper silencers on supply and exhaust – these are big (typically 900-1000 mm long and 100 mm larger diameter than the ductwork) so if possible fit them between ceiling joists. Branched systems may also require cross-talk attenuators on the supply to prevent noise transmission between rooms via the duct.

The Passivhaus standard requires design figures of < 25 dB(A) for ventilation noise in living and bedrooms (including in non-residential buildings), and <30 dB(A) for extract air rooms and other rooms in non-residential buildings. These figures are used to calculate attenuation required – they are too low to easily measure accurately, at least during the day on a busy building site. Once the system is running you shouldn't be able to hear the ventilation in bedrooms or living rooms – this is one reason why Passivhaus MVHR is easy to live with.



*Circular silencers.
Image: Earthwise Construction*



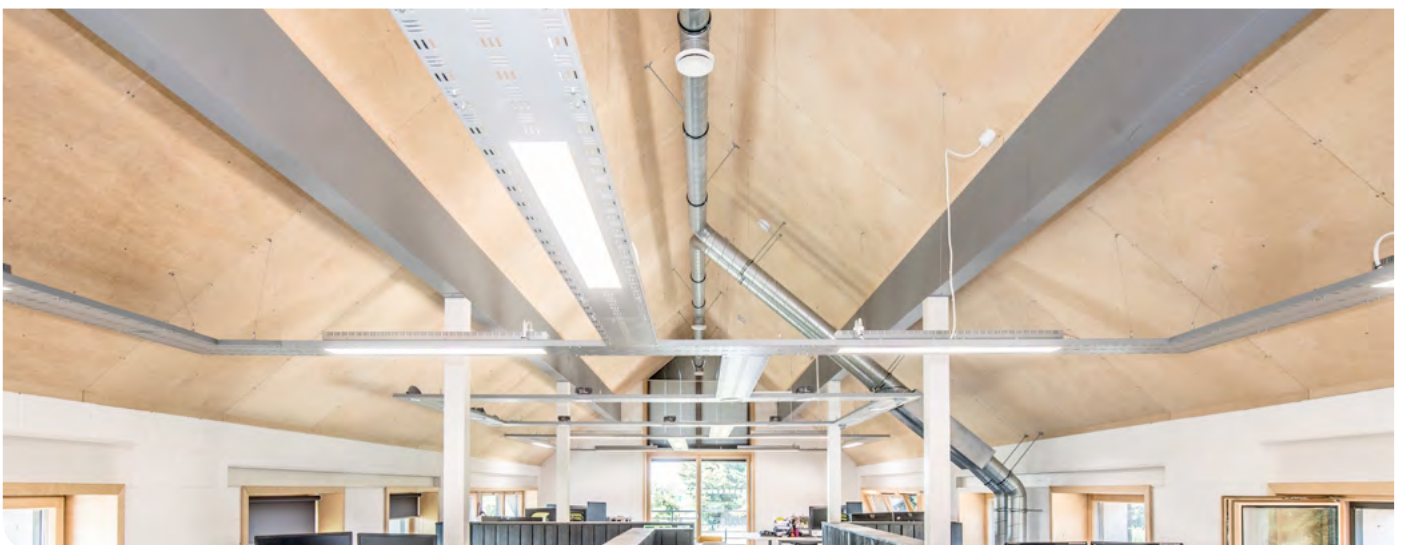
Internal terminals

Air is supplied to and extracted from spaces using room terminals. These are different for supply and extract, and there are different terminals for wall supply, directional ceiling supply, and all-round ceiling supply. The terminal type used depends on the position in the room: directional terminals are often used to throw air in from above the doorway; standard terminals need to be near the middle of the room – aligned with windows is usually good as the terminals are then unlikely to be obstructed by

high furniture. Extract terminals can be put on a wall or ceiling.

In dwellings, door undercuts are the normal way to allow air to flow out of supply rooms and into extract rooms. The minimum undercut is 10 mm above final floor finish. Non-domestic buildings often have to transfer higher airflow rates and purpose-designed transfer grilles or ducts need to be used since the gap under the door is not large enough.

Exposed ductwork with supply terminals positioned along the central spine at Swann Edwards offices. Image: Swann Edwards Architecture



SYSTEM SIZING AND FLOW RATES

There are set criteria for working out how much air to supply in a Passivhaus dwelling; these are provided in the manual for the Ventilation Final Protocol Worksheet, which is supplied with PHPP. Supply air and extract air are worked out differently, but when the system is running they must be equal, that is, in balance.

All projects have to comply with UK building regulations but design teams should always refer to Passivhaus guidance where this gives higher airflow rates. For Passivhaus we recommend supply air is set at 30 m³/h/person (the standard allows for 20 m³/h/person as a minimum but this is not expected to be sufficient for UK homes due to our mild and damp climate) – this should be calculated using the actual number of people expected to live in the house. For under-occupied houses (>40 m² per person) the backstop rate is 0.3 air changes per hour (ACH). This gives the standard airflow rate, which is usually set as the middle fan speed.

When it comes to deciding air flow to each room, the basic rules are bedroom supply of at least 15 m³/h/person, so 30 m³/h for a double bedroom and the remainder is available for living spaces. Bedrooms can have a higher ventilation rate, especially if also used as a study. Dining rooms can often rely on air passing through to the kitchen and so don't need their own supply, but this depends on the layout of a particular house.

Extract rates are set according to room type but when measured at the standard rate they will typically be lower than the regulation flow rate, with the required flow only extracted at boost rate. Note that supply and extract are both boosted together so that the supply and extract rates are the same as each other at each fan speed.

Kitchen at Ty Pren Passivhaus. Image: Nick Grant



The lower fan speed is set at 70% of the standard rate, and is used when the dwelling is either unoccupied or occupied by fewer people than usual. The higher fan speed, or boost rate, must be at least 130% of the standard rate. This boost rate also has to provide the extract ventilation rate required for kitchens and bathrooms according to building regulations.

Zehnder ComfoSense and ComfoSwitch control panels showing three fan speed rates.



Stebon Primary School
Image: Architype

Non-domestic buildings are usually designed to national ventilation standards. For schools this is according to Building Bulletin 101 which sets performance criteria in terms of CO₂ levels in classrooms. In practice this leads to design flow rates between about 22-30 m³/h/person, depending on the age of the pupils.

Measuring flow rates at Old Holloway. Image: George Mikurcik





INSTALLATION AND COMMISSIONING

Always use a fully designed ductwork system. For domestic ventilation this would usually be done by the system supplier – however, they may charge for this. Use the ductwork and fittings specified and don't allow shortcuts with flexible or shallow plastic ductwork.

Cleanliness needs to be maintained during construction. Any dirt in the ductwork system will be blown into the building later. Keep ductwork covered before use and cover open ends of ductwork during construction. Do not run the fans at all until after the final clean of the building, to avoid drawing dust and debris into the ductwork.



Above: MVHR unit during installation at Fishleys Passivhaus.
Image: Liz Taylor and Mike Hill

Commissioning is the job of setting the system up with the correct flow rates. This can only be done once the house is clean. Traditional branched systems need to have the flow rate for each room adjusted at the terminal, though good ductwork system design will give required settings (in mm). Semi-rigid systems may come designed with specific flow regulating inserts for the duct to each room. It is still necessary to check the flow at each terminal.

don't

- just stick the ventilation unit in the loft
- omit silencers
- accept shoddy ductwork installation

Hand over the system with a clear description, and demonstration of how to operate the controls and how to change filters. Also provide information on where to get replacement filters and written and illustrated instructions on how to change the filters when the time comes 3 or 6 months later.

Housing providers and landlords should highlight ventilation in resident induction, not only the first time but also for all future change-overs. A ventilation system services plan should be set up at the start of any maintenance programme, and ventilation faults should be treated as urgent. Involvement of maintenance expertise at design and specification stages is strongly recommended.

MAINTENANCE

The primary maintenance requirement is changing filters, including any extra on the kitchen grille. Some ventilation units maintain a constant air flow rate when filters get clogged, but in so doing the fans get noisier. Older or cheaper systems just supply less air. So regular filter changes are important. Write the date of install on the filter. Most control units have a filter-change indicator but this is based on number of days.

It is an advantage if occupants can periodically check the system: some systems are so quiet that occupants may not notice if the fans stop running. Over a few years dust builds up on extract valves. These can be removed and cleaned, being careful not to change airflow settings or mix valves between rooms. Terminals should be removed for redecoration, but must be labelled by room if more than one is removed.

Any MVHR error messages and faults should be reported and addressed straight away. An MVHR failure is as much an emergency as a heating system breakdown! The MVHR system should be fully serviced every four to five years.

do

- make space for the ventilation unit at the start of the design process
- provide routes for ducts – think of octopus tentacles not just typical section drawings
- fill in the Ventilation worksheet before completing the PHPP
- use a PHI certified ventilation unit
- get a complete ductwork design before starting installation
- have the system commissioned on completion – you need to do this for certification
- use a commissioning specialist familiar with Passivhaus systems

SERVICING CHECKLIST

- Confirm airflow rates are as design - either from MVHR controls or measurement at terminals
- Open MVHR unit, take out heat exchanger & inspect - clean if necessary
- Clean around filter housings
- Check fans for dirt and damage and clean if necessary
- Check bypass valve
- Check preheater and clean if necessary
- Take out condensate trap and wash through, then check condensate is draining properly from MVHR.

RETROFIT

The ventilation system in an EnerPHit building has to have the same performance as that in a new build Passivhaus. Fitting ducts into an existing building is often tricky and for schools it may be appropriate to use one MVHR in each classroom. Some classroom MVHRs are designed to work without ducts, other than short intake and exhaust through the wall.

For dwellings, the lack of designed-in space for ducts can be addressed with lowered ceilings in circulation areas, providing access to adjoining rooms, or routes above kitchen cupboards and through bathroom riser spaces.

CASE STUDY: Carlton Chapel House



This deep retrofit of a three-storey block of 15 flats in Kentish Town to the EnerPHit standard has improved the lives of its residents and contributed to positive climate action.

With energy demand massively reduced, the projected heating load for each flat was just 500W. Direct electric heating using simple wall-hung panel heaters was chosen to heat the homes, avoiding either the expense of a heat pump to meet such a low demand and pipework distribution losses in a communal heating system. Choosing electric over gas made sense for other reasons too: the standing charge alone for gas would have equalled the actual cost of heating. And the roof houses a modest array of solar photovoltaics which provide electricity directly to each flat for domestic hot water.

With the limited space available, careful planning to ensure a compact services installation was key. The flats had no ceiling voids and headroom below 2.4 metres so ductwork routes were limited. The approach used was to run ducts above kitchen wall units, with MVHR in one of the cupboards, and through the bathroom riser space to the bedroom at the rear of the flat.



Left: PV installation on the roof

Below: small MVHR unit (on the far left) and ductwork discreetly built into and over kitchen cupboards.





Heating & hot water

INTRODUCTION

A Passivhaus building can in theory be heated via the fresh air supply – by heating supply air to around 50°C. This idea of doing away with a conventional heat distribution system helped define the Passivhaus standard, but isn't necessarily the best choice. Any heating system can be used; a standard heat distribution will be smaller and cheaper in a Passivhaus than a conventional building. This also results in less embodied carbon.

HEATING LOADS

PHPP works out an overall heat load, in terms of watts and W/m² at the coldest expected conditions. This gives a continuous heat load – ie, assuming the heating is on 24/7 – and is aimed at seeing if heating via the air is possible. Continuous heating also suits heat pumps, and the small overnight temperature drop in a Passivhaus means there is minimal penalty to heat continuously, even in non-residential buildings.

When using air source heat pumps in domestic buildings there is actually a problem that Passivhaus heat loads are lower than the output of the smallest heat pumps. So in this instance the heat pump may not be running 24 hours a day after all, and also the radiators have to be sufficiently large to dispense the minimum output of the heat pump.

For designing radiator systems, you can use a simple room-by-room steady state heat loss model. Instead of using default ventilation rates (as often used in MCS calculations) it is important to take account of airtightness and MVHR. You can read off the effective ventilation and infiltration rate from the PHPP heat load sheet – for example, a ventilation rate of 0.5 ACH and 90% heat recovery is effectively $0.5 \times (1 - 0.9) = 0.05$ ACH for heat loss purposes.



Passivhaus reduces the need for services! Photo: Nick Grant

To work with renewable generation and minimise carbon emissions, heat pumps are the usual heat source. Small heat loads may be met with direct electricity – especially where this avoids disproportionate heat distribution losses – but running cost needs to be considered.

HEATING

In residential buildings annual hot water demand can exceed heating demand. However, the winter peak of heating demand may be the hardest to address in a renewable energy system. With the PER approach, PHPP weights the impact of different uses according to seasonal renewable energy availability, so despite the low heating demand in a Passivhaus, heat must still be generated efficiently.

Water tank located in a cupboard. Photo: Alan Clarke



HEATING SYSTEMS

Radiators make for a cheap heating system, and are also an excellent match for a Passivhaus. Thanks to triple-glazed windows, you can put them where you like in a room as there are no draught comfort issues. Individual room control with Thermostatic Radiator Valves (TRVs) allows the system to respond to solar and internal heat gains, which provide a far greater fraction of the heating than in a conventional building.

With a heat pump, the heating system temperature is critical to performance – energy usage is roughly proportional to the difference between outside and heating system temperatures, so low temperature systems are better. But the size of radiators needed increases as flow temperature goes down, so a balance must be struck, usually in the 30-40°C range.



Dining room with radiator at the Woodhouse Passivhaus in Presteigne. Image: Charles Grylls

Underfloor heating certainly works in a Passivhaus, but can be an expensive option – it is a radiant source so you need to install it in every room you want to be warm. This then means that the system

can be too powerful for the heat load – and needs careful control to avoid overheating. If you can keep the floor temperature down to a degree or two above room temperature, however, the heat output will be self-limiting – once the room is warmer than the floor it will stop heating the room.

The advantage of underfloor heating when using a heat pump is that the flow temperatures required are usually less than 30°C (depending on floor construction and floor coverings) which maximises the performance of the heat pump. This requires no temperature reducing mixing valves at manifolds, which were generally needed in high temperature boiler systems.

COMMUNAL HEATING SYSTEMS

The traditional approach to heating in multi-residential buildings has been to pipe hot water from central boilers (or district heating) to all flats. Here it was used for space heating and to generate hot water via a heat exchanger. The heat loss from the distribution pipework was a significant heat loss for conventional buildings and is proportionally worse for Passivhaus ones, and is sufficient to cause overheating.

Moving away from high temperature heat sources has led to a range of options. Instead of sticking with the conventional approach and trying to minimise the temperature, some are separating out heating and hot water – then the heating circuit can switch to cooling in summer. One new approach is to connect a small water-to-water heat pump to the heating circuit in each flat to generate hot water. More usually, hot water is provided by direct electric heating or central hot water generation. Alternatively, the “ambient loop” approach distributes water at around room temperature for local heat pumps to use to generate both heat and hot water.

Underfloor heating at Bristol Passivhaus Plus. Image: Greenheart Construction



HEATING SOURCES

Air source heat pumps (ASHPs) are a practical option for a Passivhaus. These have an outdoor unit which uses a refrigerant system to generate heat from lower temperature outdoor air. Modern heat pumps are able to heat hot water without needing direct electric boost (immersion heater), which is important considering that hot water often forms the majority of their annual usage in domestic Passivhaus.



Air source heat pump. Image: Alan Clarke

This weighting towards hot water generation also means that the overall performance of the heat pump may be lower than advertised when assessed according to standardised annual usage profiles (as used for the SCOP rating). Remember that some of the heating demand is also met by hot water losses too.

PHPP has a detailed section for heat pumps which takes the test data normally quoted by manufacturers for a range of operating temperatures, and uses the calculated heating and hot water loads specific to the building to estimate the electricity consumption and seasonal performance (SPF) of the heat pump with more accuracy than a standardised SCOP rating.

Ground source heat pumps are also possible, though the advantage of using the ground (as opposed to air) as a low grade heat source in winter is reduced since heating is a small proportion of the total demand, and summer hot water efficiency will be lower than an equivalent air source heat pump. So the high cost of installing ground collectors is not normally justified.

Heat pumps can also heat the air in a room directly – this is actually the usual method around the world, since it lends itself to either heating or cooling, and is generally cheaper than installing a separate water filled heating system. This is known as a split system (see image right), since the refrigerant is transferred from the outdoor unit to the indoor – unlike the usual “Monobloc” air source

heat pump which contains all the refrigerant pipework in the outdoor unit. There are also multi-split systems for multiple indoor units connected to one outdoor unit. Large systems for non-domestic buildings may use variable refrigerant flow (VRF) systems, which can be designed to heat some areas at the same time as cooling others.

Hot water can be heated separately by its own heat pump – this is usually using a hot water cylinder packaged with a small air source heat pump on top. Usually in the UK outdoor air is ducted to and from the unit, but in warmer climates the unit may just draw air, and hence heat from the room it is in, so providing some cooling as well.

On a larger scale, hot water is most efficiently generated with a heat pump using CO₂ as a refrigerant. As well as having insignificant global warming potential compared with conventional hydrofluorocarbon refrigerants, CO₂ has unusual properties which lead to high efficiency heating water all the way from cold up to hot, and they can heat to 70°C. There is some scope for using these heat pumps for communal systems, provided the heating return temperature can be kept very low – in effect such systems are often primarily hot water systems anyway.

CO₂ has also been offered as a refrigerant for domestic air source heat pumps. But its poorer performance for heating compared with standard heat pumps has made them unattractive even for Passivhaus. Another refrigerant making a comeback is propane – this had fallen out of favour due to flammability concerns, but is now used in Monobloc heat pumps, including some hot water heaters. This refrigerant has the advantage of very low global warming potential, and can also heat hot water to higher temperatures than the usual refrigerants.



2.5 kW Mitsubishi minisplit installed at Old Holloway – sufficient to heat the whole building. Image: George Mikurcik

HOT WATER SYSTEMS

Hot water often uses more energy than heating in a Passivhaus so it pays to concentrate on designing an efficient system. Use the DHW sheet in PHPP at an early stage in design, since the losses here are included in the summer overheating calculation.

Hot water heat losses fall into three categories:

- cylinder
- continuous (secondary) circulation
- and draw-off dead leg.

In addition there is the primary heat loss between the heat source and the cylinder. Cylinder heat loss is usually quoted in terms of kilowatt hours every 24 hours, so to convert this into watts, multiply by 1000 and divide by 24.

Circulation loss only applies if you have a pumped hot water loop – not normally necessary in houses. If you do have one, then insulation levels and pipe length are the important factors to optimise. Circulation losses can be very significant in non-domestic systems and communal systems. PHPP has detailed entries to ensure this is estimated as accurately as possible. Rule of thumb or percentage losses are not used in Passivhaus designs.

Draw-off losses represent the cooling down of the hot water pipe and its contents after each period of use. Here insulation is not significant – the pipe will cool down anyway: the important factors are pipe diameter and length. Shorter pipes have the advantage of lower pressure loss, which means it may be possible to use smaller diameter pipework. See AECB water standards guidance for details of radial small bore distribution systems.

Cylinder location is the key to efficient hot water distribution. It needs to be as central to the various hot taps as possible in order to minimise the total draw-off pipe length and avoid the need for secondary circulation. A house layout which groups kitchen, utility and bathrooms in one area is also very beneficial. To minimise the primary pipework heat loss the boiler or heat pump also needs to be as near to the cylinder as practical.

Multi-residential schemes with communal heating usually use a heat interface unit (HIU) to generate instant hot water in a similar way to a combi boiler. HIUs traditionally had a bypass valve which let some of the heating system water flow through continually to keep the pipework warm and ready to generate hot water. But this causes high distribution losses, so HIUs with electrically controlled bypass, ie only operated when really needed, are better.

Non-domestic buildings often have much smaller hot water demands than dwellings so for a WC hand wash basin, or a kitchenette sink, it is likely

to be more efficient to use local direct electric hot water heating. Instantaneous heating is ideal since there are no storage losses, though the flow rate is limited from such heaters; a cleaner's sink would usually require some storage.

Despite having a large number of taps, schools still have low hot water usage. An example of a Passivhaus school using direct electric hot water heating is Wilkinson primary school. This used a number of storage water heaters, with small bore pipework to distribute to nearby basins and classroom sinks to avoid the proliferation of heaters at every sink.

HOT WATER HEAT RECOVERY

One of the largest uses of hot water is for showers and here there is a good option for energy efficiency. A heat exchanger carrying the waste warm water to the drainage can transfer heat to incoming cold water. This warmed water runs to the shower to reduce the heating needed. With an electrically heated shower the plumbing is simple, but with central hot water storage the pre-heated water would run to the cold tap of the shower and maybe also to the cold feed of the hot water tank – though the latter can only be done for one heat exchanger in the system.

The heat recovery unit may be a long vertical pipe installed in the storey below the shower, or a flat heat exchanger installed under a shower tray – these are less efficient but the only option for flats or ground floor showers. This type may also go under a bath, but they only recover heat when a shower is used over the bath, since they rely on the use of hot water at the same time as the waste water goes to the drain.



Shower save WWHR installation

do

- use a cheap and simple heating system – the investment belongs in the building fabric
- consider radiators as your first option
- arrange the layout to keep hot water outlets in a reasonably compact form
- put the hot water storage in the same area as the hot water outlets
- take care not to oversize heat pumps
- fill in the PHPP DHW section before looking at overheating

don't

- use fossil fuels
- spend a lot of money on a biomass boiler or bore-hole heat pump system
- waste money on a wood burner



Solar thermal and photovoltaics (PV)

PHOTOVOLTAICS

In the past photovoltaic generation was very expensive thanks to the cost of the high tech semiconductor factories needed to make the collectors. Now mass production has brought costs down to a level where PV makes economic sense.

Smart meters allow for measurement of PV export and new tariffs may make simply exporting surplus generation a worthwhile approach for installations with such meters. Traditional electricity meters don't do this, so it has been in the householder's interest to use as much as possible of their PV generation. Immersion heater controllers are available which divert any surplus power to an immersion heater, providing effectively a cheap solar hot water system. However, this approach is not possible if a heat pump uses the immersion heater for high temperature boost. In fact the solar energy would be more efficiently used generating hot water by powering the heat pump. This is now possible, using heat pump controls that link to the PV systems, telling the heat pump to stock up on hot water when the PV system is generating surplus power.

Another way to increase self-consumption of PV generation is to use a battery, especially when decent export tariffs are available, though this is an expensive option and the environmental benefits are debatable. PHPP doesn't account for use of PV power for hot water or in batteries directly, though does allow a fixed share of PV generation to contribute to the PER total.

SOLAR THERMAL

Solar thermal has traditionally been the cheaper and lower technology way to harvest solar energy, though with current PV prices this is no longer the case. The heat can only be used in the house where it is installed. PHPP recognises this as contributing to reduced primary energy consumption because less fuel is needed for water heating. Systems need to be carefully designed to provide as much useful hot water as possible while avoiding overheating in summer. They may suffer loss of performance if not regularly maintained.

EMBODIED CARBON

While it is important to consider the embodied carbon associated with the energy-intensive manufacture of PV collectors and batteries, it is equally essential to recognise the potential benefits of PV systems at the building level. PV installations enable on-site renewable energy generation, reducing reliance on grid electricity and promoting self-sufficiency. They contribute to the transition towards cleaner and more sustainable energy sources. The suitability of PV systems should be evaluated in the context of local conditions, energy needs, embodied carbon and the broader sustainability goals of the project.

do

- consider the embodied carbon associated with PV collectors and batteries
- consider PV in the roof layout, even if not installing panels straight away. Eg. move any plant and roof access to the north where possible, and minimise parapet heights
- use PV panels that have a linear power warranty for at least 10 years
- include power optimisation on any PV panels that have any partial shading, or that will be difficult to clean
- use MCS products and installers
- ensure solar thermal has regular inspection and maintenance

don't

- lay PV panels flat – air born dirt will quickly reduce the generation
- undersize your PV inverter
- specify an over-sized solar thermal system

PV or solar thermal - or both? Renewables installation on a green roof at Ostro Passivhaus. Image: Paper Igloo





Other electrical use

Unlike other UK energy assessment methods, PHPP includes all energy use within a building. For a dwelling this includes appliances as well as lighting, though if these are not supplied with the build, then default figures are used. For non-domestic buildings the assessment of this energy use is both important and onerous. In Passivhaus schools the heating and hot water are a small part of the total energy use. Other key areas are lighting, IT systems, and kitchens. In the latter, the

energy use per meal can be assumed as a default but refrigeration needs to be assessed, requiring selection of efficient appliances which are likely to be supplied outside the building contract. Similarly, IT energy consumption will depend on specifications and installations from specialists who work for the client rather than the builder, and often coming to the design relatively late in the process.

Smaller areas of energy use such as fire detection and alarm systems, sprinkler frost protection, lifts, communications systems and controls do usually come under the main contract but are supplied by specialists who aren't used to addressing energy use and are often not engaged when construction contracts are signed. However, this is a stage when everyone would like to know if the design has reached the Passivhaus standard, so some estimates may be required, along with performance specification to limit the energy use of higher consumption systems.

do

- consider all energy uses in the building at an early stage
- issue an "information request schedule" to contractor and client to specify what information is needed and by when, in order to enable the Passivhaus model to keep up with the design and procurement timetable

don't

- assume Passivhaus is the same as SBEM

CASE STUDY: Foleshill Health Centre



Foleshill Health Centre in Coventry is the UK's first Passivhaus certified healthcare facility and is the most energy-efficient public building in the NHS estate. Point-of-use heaters were chosen to provide the hot water, so reducing hot water circulation and the concomitant risk of overheating, and also lowering the burden of maintenance for legionella prevention. To get the ventilation strategy right, a detailed room-by-room analysis was undertaken, leading to the solution of separate MVHR systems for each of the two treatment rooms.

The simpler and smaller services required for this Passivhaus health centre have a smaller capital cost than business-as-usual, as well as reducing energy bills. A 40-year cost analysis undertaken by AECOM predicts that the building will save the NHS £450,000 over that period.



Foleshill Health Centre. Images: Tooley Foster.



INTRODUCTION

With a sound design already in place, exemplary standards of on-site quality assurance are required in order for buildings to perform appropriately. This means diligence and excellent standards of craftsmanship must be employed at all times. For a project to be completed successfully, with minimised costly rework and remediation, experience suggests that there is a learning curve that needs to be addressed. This learning curve is not confined to the site office; it affects all trades on site.

KEY STAGE DESIGN REVIEWS

Structured reviews, preferably with the contractor, that interrogate a design are hugely beneficial, particularly for less experienced project teams. These reviews analyse each and every aspect of the design and the construction. Key characteristics:

- they are not simply 'compliance reviews' – they dig into the details and examine the buildability, sequencing and construction programme
- their purpose is to identify and expose risks that could impact upon project delivery
- they allow the project team the opportunity to formulate strategies that resolve the risks in an open and collaborative fashion.

TRAINING OF SITE OPERATIVES (TOOLBOX TALKS)

On some building sites, traditional attitudes can be entrenched and effort must be made to bring all site operatives on board. Each trade, from foundations to roof, has a part to play in the successful completion of a project. One of the best techniques is to hold a series of toolbox talks that



An Airtightness Team, comprising two Airtightness Champions, two Sealing Operatives also including the Site Manager, on the Erneley Close project. Photo: Paul Jennings

all site personnel are required to attend. New or replacement personnel should also receive an induction. Typically, these toolbox talks examine each detail, the sequence in which it will be constructed, the technologies that will be used when forming the detail, and trades that will impact upon the successful completion of the detail.

An atmosphere of collaboration helps these talks to work best and allows any sceptics to voice their concerns. Inviting sceptics to speak up is vital because:

- they may have a valid point and the sessions create an opportunity to address relevant concerns
- there is an opportunity to address their concerns and to bring them on board
- in the most extreme cases there is the opportunity to identify people that will obstruct the success of the project. They can then be replaced by more appropriate trades people.

GOOD PRACTICE EXAMPLE

PHPP calculations suggested that the design had the potential to satisfy the Passivhaus Standard. The challenge lay in the actual implementation of the delivery process.

A series of Key Stage Design Reviews, training of the dedicated, on-site Quality Assurance Champion, toolbox talks and site inspections and feedback underpinned the successful completion of the Certified Passivhaus homes at

St. Mary's, Oldham. The result of these processes resulted in buildable details and simplified construction processes.

Site manager Anthony Kavanagh says: *'Compared to the Code 6 homes that are also on site I would prefer to build to the Passivhaus standard. The construction details that we have developed with Mark are proving to be far more practical and easier to build.'*

Training of site operatives, undertaking detailed site inspections, provision of robust feedback and the inclusion of a dedicated, on-site quality assurance champion are all constituent parts of a successful delivery process. This chapter outlines those areas that require specific attention.



A DEDICATED, ON-SITE QUALITY ASSURANCE CHAMPION

The role of a dedicated, on-site Quality Assurance Champion is to make sure that the building satisfies its airtightness target. The role is to also help ensure that the correct insulation system is installed properly. This means ensuring that:

- the interface between insulation products has a good butt joint
- the insulation is encapsulated between the air barrier and the wind barrier
- the wind barrier system is completed in accordance with the design intent.

Their role is also to:

- organise and arrange pressure tests
- arrange for timely toolbox talks for all trades
- review the buildability of the air barrier, wind barrier and insulation systems – this means reviewing the sequencing of the build process and considering where challenges may arise before they become an obstacle
- undertake day to day site inspections of the work and instruct remediation as required
- ensure that only the specified materials are used in the construction i.e. avoid ad hoc substitutes, without prior consultation and agreement with the designer.

SITE REVIEW REPORT Inspection undertaken by Mark Siddall

Project number: 117 Project name: Date: 31.08.12

Weather: Clear Sky Site: Review: 004

Introduction:

The purpose of this report is to:

- Identify risks that may influence whether the building will achieve the required standards of performance
- Identify risks that may impair the successful formation of the air barrier
- Provide a photographic record of the typical and atypical faults that may impede the delivery of a successful air barrier

Within the scope of the work, time constraints and availability of access it was not possible to physically inspect all aspects of the building envelope. As site inspections are intermittent, with periods of time between visits, it is recognised that the formation of the air barrier may be incomplete and that there may be defects that have been concealed during the construction process.

Progress:

- Scaffold lift up to second floor.
- Second floor structure installed

Risk Status:

HIGH	MEDIUM	LOW
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Observations:

Figure	Observation	Recommendations	Owner
1	Party wall insulation was found not to be filling the cavity. Concerns regarding party wall bypass.	Remove and reinstall	Bramall
2	The party wall insulation was found not to conform with the specification.	Remove and reinstall with appropriate material	Bramall
3	Horizontal gaps in the insulation are apparent.	Assess cause of gaps and remediate	
4	Insulation is not installed tightly against the warm side of the cavity. Thermal bypass risks incurred.	Ensure that insulation is tightly installed against the warm side of the cavity.	
5	Corners are being formed in a neat fashion but are not being interlaced as they turn the corner.	Consider improving practice: greater interlacing of the batts as they turn the corner would reduce the risk of gaps being formed.	
6	Insulation generally appears to be installed tightly against the warm side of the cavity. Insulation at external face sometimes compressed by 10mm due to mortar joints.		
7	Insulation to window/door openings has been installed in sloppy manner. Insulation to window/door openings has sagged.	Rework will be required to allow install of window/doors. Care will need to be taken to avoid gaps between insulation and frames.	

Distribution

☒ Site Manager (Construction)
 ☒ Project Manager (Construction)
 ☒ Contract Manager (Construction)
 ☒ Mech. Eng.
 ☒ Architect
 ☒ File
 ☒ Passivhaus Overlay

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Example of Site Report used by Mark Siddall for a Passivhaus development.

don't

- hope for the best
- fail to plan for success
- rely upon standard industry practice when planning construction programmes
- forget to undertake site inspections and feedback

do

- allow time for planning (it will pay back in the future)
- agree and undertake key stage design reviews with the Passivhaus Designer and Passivhaus Certifier
- train site operatives
- ensure that there is a dedicated, on-site Quality Assurance Champion
- remediate mistakes ASAP (when you still have the chance).
- adopt the Passivhaus Overlay for the RIBA Plan of Work

CORE COMPONENTS OF A PASSIVHAUS PROJECT MANAGEMENT CHECKLIST

By failing to recognise the boundary between a certification process and the design and construction process, individual projects are at risk and may encounter difficulties. In part this may be because the industry has become used to certification systems that are less rigorous and have permitted this gap to go unobserved.

Listed below are the major components that are included on a Passivhaus Project Management

checklist. Each component, and its constituent sub-components, should be coordinated with relevant BS EN standards. When a building is to be certified it is vital that this checklist is developed and agreed with an approved Passivhaus Certifier.

Photographic evidence of the construction should be gathered at key stages by appropriate members of the design and construction team. The Passivhaus Designer/Consultant should assist with agreeing the regime for compiling evidence, on a project specific basis.

TRAINING/TOOLBOX TALKS (PRE-START)

- Site storage
- Workmanship
- Activities to be undertaken
- Sequencing of activities



Ductwork protected from site debris during storage.

AIRTIGHTNESS – MATERIALS AND WORKMANSHIP

- Primary air barrier system
- Window installation
- Service penetrations



Air barrier installed in a manner that allows easy inspection and remediation during construction.

INSULATION INSTALLATION – MATERIALS AND WORKMANSHIP

- Walls, roof, floor, windows
- Junctions
- Services



Check construction tolerances. Insulation is encapsulated tightly between internal and external leaf to avoid thermal bypass.

SERVICES – MATERIALS AND WORKMANSHIP

- MVHR unit installation
- MVHR ductwork & silencers
- DHW
- Pipes and plumbing
- Heat sources
- Controls



Ductwork protected from site debris during installation.

WINDTIGHTNESS – MATERIALS AND WORKMANSHIP

- Primary wind barrier system
- Window installation
- Service penetrations



Wind barrier installed in a manner that allows easy inspection and remediation during construction.

BUILDERS' WORK

- Joinery (door over/undercuts etc)



Air transfer provision is checked against design drawings and specification.

CASE STUDY: Closeburn Passivhoos



Above and left: the completed homes. Images: Tom Manley Photography

Below: under construction. Image: John Gilbert Architects



These three homes at Castle Crescent, Closeburn are the first community-owned Passivhaus homes in Scotland. They are owned in perpetuity by the Closeburn community, via Nith Valley Leaf Trust, and rented to people with a local connection, on an affordable basis.

Closeburn Passivhaus was the first project to benefit from the development of a construction method designed to lower costs, improve quality, speed up construction on site, and to be replicable at scale. In summer 2019 the team of John Gilbert Architects and contractor Stewart and Shields worked together to build a full-size Passivhoos prototype in a factory unit in Port Glasgow. This allowed them to develop the stick built timber frame, test the sequence of construction and discuss further simplification of the airtightness layer. It was used to train all of Stewart and Shields apprentices and staff. Site meetings allowed the team to refine details to enable the houses to be built quicker and better on site.

The I-joist timber frame with airtight OSB internally was insulated with mineral wool plus 50mm of phenolic insulation board externally. Balloon frame construction avoided a thermal bridge at the first floor to external wall junction. The timber frame construction and timber cladding reduced embodied carbon of construction, and embodied carbon calculations measured progress in line with RIBA 2030 challenge targets. Tenants moved into their new homes in 2020, and Passivhaus certification followed a few months later.

INTRODUCTION

Buildings don't use energy: people do. Occupants play an important part in determining the energy performance and comfortable operation of a building and it is highly recommended that a soft landings programme* is planned in from the outset of any project. This is not specific to Passivhaus but a universal best practice approach.

A soft landings programme offers multiple benefits: user satisfaction, fine tuning the building systems, and ensuring the designed energy performance has been met. The soft landings and post-occupancy evaluation process also close the feedback loop, providing invaluable information for the client and the construction team alike.

This chapter outlines a soft landings approach with a focus on Passivhaus. Some aspects of using a Passivhaus can be different to occupying a conventionally designed and serviced building. The handover process and maintenance regime will need to be tailored to suit different heating, ventilation and cooling methods, different equipment and component demands, and different occupant types. Occupants should be appropriately inducted at handover stage and supported during the post occupancy stage.

OCCUPANT COMFORT AND ENGAGEMENT

Depending on the size, location and function of the building, the operation of a Passivhaus could involve different levels of user engagement. Although Passivhaus methodology doesn't require the occupants to make a significant behaviour or lifestyle change, it is still important that the users are aware of the principles and particularities of operating a Passivhaus, where certain behaviour adaptations are welcomed to ensure optimum energy performance and comfort. At the same time, appreciating that the building is carefully designed to keep them comfortable and healthy at a low cost, and how it achieves this, will encourage residents to actively engage with its ongoing use and maintenance.

As always, understanding is the key. **For the occupant**, a clear understanding of the controls and techniques available to them will ensure they can achieve a comfortable environment with very low energy demands. **For those responsible for maintenance**, an understanding of the sensitivities and contribution of components to the overall performance will ensure the comfort and energy qualities are delivered as designed.

The following pages give a simple 4-step guide for the delivery team to provide a high level of support, which will be most useful for social housing providers, but can be adapted to suit other projects.

LEARN MORE



pht.guide/benefits

*<https://www.bsria.com/uk/consultancy/project-improvement/soft-landings/>

Tree planting at Sharnbrook



Occupants are key to the energy performance and comfortable operation of a building, and implementation of a soft landings programme and post-occupancy evaluation is best practice on any project. In a Passivhaus in particular, the handover process and maintenance often have to be tailored to suit the specific services installation.



STEP 1: UNDERSTANDING THE PRINCIPLES – WHAT TO EXPECT

The prospective occupants should be introduced to the basic principles of Passivhaus prior to their move-in day. This could be through a series of client meetings and user consultations from the initiation of the project all the way to the construction and handover, or through a simple illustrated guide explaining how Passivhaus provides comfort and reduces energy consumption, and how using Passivhaus might be slightly different from the kind of conventional building that they are accustomed to. Most of the time, a combination of both is the most effective way to establish this basic understanding.

If energy performance estimation is required by the occupants, this should be supplied as a range, and should be explained that the energy consumption will partly be determined by how the occupants use the building.

It is highly recommended that a soft landings framework is agreed upon from the outset of the project, including Extended Aftercare and Post Occupancy Evaluation (POE). The specific process could be tailored to suit each project.

“ Stunning. As soon as you walked in you just felt at home. It was lovely... We have a system which takes out the bad air and brings in good air, and it also rotates heating around the property so you save on heating because they're also triple glazed and well insulated. We rarely have the heating on, all the time.”

**Resident at Primrose Park,
Passivhaus development in Plymouth**

STEP 2: FAMILIARISATION WITH THE CONTROLS – ‘QUICK START’

In preparation for the move-in day, the building system should be ready-set and functioning. Consider the provision of stickers or laminated graphic instructions affixed to plant where maintenance is to be carried out by occupants or untrained persons.

On the move-in day, provide the occupants with a very simple ‘Quick Start’ guide to the various controls and techniques, checked and proof-read by a non-technical person, ideally someone fluent in the occupant’s language. It should contain a short simple overview followed by a number of simple headings such as ‘what to do if I’m feeling too cold’, etc. The advice should be limited under each section to no more than a couple of paragraphs. Including photographs or graphics of the controls being discussed, with simple descriptions of each function, will be helpful.

This ‘Quick Start’ guide should be both physical and electronic. Consider making a short yet specific ‘how to’ video of each topic for posting online, which will cater for a wider generation and more neurodiverse occupants. The occupants can take their time to go through the ‘Quick Start’ guide by themselves after they have settled in, without being overloaded on the move-in day with further demonstrations and explanations.

Identify who to call when faults or more complex maintenance tasks require action in the guide, and if necessary, include a script to use. Complications often arise with inexperienced call centres who may not know what a Passivhaus is or what MVHR stands for.

More technical or complex documents should be included as appendices within the overall building manual or log book.

Quick Start Guide Image: John Gilbert Architects





STEP 3: LEARNING AS A COMMUNITY – TRIAL AND ERROR

During the initial occupancy stage, a learning/feedback session should be arranged no later than two weeks after the occupants have moved in. Consider who should lead the session – it is important not to overwhelm or frighten the occupants with hoards of technical experts. In a relaxed atmosphere, work through the 'Quick Start' guide (perhaps using a checklist) and explain each of the main functions. Involve occupants in hands-on experience, and avoid the temptation to demonstrate. Learning by doing through trial and error is far more effective and confidence-building. Questions often occur after the session so provide a further point of contact.

Repeating a similar session at the following change to winter/summer provides an opportunity to catch up with how the users have coped so far as well as to go over how users might control for cooling instead of heating, for example. Those sessions are important in fine tuning the systems and ensuring proper operation in the first year of occupancy. There should be a designated project team that oversees those sessions and responds to issues raised promptly.

It is recommended to involve as many occupants as possible collectively in this learning process. Learning as a community has proven benefits in encouraging sustainable behaviours and improving the lived experience.

STEP 4: CONTINUING THE SUPPORT – MONITORING AND TROUBLESHOOTING

It is important to continuously support the occupants through the extended aftercare process. Make sure that they know who to contact in case of fault/problem and that the maintenance and fault response team is well-informed and approachable (see chapter 8 above for detailed advice on MVHR maintenance). At the same time, keep the communication channel open for occupants who want to go the extra mile to save energy.

In addition to the two learning/feedback sessions, consider running an informal quarterly energy audit session with the occupants where collective problems can be resolved and experiences/best practices can be shared. Results can be shared with occupants via a regular newsletter which also gives an opportunity to remind residents at season changes of the best ways to optimise their own comfort in their homes – for example, closing blinds during the day at the peak of summer to reduce solar gain and opening windows at night time for cooling purge ventilation.

As part of the extended care, conducting Post Occupancy Evaluation (POE) is an effective way to gather performance and comfort data and occupants' feedback to improve energy efficiency. This could include monitoring energy and internal comfort data as well as BUS (Building Use Studies - busmethodology.org.uk) surveys and thermal imaging reports. This should form part of the soft landings agreement from the start of the project to be sure that the necessary equipment will be in place for the POE.

Castle Crescent, Closeburn. Image: © Tom Manley



do

- provide advance information. The earlier the journey starts, the sooner it finishes
- set the building up to function prior to 'move in' day
- provide a very simple 'Quick Start' guide, stickers or posters where appropriate
- consider accessible video guides
- familiarise occupants through hands-on experiences
- re-familiarise at change of season
- ensure maintenance teams and fault response teams are well informed
- provide continuous support throughout the soft landings process
- involve maintenance teams in the design

don't

- over complicate user guides
- wait till the 'move in' day
- dump all information in one go
- frighten occupants with complicated or technical explanations
- restrict occupants' behaviour or design out the role of the occupants
- leave the maintenance team in the dark



Projects featured in this guide

Unless otherwise mentioned, all projects featured or pictured in this guide are certified Passivhaus or EnerPHit buildings.

Find links to full case studies at pht.guide/howto, including details of project teams and performance data. Thanks to all Passivhaus Trust members who have contributed images and illustrations.

Next steps

- Find a Passivhaus expert to help you with your project:
bit.ly/PHTmembers
- Get inspiration from the UK Passivhaus Awards
bit.ly/PHTawardshalloffame
- Explore Passivhaus for particular sectors and building types
bit.ly/PHTprojectsectors
- Extend your own Passivhaus knowledge with further training and events
<https://www.passivhaustrust.org.uk/events.php>



Join the Passivhaus Trust!

Membership of the Passivhaus Trust is open to all and gives you the opportunity to:

- Influence policy-makers, developers, clients and other key players
- Network with leading Passivhaus experts, practitioners and academics
- Access the latest technical information and experience relating to Passivhaus in the UK
- Connect with the broader international Passivhaus community, through iPHA (the International Passive House Association)

The Passivhaus Trust is an independent, non-profit organisation that provides leadership in the UK for the adoption of the Passivhaus standard and methodology.

Passivhaus is the leading international low energy design standard, backed with over 30 years of building performance evidence. It is a tried & tested solution that enables a meaningful transition to net-zero now. Over 65,000 buildings have been certified to this standard worldwide. The Trust promotes Passivhaus as a robust way of providing high standards of occupant comfort and health AND slashing energy use and carbon emissions from buildings in the UK.

Please find us on Twitter, LinkedIn, Instagram, & Facebook @PassivhausTrust. Keep up to date with all things Passivhaus by joining our mailing list.

www.passivhaustrust.org.uk



The UK Passive House Organisation